

Orbitrap Astral Operating Manual

Operating Manual

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Orbitrap Astral

Mass Spectrometer

Operating Manual

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Technical data of Orbitrap Astral systems

The following table summarizes the most important technical data of the Orbitrap Astral system. See the respective chapters of the manual for details and additional instrument properties.

Table Technical data of Orbitrap Astral systems

Parameter	Specification		Value
Instrument properties			
Mass spectrometer	Length × width × height		1,963 mm × 952 mm × 1,394 mm
	Weight		680 kg
Forepump ¹⁾ _{.....}	Length × width × height		654 mm × 420 mm × 344 mm
	Weight		78 kg
Complete system (incl. data system)	Noise emission		< 70 dB(A)
	Heat output		Operation: 1650 W, Bakeout: 2750 W
Power requirements			
Mass spectrometer	Input	Nominal voltage	2 × 208–240 Vac (± 10%), 50/60 Hz, 10 A
		Fuse ²⁾ _{.....}	2 × 15/16 A (tripping characteristic C)
		Operation	Apparent power: 1000 VA; effective power: 900 W
		Bakeout	Apparent power: 2500 VA; effective power: 2000 W
	Output	4x	208–240 Vac, 50/60 Hz, 3 A total
	Protection degree		IP20
	Protection class		Class I
	Overvoltage category		II

Parameter	Specification		Value
Forepump	Input	Nominal voltage	208–240 Vac, 50/60 Hz
		Power	650 W
		Fuse ²⁾	15/16 A (tripping characteristic C)
Data system	Input	Nominal voltage	100–240 Vac, 50/60 Hz
		Power	100 W
		Fuse	15/16 A
Gas requirements			
Source gas (minimum requirement)	Type		Nitrogen
	Purity		99% or greater (high purity)
	Supply rate		Max. 45 NL/min
	Pressure		0.6 ± 0.05 MPa
HCD gas / reagent carrier gas (mandatory)	Type		Nitrogen
	Purity		99.999% or greater (ultra high purity)
	Supply rate		Max. 0.07 NL/min ³⁾
	Pressure		0.6 ± 0.05 MPa
Operating environment			
	Laboratory temperature		18–27 °C
	Max. temperature fluctuation		0.5 °C/10 min
	Humidity		20–80%, non-condensing and non-corrosive atmosphere
	Pollution degree		2
	Max. altitude		3,000 m above sea level
Pump exhaust requirements	Inrush flow rate		10 L/min
	Continuous flow rate		7 L/min

Parameter	Specification	Value
Ion source exhaust requirements	Maximum flow rate	45 L/min

TIP

The unit *NL* refers to liters at normal conditions, measured at a temperature of 0 °C and a pressure of 1013.25 hPa(a). These reference conditions are indicated by the added letter “N” in the unit of volume used.

About this manual

IMPORTANT

Export classification: not listed / Exportklassifizierung: nicht ausfuhrgenehmigungspflichtig

The Orbitrap Astral™ system is a member of the Thermo Scientific™ family of mass spectrometers that are powered by Orbitrap™ technology.

Contents

- Special notices and safety symbols
- Typographical conventions
- Access documentation
- Reference documentation
- Training

This Operating Manual contains precautionary statements that can prevent personal injury, instrument damage, and loss of data if properly followed. It also describes the modes of operation and principle hardware components of your Thermo Scientific instrument. In addition, this manual provides step-by-step instructions for cleaning and maintaining your instrument.

Designed, manufactured and tested in an ISO9001 certified facility, this instrument has been shipped to you from our manufacturing facility in a safe condition. This instrument must be used as described in this manual. Any use of this instrument in a manner other than described here may result in instrument damage and/or operator injury.

3.1 Special notices and safety symbols

Make sure you understand the special notices, symbols, and caution labels in this guide. Most of the special notices and cautions appear in boxes; those pertaining to safety also have corresponding symbols.

NOTICE

Highlights information necessary to prevent damage to software, loss of data, or invalid test results; or might contain information that is critical for optimal performance of the product.

NOTE

Highlights information of general interest.

TIP

Highlights helpful information that can make a task easier.

3.2 Typographical conventions

Different typographical conventions have been established for Thermo Fisher Scientific manuals.

Viewpoint orientation

The expressions *left* and *right* used in this manual always refer to the viewpoint of a person that is facing the front side of the instrument.

Data input

Throughout this manual, the following conventions indicate data input and output with the computer:

- Messages displayed on the screen are represented by capitalizing the initial letter of each word and by italicizing each word.
- Input that you enter by keyboard is identified by quotation marks: single quotes for single characters, double quotes for strings.
- For brevity, expressions such as “choose **File > Directories**” are used rather than “pull down the **File** menu and choose **Directories**.”
- Any command enclosed in angle brackets < > represents a single keystroke. For example, “press <F1>” means press the key labeled *F1*.
- Any command that requires pressing two or more keys simultaneously is shown with a plus sign connecting the keys. For example, “press SHIFT + F1” means press and hold the <Shift> key and then press the <F1> key.
- Any button that you click on the screen is represented in bold face letters. For example, “click **Close**.”

3.3 Access documentation

You can access product documentation using one of the following ways:

Visit the Help Portal

Full documentation (including release notes, installation instructions, operation manuals, and so on) is maintained on the Thermo Fisher Scientific Chromatography and Mass Spectrometry Help Portal. This documentation portal is regularly updated with the latest content, including video tutorials. Go to docs.thermofisher.com.

With a Thermo Fisher Scientific account, you can use features on the documentation portal, such as bookmarking documents, saving searches, and compiling documents into collections for easy reference.

Procedure

1. Go to docs.thermofisher.com/p/TechDocAccess to display a registration form.
2. Fill out the form to request access as a customer. Among other data, you have to provide the serial number (S/N) of your instrument. You can find the serial number of your instrument on the name plate. Refer to the Operating Manual of your instrument for information about its location.
3. Click **Submit Form**. The form contents will be sent using your default email client. You will receive an email with additional login information.

Result

Upon submitting the form, you will be directed to home page of the Help Portal.

View the product documentation on the data system computer

To view the product manuals, do one of the following:

- From the Windows™ taskbar, select **Start > Thermo Instruments**, and then open the applicable PDF file.
- From the Windows explorer, navigate to the following path:
C:\Program Files\Thermo Scientific\Instruments\TNG*instrument mode\version\system\Manuals*

3.4 Reference documentation

This *Orbitrap Astral Operating Manual* represents the Original Operating Instructions. In addition to this manual, Thermo Fisher Scientific provides other documents for the Orbitrap Astral mass spectrometer that are not part of the Original Operating Instructions. Reference documentation for the Orbitrap Astral MS includes the following:

- *Orbitrap Astral Preinstallation Requirements Guide*

This manual contains information on the necessary environmental conditions in the intended location for the instrument.

- *Orbitrap Astral Performance Maintenance Manual*

This manual describes the user maintenance for the quadrupole and the bent flatapole.

- *Orbitrap Astral Software Manual*

This manual describes features of the instrument software.

You can access PDF files of the documents listed above and of this manual from the data system computer. Refer also to the user documentation that is provided by the manufacturers of third-party components:

- Forepump
- Turbomolecular pumps
- Syringe pump
- Data system computer and monitor
- Safety data sheets

3.5 Training

Thermo Fisher Scientific offers worldwide training on instruments and software. Experience has shown that maximum results can be obtained from a scientific instrument if the instrument operators receive an adequate training.

We recommend that the key users participate in a basic operator training. For information on training courses and enrollment, contact your local Thermo Fisher Scientific office.

Scope of delivery

The Orbitrap Astral mass spectrometer is shipped with the following components:

- Orbitrap Astral mass spectrometer
- Dry pump
- API Source with HESI probe
- Calibration ESI probe
- Data system computer with monitor
- Syringe pump
- Installation Kit including
 - Equipment for connecting the above components (hoses, cables)
 - Computer equipment
 - Tools for installation and maintenance
 - Spare parts

In addition to this Operating Manual, other documents for the Orbitrap Astral mass spectrometer are available as PDF files on the data system computer.

Functional description

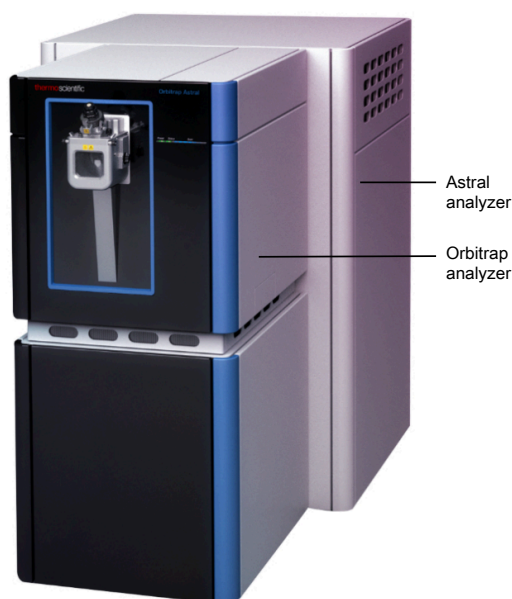
The Orbitrap Astral mass spectrometer is a stand-alone instrument with an atmospheric pressure ionization (API) source for liquid chromatography (LC) mass spectrometry (MS) high-throughput applications. The ions created by the API source can be analyzed with the Orbitrap™ analyzer and the Asymmetric Track Lossless (Astral) analyzer.

Contents

- Layout of the Orbitrap Astral mass spectrometer
- Operating modes of the Orbitrap Astral mass spectrometer
- Instrument front side
- Left instrument side
- Right instrument side
- Instrument rear side
- Ion optics
- Curved linear trap
- Ion-routing multipole
- Orbitrap analyzer
- Astral analyzer
- Vacuum system
- Electronic assemblies
- Cooling fans
- Additional hardware

5.1 Layout of the Orbitrap Astral mass spectrometer

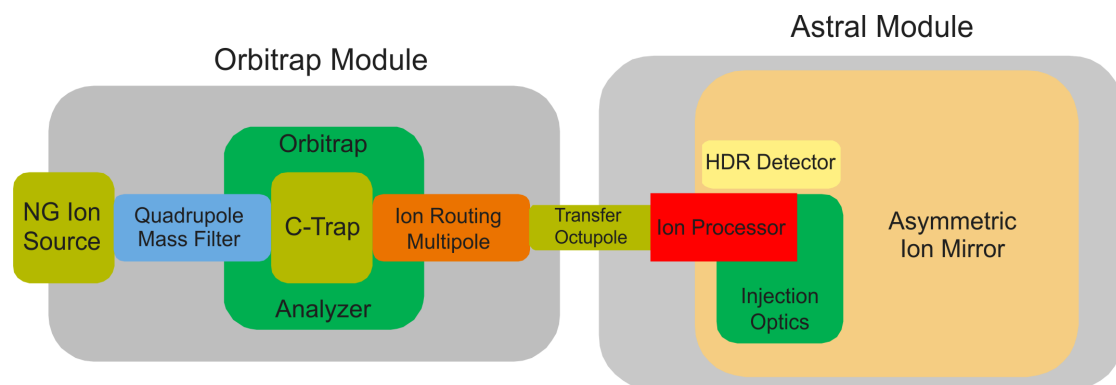
Figure Orbitrap Astral MS



The Orbitrap Astral mass spectrometer consists of the following main components:

- Thermo Scientific™ EASY-IC™ ion source for improved mass-to-charge ratio (m/z) assignment
- Injection filter with mass resolving capabilities
- Quadrupole mass filter for high selectivity and high ion transmission
- Ion-routing multipole for trapping ions and performing HCD (Higher Energy Collisional Dissociation) experiments
- Intermediate storage device (C-Trap) for short pulse injection
- Orbitrap analyzer for high dynamic range HRAM MS
- Asymmetric Track Lossless (Astral) analyzer for fast and sensitive HRAM MS/MS

Figure Orbitrap Astral MS block diagram



5.2 Operating modes of the Orbitrap Astral mass spectrometer

The samples can be introduced into the Thermo Scientific OptaMax NG™ API source at the front side by a variety of methods that include direct infusion (with a syringe pump) or a U-HPLC system (for example, Thermo Scientific Vanquish™).

The IC source delivers calibrant ions into the stream of analyte ions. The instrument control software uses the precisely known mass-to-charge ratio of the calibration mass peak to fine-adjust the instrument's mass-to-charge ratio calibration.

The injection filter transmits the ions from the source to the quadrupole mass filter or the transfer octupole. The injection filter also performs coarse pre-filtering of the ions according to their mass-to-charge ratios.

The quadrupole rod assembly operates as an ion transmission device with the possibility to filter the transmitted ions according to their mass-to-charge ratios. By applying RF voltages and DC voltages to the rods, the filter characteristic is configured.

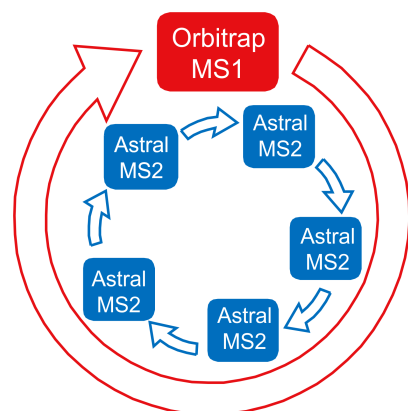
The ions are transferred into the C-Trap through four stages of differential pumping. The ions are passed through the C-Trap into the ion-routing multipole (IRM).

The IRM adds a Higher Energy Collision Induced Dissociation capability to the instrument. In combination with the quadrupole mass filter, this allows MS/MS experiments or all-ion fragmentation (AIF) in case of a broad range of selected mass-to-charge ratios. After the ions have been fragmented in the IRM, the IRM voltages are ramped up and the ions are transferred back into the C-Trap.

From the C-Trap, the ions are then injected through three further stages of differential pumping by a lens system (Z-lens) into the Orbitrap analyzer where mass spectra are acquired by image current detection. The vacuum in the Orbitrap analyzer is maintained below 1E-9 mbar.

The Astral mass analyzer adds fast (up to 200 Hz) and sensitive MS2-level detection that is fully parallelized with Orbitrap mass analyzer scanning.

Figure Orbitrap Astral workflow



The properties of the two analyzers complement each other:

Performance Criteria	Orbitrap Analyzer	Astral Analyzer
Mass Accuracy	+++++	++++
Sensitivity	+++	+++++
Resolution	+++++	++++
Repetition Rate	+++	+++++

5.2.1 Performance specifications of the Orbitrap Astral mass spectrometer

Table Performance specifications – Orbitrap Astral mass spectrometer

Parameter	Value
Orbitrap Analyzer	
Orbitrap mass range	Standard mass range m/z 40–6000, m/z 40–8000 with BioPharma option
Orbitrap resolution	Standard resolution settings range from 7500 to 480,000 (FWHM) at m/z 200, with isotopic fidelity up to 240,000 FWHM
Acquisition rate ¹⁾	Orbitrap MSn acquisition rates up to 40 Hz when measured using a data-dependent experiment with Pierce™ FlexMix™ calibration solution, MS and HCD MS/MS resolution setting of 7500 @ m/z 200
Orbitrap mass accuracy ¹⁾	External calibration achieves <3 ppm RMS drift over 24 hours, measured with FlexMix calibration solution; internal calibration achieves <1 ppm RMS drift over 24 hours, measured with FlexMix calibration solution
MS/MS Electrospray Ionization (ESI) Orbitrap sensitivity	MS/MS: 50 fg reserpine on column S/N 100:1 SIM: 50 fg reserpine on column S/N 150:1
Dynamic range	>5000 within a single Orbitrap mass spectrum
Astral Analyzer	
Astral mass range	Standard mass range m/z 40–6000
Astral resolution ¹⁾	80,000 FWHM at m/z 524

Parameter	Value
Acquisition rate ¹⁾ _{.....}	Astral MS ² acquisition rates up to 200 Hz with a maximum injection time of 3 ms
Astral mass accuracy ¹⁾ _{.....}	External calibration achieves <5 ppm RMS drift over 24 hours, measured with FlexMix calibration solution
Dynamic range	>1000 within a single microscan Astral mass spectrum

5.3 Instrument front side

The system status control LEDs and the API source are at the front side of the Orbitrap module.

5.3.1 System status LEDs

Three system status LEDs at the front side indicate the main functions of the system. The table explains the function of the LEDs.

LED	Status	Information
Power	Green	Instrument is receiving power.
	Off	Instrument is not receiving power.
Status	Green	Instrument is in operating mode.
	White flashing	Procedure or bake-out in progress.
	Yellow	Warn status, does not prevent measurement, but proper operation could be affected.
	Yellow flashing	Grave failure detected that does prevent measurement.
	Off	The instrument is in service mode. Pumps are running but electronics is switched off.
Scan	Blue animation ¹⁾ _{.....}	Instrument is scanning in MS1 mode.
	Purple animation	Instrument is scanning in MS2 mode.

LED	Status	Information
	White bar graph	Progress bar for procedure.
	Same as Status LED (in Standby Mode)	Same as Status LED.
	Off	Instrument is powered but in off mode.

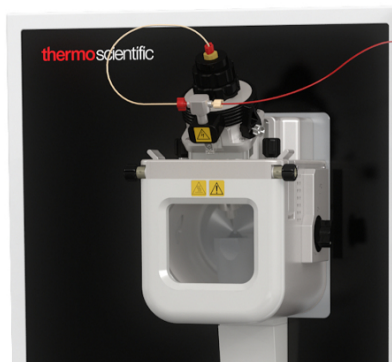
NOTE

The system status LEDs give a overview of the general system status. They do not have any function for the safety status of the instrument. It is not sufficient that the Power LED is off because it might be defective. Before you perform maintenance on the instrument, make sure that the main power circuit breaker switch (labeled Main Power) is in the Off (O) position and that the power cords of the mass spectrometer and the forepump are disconnected.

5.3.2 Atmospheric pressure ionization source

The atmospheric pressure ionization (API) source forms gas-phase sample ions from sample molecules that are contained in solution. The API source also serves as the interface between the LC and the mass spectrometer. The Orbitrap Astral mass spectrometer is shipped with the Thermo Scientific OptaMax NG™ API source and a HESI probe.

Figure API source with HESI probe attached



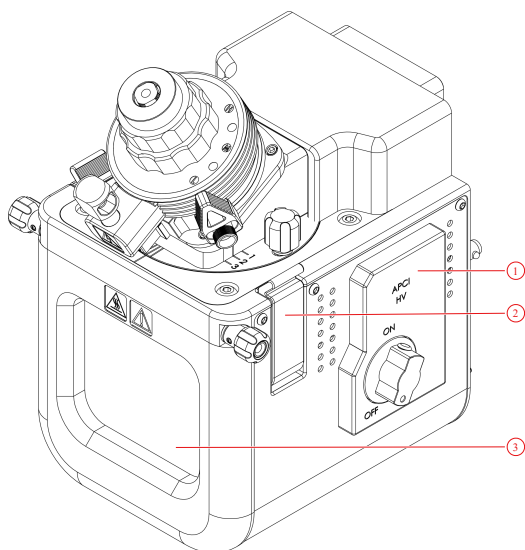
5.3.2.1 API source

The API source is the part of the ion source that is at atmospheric pressure. You can configure the API source to operate in any of several API modes, including heated electrospray ionization (HESI), atmospheric pressure chemical ionization (APCI), and atmospheric pressure photoionization (APPI). For APCI and APPI, additional parts are required. A separate nanospray ionization (NSI) probe is

necessary to operate in NSI mode. The ions produced in the API source are transmitted by the ion optics into the Orbitrap mass analyzer, where they are separated according to their mass-to-charge ratios.

The API source housing enables you to quickly switch between ionization modes without the need for specialized tools. A safety relay switches off the high voltage supply (8 kV) when you remove the source from the mass spectrometer. A window at the front side of the ion source housing permits viewing the probe position during ESI operation and the addition of accessories.

Figure OptaMax NG API source (HESI mode)



No.	Description	No.	Description
1	APCI components (built-in)	3	Window
2	Locking lever (locked position)		

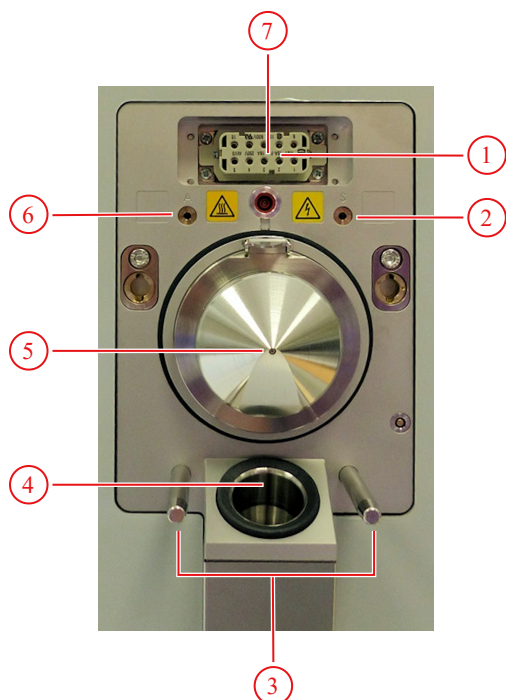
5.3.2.2 Ion source mount

The ion source mount on the front side of the mass spectrometer allows interchanging HESI, APCI, APPI, and calibration probes without using tools. The mount has high-voltage electrical connections for the electrospray needle (ESI and HESI) and for the vaporizer and corona discharge needle (APCI). When you remove the ion source, a high-voltage safety interlock switches the statuses of various instrument components:

- ESI and HESI spray voltage (or APCI corona discharge voltage) is switched off.
- All API source and lens voltages, including the ion transfer tube offset voltage are switched off.

- Gases are switched off.
- The instrument is switched to standby condition if it was on before.

Figure Ion source mount



No.	Description	No.	Description
1	Connector for heater power and signal connections (readbacks and safety interlock)	5	Sweep cone
2	Sheath gas outlet	6	Auxiliary gas outlet
3	Ion source housing guide pins	7	High voltage (8 kV) socket
4	Ion source drain		

5.3.2.3 Solvent drainage

The OptaMax NG API source can accommodate high flow rates. Therefore, you must collect the waste solvent in a manner that does not build up pressure in the source. The Orbitrap Astral mass spectrometer has a built-in drain. It routes the solvent waste from the API source to the solvent waste container that is connected to the source drain port.

5.3.3 Ion source interface

The ion source interface consists of the components of the API source that are held under vacuum (except for the atmospheric pressure side of the ion sweep cone).

API source interface

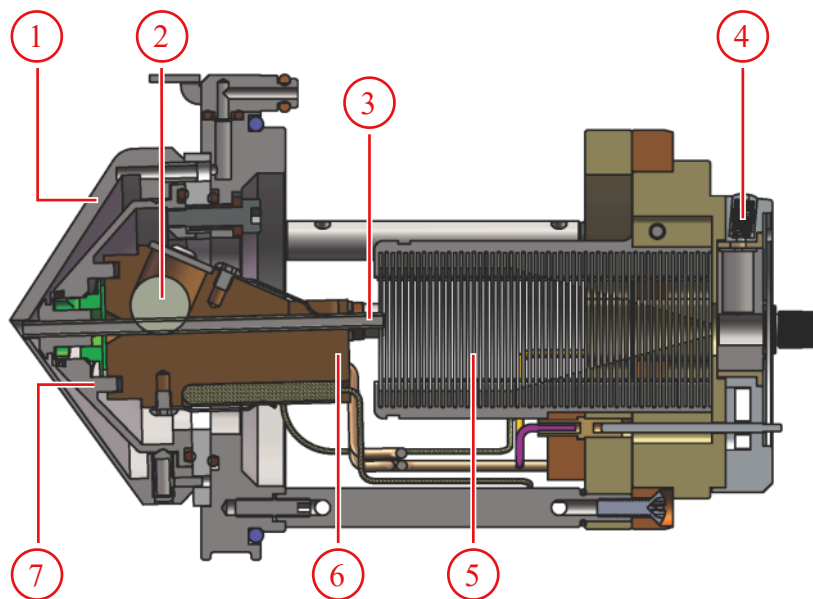
The API source interface includes a sweep cone, an ion transfer tube, two cartridge heaters, a heater block, a sensor, a vent prevent ball, and the ion funnel.

The sweep cone is a metal cone over the ion transfer tube. The sweep cone channels the sweep gas toward the entrance of the ion transfer tube, acts as a physical barrier that protects the entrance of the ion transfer tube, and increases source robustness. The net result is a significant increase in the number of samples to analyze without a loss of signal intensity.

TIP

In addition, keeping the ion transfer tube entrance as clean as possible reduces the need for frequent maintenance. Install the sweep cone to improve ruggedness when you analyze complex matrices such as plasma or nonvolatile salt buffers. Remove the sweep cone before you perform NSI experiments.

Figure API source interface with ion funnel



No.	Description	No.	Description
1	Sweep cone	5	Ion funnel
2	Vent prevent ball	6	Heater block

3	Ion transfer tube	7	Spray cone (directly behind the sweep cone)
4	IC unit		

The ion transfer tube is a metal tube that has a rectangular orifice. It assists in desolvating ions that are produced by the API spray insert while transferring them into the vacuum system.

The heater block contains two heater cartridges, surrounds the ion transfer tube, and heats the tube to temperatures up to 400 °C (752 °F). A thermocouple measures the temperature of the heater block. Typical temperatures of the ion transfer tube are 320 °C (608 °F) for HESI and 275 °C (527 °F) for APCI, but these temperatures vary with the flow rate and the mobile phase composition. A decreasing pressure gradient draws ions into the ion transfer tube in the vacuum manifold. When you remove the ion transfer tube (after it has cooled to room temperature), the vent prevent ball drops into place to stop air from entering the vacuum manifold. Therefore, you can remove the ion transfer tube for cleaning or replacement without venting the system.

Ions from the ion transfer tube pass through the ion funnel and then the IC unit. The ion funnel is an ion transmission device that consists of equally spaced, stainless-steel electrodes. The mass spectrometer applies an RF voltage to the electrodes, and adjacent electrodes have voltages of opposite phase. As the RF amplitude increases, ions of progressively higher mass-to-charge ratios pass through to the IC unit and move toward the injection filter. The IC unit acts as a vacuum baffle between the higher-pressure API source interface region and the lower-pressure ion optics region of the vacuum manifold. The ion funnel and the IC unit mount to the API source interface cage.

5.4 Left instrument side

The forevacuum ports and the source drain port are located at the left side of the instrument. They are located behind a panel and not accessible for customers.

Connect the large inlet port of the forepump to the large vacuum port (38 mm [1 ½ in.] OD) at the left side of the instrument. Connect the small inlet port of the forepump to the small vacuum port (19 mm [¾ in.] OD). Corresponding sections of reinforced PVC tubing are shipped with the instrument.

The solvent waste container is installed in the left side of the compartment below the Orbitrap module.

5.5 Right instrument side

The control panel and the accessories cabinet are located at the right side of the instrument.

5.5.1 Control panel

To access the control panel, remove the magnetic lid. The lid can remain mounted even when the peripherals are connected. The bottom between the cover and the sockets is open. The cables can be led out through the gap.

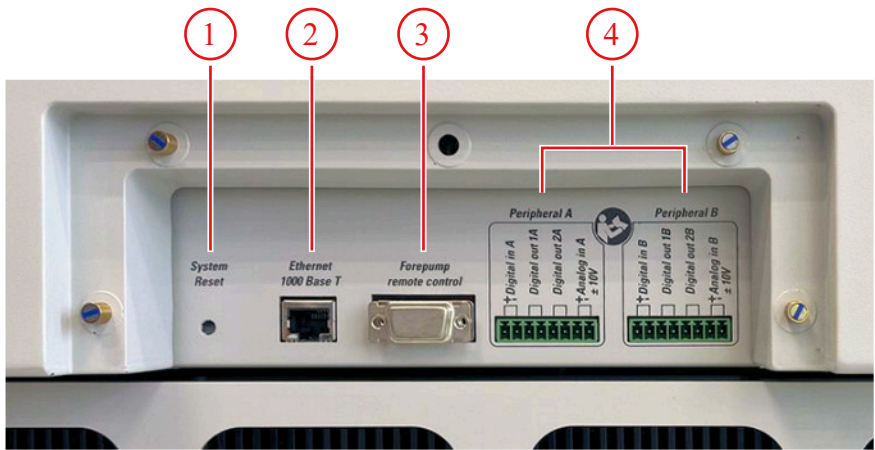
Figure Removing the lid of the control panel



TIP

The control panel is identical with the one of the Orbitrap Exploris Series instruments. Some controls are not used with the Orbitrap Astral MS. Their functions are repurposed elsewhere in the instrument.

Figure Control panel



No.	Description	No.	Description
1	System reset	3	Not used
2	Not used	4	Peripheral control connectors

5.5.1.1 Reset button

The reset button is located on the control panel. When you press the button, the instrument control software is reloaded from the data system.

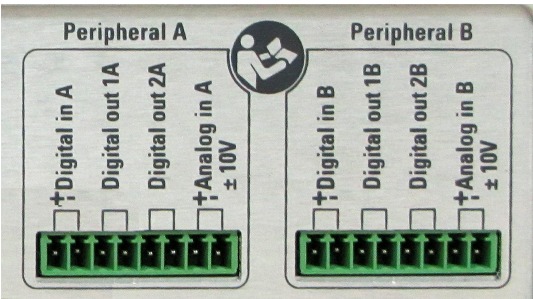
TIP

Use the reset button only if the instrument does not respond to the control program on the data system computer or if you must restart the instrument.

5.5.1.2 Peripheral connectors

The Orbitrap Astral mass spectrometer has two connectors (Peripheral A, Peripheral B) to communicate with peripherals (LC systems, for example) by digital and analog signals.

Figure Peripheral control connectors



Suitable plug connectors for the peripheral control connection are provided with the Installation Kit. Refer to the Help for information about setting up the communication with external devices.

Peripheral A

The instrument method's contact closure is tied directly to the Peripheral A Digital out 1 connector. The contact closure signals are transmitted through a trigger cable that connects the respective port to the external device.

Table Peripheral A control connectors

Port	Description
±Digital in	+ (positive, pin 1) and – (negative, pin 2) Based on the configuration, the mass spectrometer starts the data acquisition when it receives a digital contact closure signal from an external device.
Digital out 1 Digital out 2 (interchangeable)	The mass spectrometer transmits the ready status through contact closure to an external receiving device. The mass spectrometer transmits a programmable contact closure output signal to the inputs of an external receiving device (a fraction

Port	Description
	collector, for example).
Analog In ±10 V	+ (positive, pin 7) and – (negative, pin 8) The mass spectrometer can record an analog signal between –10 V and +10 V from an external device (for example, an LC detector) with the mass spectrometer data.

Peripheral B

Use the connector labeled Peripheral B to connect the cabling of the handshake lines that come with certain types of ion sources (for example, MALDI). If an external trigger is provided and handshake mode is active, the connections work as follows:

Table Peripheral B control connectors

Port	Description
Digital in B:	Trigger input, closing the connection between the two pins starts the scan. The trigger should only be applied when the Ready output is active (closed), and the trigger should be deactivated only after the ready output is deactivated.
Digital out 1B:	Diagnostic Trigger out output. This will be activated (closed) by the instrument while the selected scan event state is performed. It will be deactivated when the next state of the scan event is reached.
Digital out 2B:	Ready output. This is activated (closed) by the instrument when it is ready to retrieve the (next) trigger signal. It will deactivate when the trigger is retrieved (when the ion injection for the scan starts).

The trigger mechanism can be tested/demonstrated by connecting [Digital in B] with [Digital out 2B] by a 2-way cable. In this feedback/shortcut configuration, the instrument will run with handshake synchronization, at the maximum speed that would be possible with synchronization active.

TIP

The +/- pins are reversed compared to Exactive Series instruments.

5.5.2 Accessories cabinet

The accessories cabinet provides space for operating the syringe pump and the switching valves. The interior of the cabinet can be accessed by lifting the lid of the accessories cabinet.

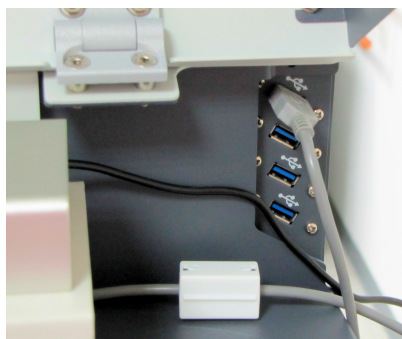
Figure Accessories cabinet



USB ports

Four USB ports allow the internal computer to communicate with other approved devices of the LC/MS system (for example, syringe pump). Do not connect any device to these ports other than dedicated Thermo Fisher Scientific devices.

Figure USB ports



5.5.3 Power outlets for peripherals

When you remove the panel from the right side of the cabinet below the Orbitrap module, you can access the peripheral power supply ports. The four ports are located on a power strip that is mounted to the bottom front side of the Astral module.

Use the power outlets to supply electric power for low power devices of the LC/MS system (for example, a syringe pump or switching valves). These outlets are rated at 208–240 Vac, 50/60 Hz, 3 A total maximum. They are controlled by the main power circuit breaker switch.

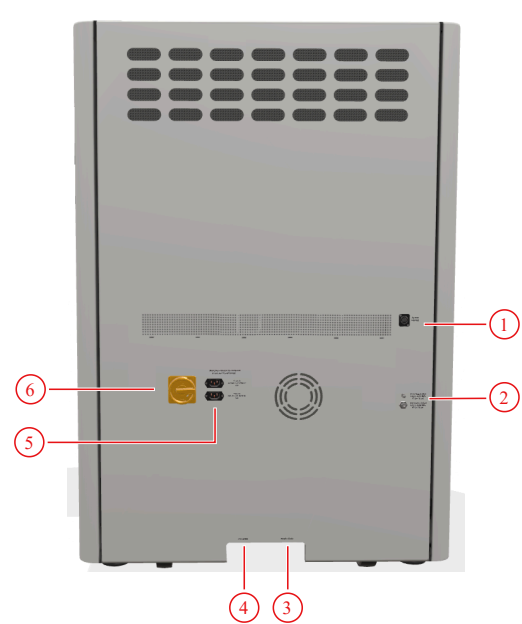
Figure Power outlets for peripherals



5.6 Instrument rear side

The back side of the instrument provides switches and connections that concern connections for power supply for the instrument and communication with peripherals.

Figure Rear side of Orbitrap Astral MS



No.	Description	No.	Description
-----	-------------	-----	-------------

1	Ethernet port (labeled System Interlink)	4	Forepump exhaust outlet
2	Gas inlets: C-Trap & HCD (top), Source & Vent (bottom)	5	Main power connectors
3	Waste/drain exhaust outlet	6	Main power switch

Use the Ethernet port to connect the mass spectrometer to the data system computer and the LC via an Ethernet switch.

The *main power circuit breaker switch* is located at the left rear side. In the OFF (O) position, the circuit breaker removes all power from the mass spectrometer, including the turbomolecular pumps. It switches off the forepump by the remote control cable. In the ON (I) position, power is supplied to the mass spectrometer. In the standard operational mode, the circuit breaker is kept in the ON (I) position.

TIP

Power is to remain on. The mass spectrometer should remain on and pumping continuously for optimum performance.

Figure Mains switch with mains supply connectors



Two power connectors for the mains supply are located next to the main power switch. The Orbitrap Astral MS operates at a nominal voltage of 208–240 V AC, 50/60 Hz.

NOTE

Both power connectors must be connected to the power supply in the laboratory to operate the instrument.

Use separate wall outlets to supply electrical power for the forepump, the data system, and the devices of the LC system if applicable.

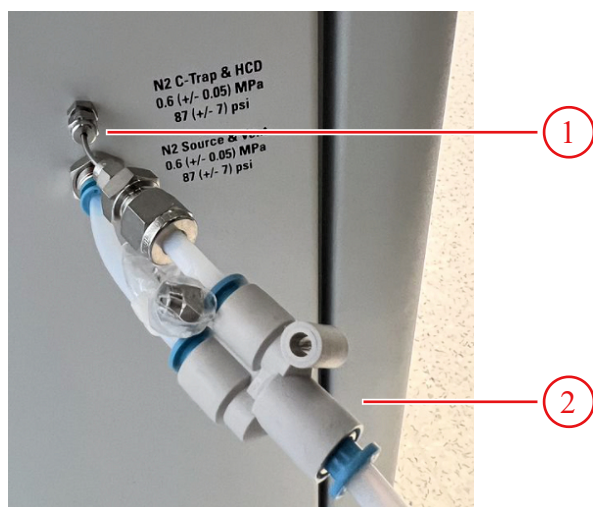
At the bottom of the rear side, a recess allows leading out the hoses for the pump exhaust and the solvent waste container exhaust.

5.6.1 Gas inlets

Two gas inlet ports are at the rear side of the instrument. The port for source gas & vent gas (bottom) has a press-in fitting for a 6 mm hose. The port for C-Trap gas & HCD collision gas (top) has a Swagelok™-type fitting for 1/16 in. tubing.

Use high-purity (HP, 99%) nitrogen for the API source gases (sheath gas, auxiliary gas, and sweep gas). The HP nitrogen is also used to vent the vacuum manifold with the vent valve if the system is shut down. The collision gas (trapping gas) of the C-Trap, the collision gas for the ion-routing multipole, the reagent carrier gas of the IC source, and the collision gas for the ion processor require ultra-high purity (UHP, 99.999%) nitrogen. The necessary gas pressure is 0.6 ± 0.05 MPa (6 ± 0.5 bar, 87 ± 7 psi).

Figure Gas connections at the rear instrument side



No.	Description	No.	Description
1	Gas inlets: C-Trap & HCD (top), Source & Vent (bottom)	2	Y-piece

NOTE

Do not connect other gases than nitrogen to the Orbitrap Astral mass spectrometer. This might affect instrument performance. The maximum pressure for the gas inlet is 0.65 MPa (6.5 bar, 94 psi).

5.6.1.1 Gas supply for the API source, IRM, and C-Trap

The gas flow from the top gas port is directed through a Teflon™ hose to the Source PCB. On the Source PCB, a valve terminal divides the nitrogen flow into three streams (sheath gas, auxiliary gas, and sweep gas) to the API source. Also, a T-piece directs part of the nitrogen flow into the vent valve.

When the instrument switches off, the vent valve opens and the nitrogen is led through stainless steel tubing to the vacuum manifold. The nitrogen gas flow ensures a clean and controlled venting of the instrument.

The gas flow from the bottom gas port is directed through 1/16 in. tubing to a pressure regulator, which keeps the gas pressure to the C-Trap and the ion-routing multipole constant. An electronic pressure regulator sets the nitrogen pressure according to the operating mode of the instrument and the chosen user setting. From the electronic regulator, a small part of the gas is led as collision gas to the ion-routing multipole behind the C-Trap. The larger part of the nitrogen stream is diverted into the source vacuum.

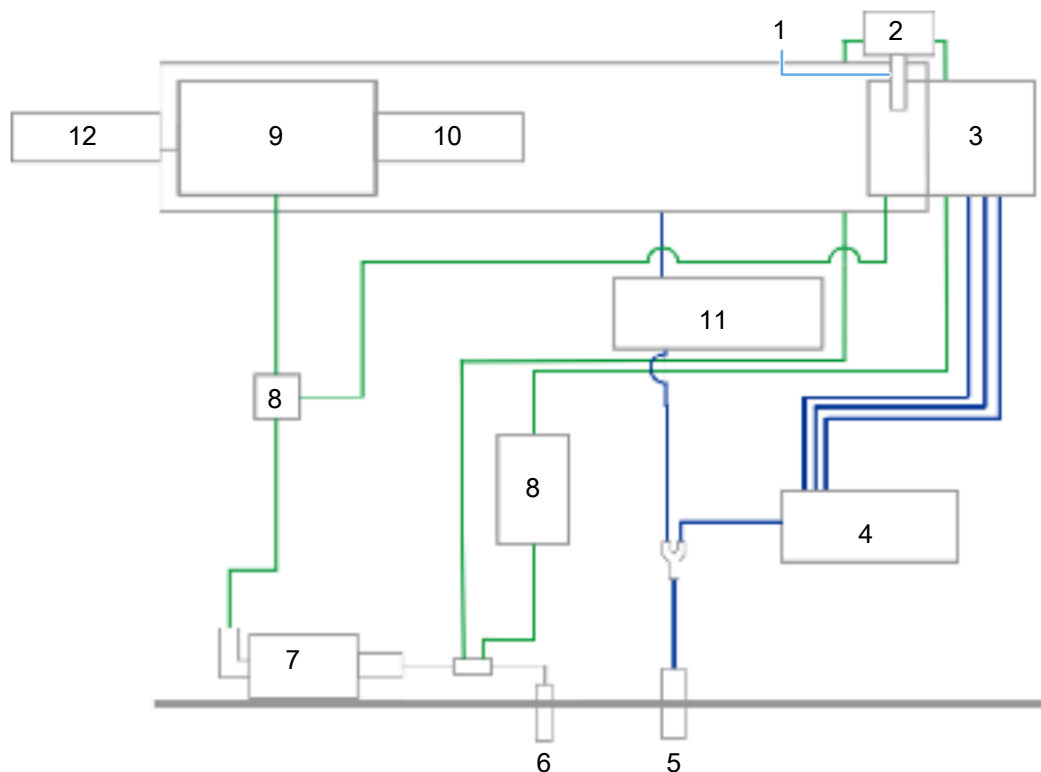
5.6.1.2 Gas supply for the IC source

Two gas streams are directed from the input line of the HCD collision gas (UHP nitrogen) to the calibrant distribution assembly next to the API source. One gas stream provides a constant vapor pressure of the calibrant species to the IC source. The other gas stream provides the correct gas pressure to the discharge region to allow for a stable discharge.

TIP

The total consumption of UHP nitrogen gas is 0.07 NL/min. It consists of the base flow (30 sccm [$\approx$ 0.03 NL/min]) for the C-Trap and the IRM and the make up flow (35 sccm [$\approx$ 0.04 NL/min]) for the IC source. When the IC source is switched off, the consumption of the gas decreases significantly.

Figure Schematic of the internal gas supplies in MS/IC system



No.	Description	No.	Description
1	IC source	7	Regulator
2	Calibrant distribution assembly (with fluoranthene oven and heated split)	8	Electronic pressure regulator
3	API source	9	Ion-routing multipole
4	Source PCB	10	C-Trap
5	Source gas & vent gas inlet (bottom)	11	Vent valve
6	C-Trap & HCD gas inlet (top)	12	Ion processor

5.7 Ion optics

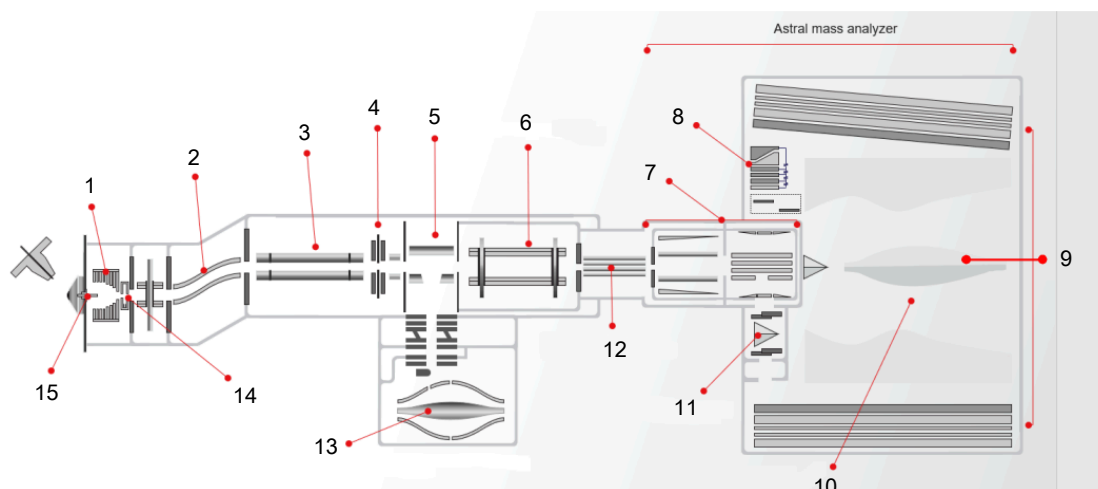
The ion optics focus the ions that are produced in the API source and transmit them to the C-Trap.

5.7.1 Inlet stack of the Orbitrap Astral mass spectrometer

The inlet stack includes a heated high-capacity ion transfer tube—with a rectangular orifice—that assists in desolvating the ions. The tube guides the ions to an ion transmission device, the ion funnel.

This device consists of a stack of electrodes to which an RF voltage is applied to transport and focus the ions to the Internal Calibrant Source. From there, the ions move toward the injection filter and the bent flatapole, which form the Advanced Active Transfer Tube Beam Guide (AABG).

Figure Schematic of the Orbitrap Astral mass spectrometer

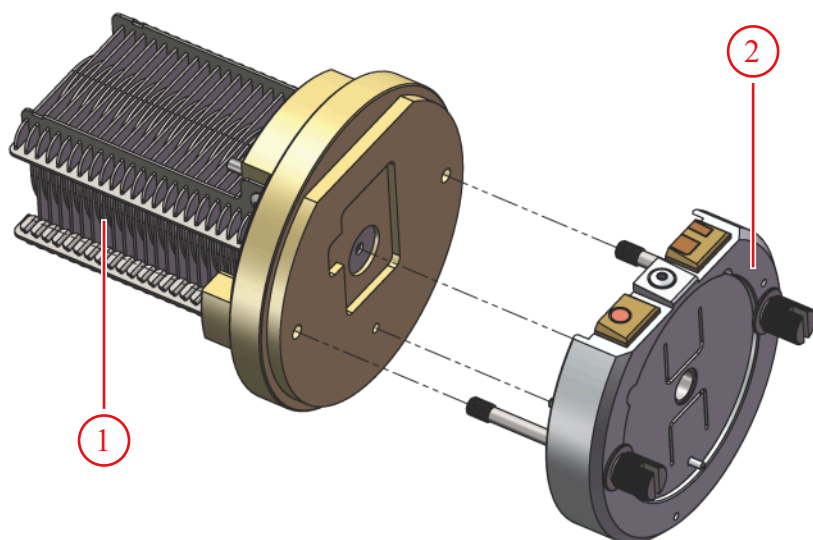


No.	Description	No.	Description
1	Electrodynamic Ion Funnel	9	Asymmetric Ion Mirror
2	Advanced Active Beam Guide	10	Ion Foil
3	Advanced Quadrupole Technology (AQT)	11	Injection Optics
4	Independent Charge Detector	12	Transfer Octupole
5	C-Trap	13	Ultra-High Field Orbitrap Mass Analyzer
6	Ion-Routing Multipole	14	EASY-IC Internal Calibrant Source
7	Ion Processor	15	High Capacity Ion Transfer Tub
8	HDR Detector		

5.7.2 Internal calibrant discharge source

The internal calibrant discharge source is installed in the IC source heater interface, which is located in the API source interface just after the ion funnel.

Figure Internal calibrant discharge source and ion funnel (rear view)



No.	Description	No.	Description
1	Ion funnel	2	IC source heater interface

The internal calibrant discharge source delivers a regulated number of calibrant ions into the much larger population of analyte ions. The internal calibrant ions are used as a lock mass (m/z) that significantly improves the mass-to-charge ratio (m/z) assignment accuracy to less than 1 ppm (up to m/z 1500) in every Fourier transform (FT) mass spectrum.

The instrument control software uses the precisely known mass-to-charge ratio of the calibration mass peak to provide real-time fine adjustment of the instrument's mass-to-charge ratio calibration, enabling a correction for otherwise uncompensated errors that occur due to temperature changes and scan-to-scan variations in the total charge of the population of ions analyzed.

5.7.2.1 Internal calibrant introduction

The internal calibrant discharge source has its own dedicated continuous introduction system that delivers a highly stable flow of fluoranthene as calibrant. The calibrant distribution assembly receives an initial charge of fluoranthene during the installation. You can expect this charge to last at least one year. Only a Thermo Fisher Scientific field service engineer can replace the calibrant when it is consumed.

A glow discharge electrode stably produces a very intense current of fluoranthene radical anions and a moderate current of fluoranthene radical cations. When you switch on the IC, it uses the radical anions (negative polarity) and cations (positive polarity) of fluoranthene as the internal calibration lock mass.

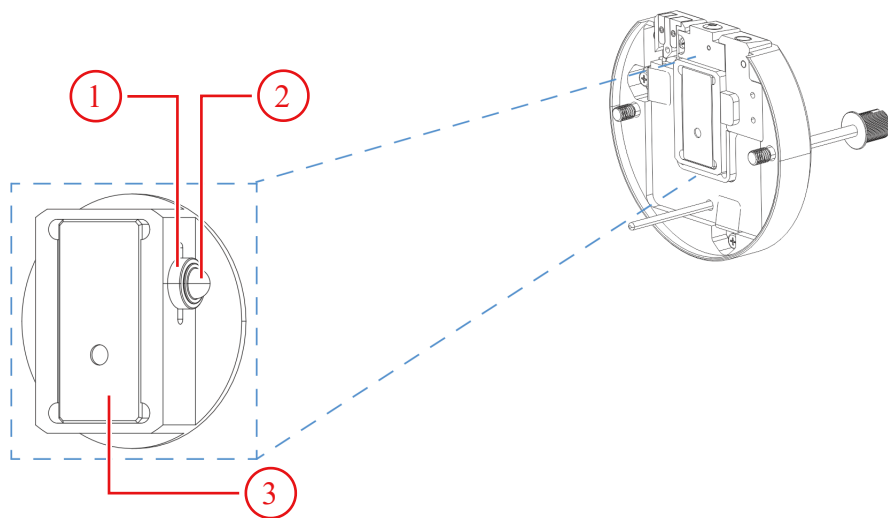
5.7.2.2 Calibrant distribution assembly

The calibrant distribution assembly connects through the vacuum manifold to the terminals of the IC source heater interface to deliver the calibrant (fluoranthene) to the calibrant ion source for ionization and further IC use. The assembly consists of:

- a pressure reducer with open split into vacuum and gas restrictors to provide a very low, constant flow of UHP nitrogen into the oven;
- the oven containing fluoranthene;
- a heated split where the make-up gas is mixed in, coming from the Porter pressure regulating unit through a gas restrictor;
- an interface to the IC source heater interface in the vacuum of the MS.

5.7.2.3 IC Source heater interface

Figure Close-up view of the calibrant ion source in the IC source heater interface



No.	Description	No.	Description
1	Ceramic insulator	3	Anode (lens/ion volume)
2	Discharge cathode		

The IC source heater interface is located in the API source interface region between the ion funnel and the injection filter. It consists of a mounting assembly that includes the calibrant ion source heater, a gas conduit, a high voltage contact, and the calibrant ion source.

The figure shows the IC source heater interface with a close-up view of the front of the calibrant ion source. The calibrant ion source contains the internal ion volume where reagent ion species are created, the discharge cathode, the anode, and a ceramic insulator.

The calibrant ion source is a consumable part.

5.7.3 Bent flatapole

The bent flatapole always acts as an ion transmission device. It guides the ions through an arc from the injection filter to the quadrupole and removes the neutral gas jet and solvent droplets that pass through the ion funnel and the injection filter. The neutral particles cannot follow the bent path of the flatapole.

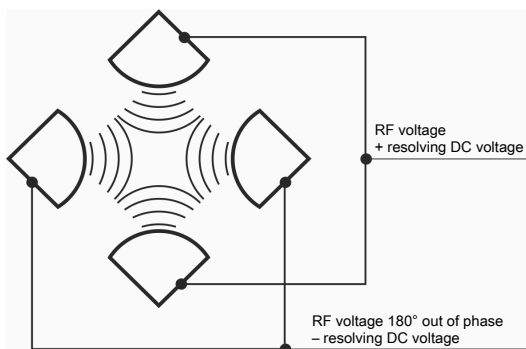
An RF voltage with a common-mode DC offset enables an efficient ion transport and the separation from neutrals. An additional axial DC gradient is applied to the bent flatapole to accommodate a fast ion transport.

5.7.4 Quadrupole mass filter

The Orbitrap Astral MS uses advanced quadrupole technology (AQT). A segmented hyperbolic quadrupole is used to provide the best transmission efficiency and rectangular transmission window shapes. The entrance segment is driven by RF voltage only, so that ions entering the quadrupole stay well focused before entering the middle section, the resolving quadrupole. Similarly, the exit section of the quadrupole also minimizes the fringe field regions and allows precise focusing. This is important for proper transfer of the ions to the C-Trap.

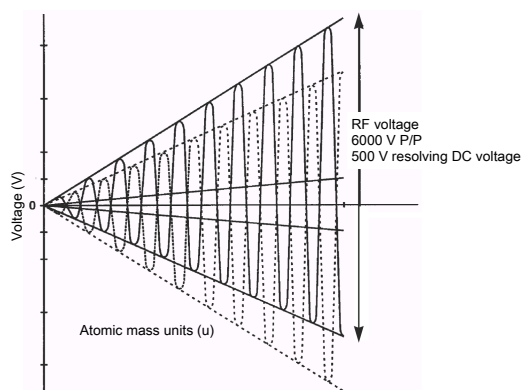
In a quadrupole rod assembly, rods opposite each other in the array are connected electrically. Thus, the four rods can be considered as two pairs of two rods each. RF voltages and resolving DC voltages are applied to the rods and these voltages are adjusted for each scan. Voltages of the same amplitude and sign are applied to the rods of each pair. However, the voltages applied to the different rod pairs are equal in amplitude but opposite in sign. The RF voltage applied to the quadrupole rods is of constant frequency of approximately 860 kHz and varies from 0 to 6000 V peak-to-peak amplitude. The resolving DC voltage varies from 0 to ± 500 V.

Figure Polarity of the RF voltages and resolving DC voltages applied



In the following figure, the solid line represents the combined RF voltage and resolving DC voltage applied to one rod pair, and the dashed line represents the combined RF voltage and resolving DC voltage applied to the other rod pair. The RF-to-DC-voltage ratios and their values determine the range of mass-to-charge ratios to be transmitted through the quadrupole mass filter.

Figure Magnitude of the RF voltages and resolving DC voltages applied between the rods of the quadrupole mass filter



In the mass spectrometer, the rods of the quadrupole are supplied with a variable ratio of RF voltage and resolving DC voltage. For each injection, controlled by the split lens, the quad RF amplitude and resolving DC voltage are set to fixed values. Under these conditions, only ions of a certain range of m/z ratios are maintained within bounded oscillations as their velocity carries them through the mass filter. At the same time, all other ions undergo unbounded oscillations. These ions strike one of the rod surfaces, become neutralized, and are pumped away.

The quadrupole offset voltage is a DC potential applied to the quadrupole rods in addition to the filtering DC voltage. The offset voltage applied to the two rod pairs of the assembly is equal in amplitude and equal in sign. The quadrupole offset voltage accelerates or decelerates ions and, therefore, sets the translational kinetic energy of the ions as they pass the quadrupole rod assembly.

5.7.5 Independent charge detector

The independent charge detector (ICD) is positioned between the quadrupole and the C-Trap. It is used to increase the fidelity of the Automatic Gain Control (AGC) in certain operation modes. The assembly includes a split lens, which is used to start and stop the injection of ions into the C-Trap. A precise determination of the ion beam intensity and a fast switching of the ion beam are needed for the Automatic Gain Control (AGC) of the Orbitrap mass analyzer. The assembly also includes a short transport RF multipole, which is the interface between the ICD and the C-Trap.

5.8 Curved linear trap

The mass spectrometer always passes the ions through the gas-filled curved linear trap (C-Trap) before trapping them in the ion-routing multipole. The MS then passes the ions back to the C-Trap before injecting them into the Orbitrap mass analyzer.

The ions that enter the C-Trap lose their kinetic energy by colliding with the nitrogen collision gas, which dissipates their kinetic energy and cools them down to the center axis of the C-Trap.

Voltages on the end apertures of the C-Trap (entrance and exit apertures) are controlled to provide or avoid a potential well along its axis. These voltages may be later ramped up to squeeze ions into a shorter thread along this axis.

5.9 Ion-routing multipole

The ion-routing multipole (IRM) consists of a straight multipole that is mounted inside an enclosed assembly, which is connected in direct line-of-sight to the C-Trap. The MS supplies the IRM with the nitrogen collision gas to increase the multipole's gas pressure. The C-Trap is directly attached to the IRM, so part of the collision gas also flows into the C-Trap.

For Full Scan operation, the IRM is used to cool and trap the ions. A potential gradient is used to effectively transport the ions back towards the C-Trap.

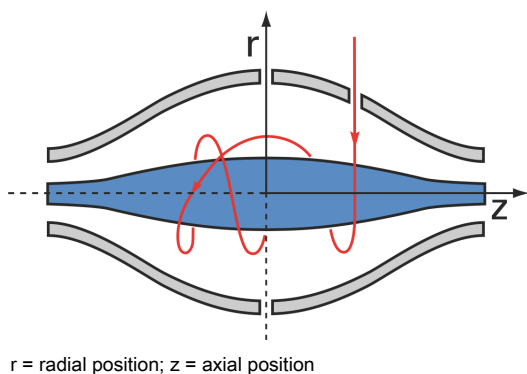
For HCD (Higher Energy Collisional Dissociation), ions pass through the C-Trap into the IRM. The offset voltage between the C-Trap and the IRM accelerates the precursor ions into the gas-filled IRM leading to fragmentation. The MS applies a potential gradient to the IRM to provide fast extraction of ions, such that it returns ions at a reliable rate.

The spectra of the fragments generated in the IRM and detected in the Orbitrap analyzer are comparable to the typical fragmentation patterns obtained on triple-quadrupole instruments.

5.10 Orbitrap analyzer

The heart of the Orbitrap™ analyzer is an axially-symmetrical mass analyzer. It consists of a spindle-shaped central electrode surrounded by an outer electrode that is split into bell-shaped halves. The Orbitrap analyzer employs electric fields to capture and confine ions.

Figure Schematic of Orbitrap cell and example of stable ion trajectory



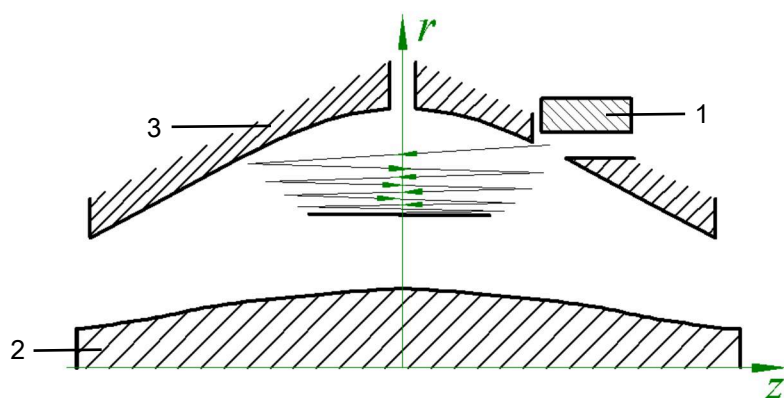
5.10.1 Extraction of ion packets

For ion extraction, the RF voltage on the rods of the C-Trap is ramped off and extracting voltage pulses are applied to the electrodes, pushing ions orthogonally to the curved axis through a slot in the inner electrode. Because of the initial curvature of the C-Trap, including the Z-Lens, and the subsequent lenses, the ion beam converges on the entrance into the Orbitrap analyzer. The lenses that follow the C-Trap (Z-lens) also act as differential pumping slots and cause spatial focusing of the ion beam into the entrance of the Orbitrap analyzer. In the Z-Lens, ions are electrostatically deflected away from the gas jet, thereby eliminating gas carryover into the Orbitrap analyzer.

Owing to the fast pulsing of ions from the C-Trap, ions of each mass-to-charge ratio arrive at the entrance of the Orbitrap analyzer as short packets that are only a few millimeters long. For each mass-to-charge population, this corresponds to a spread of flight times of only a few hundred nanoseconds for mass-to-charge ratios of a few hundred Daltons per charge. Such durations are considerably shorter than a half-period of axial ion oscillation in the Orbitrap analyzer. Because ions are injected into the Orbitrap analyzer at a position offset from its equator, these packets start coherent axial oscillations without the need for any additional excitation cycle.

The evolution of an ion packet during an increase of the electric field is shown schematically in the following figure. When the injected ions approach the opposite half of the outer electrode for the first time, the increased electric field (owing to the change of the voltage on the central electrode) contracts the radius of the ion cloud by a few percent. The applied voltages are adjusted to prevent collision of the ions with the electrode. A further increase of the field continues to squeeze the trajectory closer to the axis, meanwhile allowing for newly arriving ions (normally, with higher m/z) to enter the Orbitrap analyzer as well. After ions of all m/z have entered the Orbitrap analyzer and moved far enough from the outer electrode, the voltage on the central electrode is kept constant and image current detection takes place.

Figure Principle of electrodynamic squeezing of ions in the Orbitrap analyzer as the field strength is increased



r = radial position; z = axial position

No.	Description	No.	Description
1	Deflector electrode	3	Outer electrode
2	Central electrode		

5.10.2 Measuring principle

In the Orbitrap mass analyzer, stable ion trajectories combine rotation around an axial central electrode with oscillations along it. The Orbitrap analyzer's electrodes are shaped such that the ions experience a quadro-logarithmic potential, for which their axial oscillations are harmonic. The frequency ω of these harmonic oscillations along the z-axis depends only on the ion's mass-to-charge ratio m/z and the instrumental constant k :

$$\omega = \sqrt{\frac{z}{m}} \times k$$

The split halves of the outer electrode of the Orbitrap analyzer detect the image current produced by the oscillating ions. By Fast Fourier Transformation (FFT) of the amplified image current, the instrument obtains the frequencies of these axial oscillations and thus, the mass-to-charge ratios of the ions.

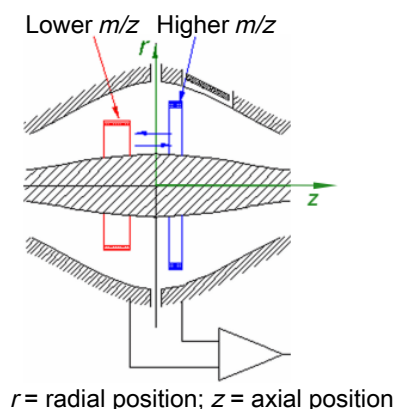
5.10.3 Ion detection

During ion detection, the central electrode and the additional electrode, which deflects the ions during the injection and compensates electric field imperfections during the measurement, are maintained at very stable voltages so that no mass drift can take place. The outer electrode is split in half at $z=0$, allowing the ion image current in the axial direction to be collected. The image current on each half of the outer electrode is differentially amplified and then undergoes analog-to-digital conversion before processing by the fast Fourier transform algorithm.

As mentioned above, stable ion trajectories in the Orbitrap analyzer combine axial oscillations along the z -axis with rotation around the central electrode and vibrations in the radial direction. For any given m/z , only the frequency of axial oscillations is completely independent of initial ion parameters, whereas rotational and radial frequencies exhibit strong dependence on initial radius and energy. Therefore, ions of the same mass-to-charge ratio continue to oscillate along z together, remaining in-phase for many thousands of oscillations.

In contrast to the axial oscillations, the frequencies of radial and rotational motion will vary for ions with slightly different initial parameters. This means that in the radial direction, ions dephase orders of magnitude faster than in the axial direction, and the process occurs in a period of only 50–100 oscillations. After this, the ion packet of a given m/z assumes the shape of a thin ring, with ions uniformly distributed along its circumference. Because of this angular and radial smearing, radial and rotational frequencies cannot appear in the measured spectrum. Meanwhile, axial oscillations will persist, with axial thickness of the ion ring remaining small compared with the axial amplitude. Moving from one half of the outer electrode to the other, this ring will induce opposite currents on these halves, thus creating a signal to be detected by differential amplification.

Figure Approximate shape of ion packets of different m/z after stabilization of voltages



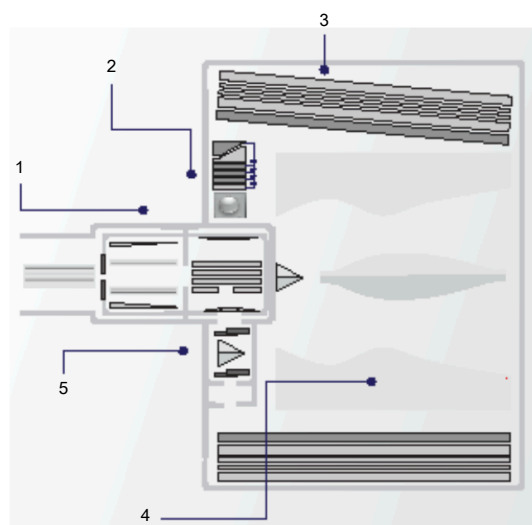
5.11 Astral analyzer

The ions are transferred from the ion routing multipole into the ion processor where they are trapped and fragmented at speeds up to 200 Hz. Then, the ions are ejected through a series of ion optics that precisely align the ion packet to improve sensitivity.

The ion packet is then accelerated into an open electrostatic trap consisting of an asymmetric track composed of asymmetric ion mirrors that provides a path length of over 25 meters to provide resolution of up to 80,000 at m/z 524 and an ion foil, which shapes and focuses the ions onto the asymmetric track.

The novel high dynamic range (HDR) detector provides sensitive single ion detection with excellent linearity over multiple orders of magnitude, low noise due to an isolated vacuum, and a long lifetime.

Figure Astral mass analyzer



No.	Description	No.	Description
1	Ion processor	4	Ion foil
2	HDR Detector assembly	5	Injection optics
3	Asymmetric ion mirror		

The Astral analyzer is divided in these subsystems:

- Ion Processor

The Ion Processor is a rectilinear ion trap that combines a high-pressure (HP) part for fragmentation and cooling with a low-pressure (LP) part used for the extraction of the ions into the Astral analyzer.

- Injection Optics

The Injection Optics shape the ion packets before they enter the analyzer.

- Asymmetric Ion Mirror

The Asymmetric Ion Mirror provides a long and folded (>20 m) flight path. The ions are reflected between two opposing mirrors. The mirrors converge with a distinctive tilt. The resulting field gradient reflects the ions towards the entrance. The tilt also focuses the ion beam in the drift direction.

- Ion Foil

The Ion Foil shapes and focuses the ions onto the asymmetric track with minimal loss.

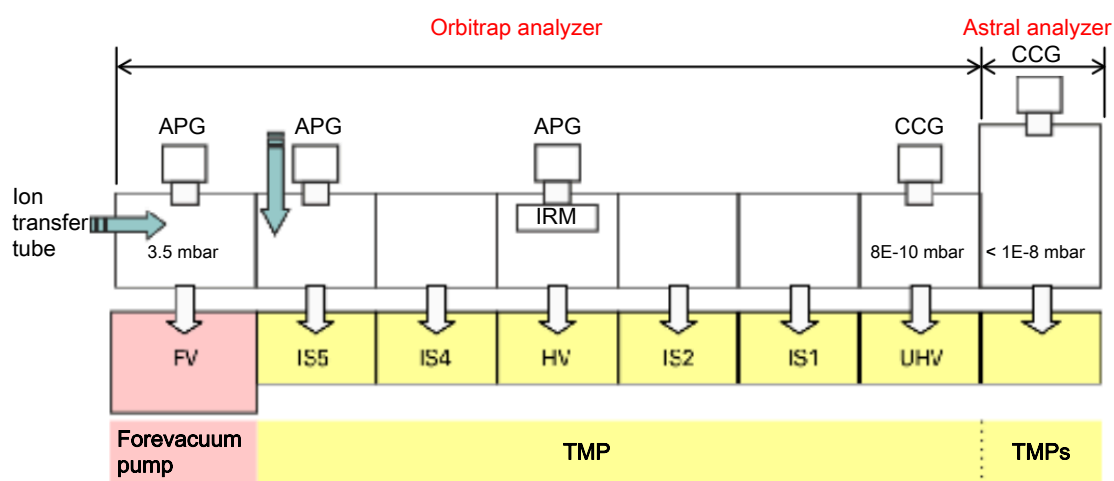
- HDR Detector

The High Dynamic Range Detector aligns the ion packet for extremely sensitive detection with low noise.

5.12 Vacuum system

The vacuum manifold encloses the ion source interface, the ion guides, the C-Trap, and the Orbitrap analyzer. The vacuum manifold consists of thick-walled aluminum chambers with machined flanges on the front, the sides, and the bottom, and various electrical feedthroughs and gas inlets. The vacuum manifold is divided into seven chambers. The region in the first chamber, called the funnel region, is evacuated by an external roughing pump (source vacuum pump). An internal six-stage turbomolecular pump (TMP) applies a vacuum of increasing quality along the other six regions, the sixth region being evacuated by the UHV stage of the TMP.

Figure Schematic of vacuum system of Orbitrap Astral mass



APG = Active Pirani Gauge, CCG = Cold Cathode Gauge, FV = Forevacuum, HV = High Vacuum, IS = Interstage, IRM = Ion-Routing Multipole, UHV = Ultra High Vacuum

Table Vacuum regions overview

Region in instrument	Vacuum [mbar]	Gauge type	Applied by
Ion source housing	1E+03	Pirani	Atmosphere
Funnel region	3.5		Forevacuum pump
Injection filter	4E-05	Pirani	TMP IS5
Bent flatapole			TMP IS4
Quadrupole			TMP HV
Z-lens		Cold cathode	TMP IS2
Region between Z-lens and Orbitrap analyzer			TMP IS1
Orbitrap analyzer	< 8E-10		TMP UHV
Ion-routing multipole	1.1E-2	Pirani	Collision gas
Astral analyzer	< 1E-8	Cold cathode	TMP

5.12.1 Vacuum gauges

Five vacuum gauges monitor the vacuum:

- The vacuum in the funnel region is monitored by an Active Pirani gauge that is connected to the forevacuum line.
- The vacuum in the injection filter region is monitored by an Active Pirani gauge.
- The vacuum in the ion-routing multipole is monitored by an Active Pirani gauge.
- The vacuum in the Orbitrap analyzer chamber is monitored by a Cold Cathode Gauge. This gauge is directly mounted to the turbomolecular pump in the Orbitrap analyzer chamber.
- The vacuum in the Astral analyzer chamber is monitored by a Cold Cathode Gauge. This gauge is directly mounted to the turbomolecular pump in the Astral analyzer chamber.

5.12.2 Forepump

The Orbitrap Astral mass spectrometer is shipped with a high-throughput vacuum dry pump. It evacuates the funnel region of the vacuum manifold and supplies the forevacuum for the TMPs. The pump has a nominal pumping speed of $\geq 110 \text{ m}^3/\text{h}$. The pump is placed on the floor below the Astral compartment.

The forepump is switched on and off by a cable that is plugged into the instrument control board of the MS. The power supply cord of the forepump is plugged into a separate wall outlet. The pump automatically shuts down when the communication with the mass spectrometer stops.

For a detailed description of the forepump, refer to the handbooks of the manufacturer.

5.12.3 Turbomolecular pumps

A six-stage turbomolecular pump (TMP) provides the high vacuum for the Orbitrap module. The air cooled TMP (Split Flow XL) is attached to the bottom of the vacuum manifold. A pump controller supplies power to the TMP and controls it. The controller sends status information of the TMP (such as temperature or rotational speed) to the MS and the data system. The main power circuit breaker switch turns off the TMP.

In the Astral analyzer, the vacuum in the inner chamber is established by a six-stage turbomolecular pump (Split Flow XL). It is backed by the drag stage of a single-stage turbomolecular pump (HiPace 80 Neo) in the transfer region, which also provides additional pumping for the ion processor region.

For a detailed description of the TMPs and for instructions on user maintenance, refer to the handbooks for the pumps.

5.12.4 Vent valve

The vent valve is a solenoid-operated valve that allows the vacuum manifold to be vented. The vent valve is closed when the solenoid is energized.

The vacuum manifold is vented when external power is removed from the instrument. (Power is removed from the instrument by a power failure or by placing the main power circuit breaker in the Off (O) position.) After the external power is removed, power is provided to the vent valve until the rotational speed of the TMP is reduced sufficiently to allow controlled venting of the instrument. If external power is not restored to the instrument in this time, power to the vent valve solenoid is shut off. When power to the vent valve solenoid is shut off, the vent valve opens and the manifold is vented with nitrogen. The vent valve closes after power is restored to the instrument.

5.12.5 System bake-out

After the system has been open to the atmosphere (for example, during maintenance work or a power outage), the vacuum deteriorates due to contaminations of the inner parts of the vacuum system caused by moisture. These contaminations must be removed by heating the vacuum system: a system bake-out. See Bake out the system for instructions on performing a manual system bake-out.

Several heating elements inside the vacuum chamber provide the high temperatures that are necessary to perform a system bake-out.

5.13 Electronic assemblies

The electronic assemblies that control the operation of the mass spectrometer are distributed among various printed circuit boards (PCBs) and other modules, in the embedded computer, and on or around the vacuum manifold of the mass spectrometer. Dedicated electronic assemblies control the operation of the components of the Internal Calibration source such as heaters and pressure regulators. You cannot service the electronic assemblies.

TIP

If you need assistance, contact your local Thermo Fisher Scientific field service engineer. Before you call a service engineer, try to find the defect by means of errors indicated in the Tune application. A precise description of the defect will ease the repair and reduce the costs. Running the appropriate diagnostics can provide further insights.

5.14 Cooling fans

Several fans, including those in the power supply subassemblies, provide internal cooling for the Orbitrap Astral mass spectrometer. Cooling air enters through the three main air intake fans on three sides of the MS. Exhaust air leaves the instrument from the ventilation slots at the rear side and the top side.

The only user-serviceable part are the air filters in front of the air intake fans. For the recommended maintenance schedule, see Maintenance of the fan filters.

In addition to the fans described in this topic, various printed circuit boards are equipped with individual fans.

5.15 Additional hardware

In addition to connecting the Orbitrap Astral mass spectrometer to one or two LC systems, you can use the instrument with a syringe pump or switching valves. With the accessories cabinet, the mass spectrometer provides a place for the operation below a lid at the top right side of the instrument.

5.15.1 Syringe pump

You can connect the Orbitrap Astral mass spectrometer to a syringe pump. The syringe pump delivers sample solution from the syringe into the API source. When the syringe pump is operating, a motor drives a pusher block that depresses the plunger of the syringe at a user selectable rate. Liquid flows out of the syringe needle and into the sample transfer line as the plunger is depressed. For instructions on setting up the syringe pump, see *Set up the syringe*. A suitable syringe pump is shipped with the instrument.

The instrument software can control established syringe pumps by a USB interface in the accessories cabinet. The MS can supply electric power to the syringe pump. Or, use a properly grounded wall outlet. *Set up the inlet for direct infusion* gives instructions on connecting the syringe pump to the MS.

You can set syringe pump parameters with the Tune application. Use the Tune application to switch on/off the syringe pump by an instrument method or manually. Refer to the Help for details.

5.15.2 Switching valves

You can connect the Orbitrap Astral mass spectrometer up to two external switching valves. Suitable switching valves (Rheodyne™ MX Series II™) are available from Thermo Fisher Scientific. The instrument software can control established switching valves by a USB interface in the accessories cabinet. The Universal Power Supply of the valve can be operated from inputs of 100–240 V AC, 50–60 Hz. The MS can provide electric power to the valves. Or, use a properly grounded wall outlet. *Establish power supply and communication for the switching valve(s)* gives instructions on connecting the switching valve(s) to the MS.

You can configure (plumb) a switching valve as a loop injector for flow injection analysis or as a divert valve for direct infusion, high-flow infusion, or LC/MS experiments. Procedures for plumbing the valve in the loop injector or divert valve configuration are given in *Set up the inlet for direct infusion*.

You can control the switching valves with the Tune application. Refer to the Help for instructions about operating the switching valves.

You can also use the switching valve button to divert the LC flow between the mass spectrometer and waste when the valve is in the divert valve configuration. Or, you can switch between load mode and inject mode when the valve is in the loop injector configuration.

Safety

This chapter contains information that is important for your own safety or the safety of others, and that prevents damage to the instrument. Read this chapter carefully before you install or operate the instrument and its accessories, or come into contact with it.

NOTE

To comply with safety and warranty requirements, the instrument and accessories described in this manual are designed to be used only by properly trained personnel.

Any installation, adjustment and repair of this equipment must be carried out only by a certified Thermo Fisher Scientific service representative who is aware of the hazards involved.

To protect our operating personnel, we ask you to adhere to special precautions when you send back parts to the factory for exchange or repair.

Contents

- Safety symbols and signal words in this manual
- Safety symbols on the instrument
- Intended use
- Electric safety precautions
- In case of an emergency
- Residual hazards
- Personal protective equipment

6.1 Safety symbols and signal words in this manual

Notices concerning the safety of the personnel operating the instrument appear different from the main flow of text:



Always be aware of what to do with, and the effect of, safety information.



CAUTION

Points out a hazardous situation that can lead to minor or medium injury if it is not avoided.



WARNING

Points out a hazardous situation that can lead to severe injury or death if it is not avoided.



DANGER

Points out a hazardous situation that will lead to severe injury or death if it is not avoided.

6.1.1 Observe this manual

Always keep this manual near the instrument to have it available for quick reference.



Be sure to read and comply with all precautions described in this manual.

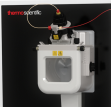
System configurations and specifications in this manual supersede all previous information received by the purchaser.

6.2 Safety symbols on the instrument

The table lists all safety labels on the instrument and their positions. Please carefully read and take note of all safety notices to prevent harm to the operator and to protect the instrument against damage. If present, read and follow the instructions on the labels.

Safety labels on the instrument

Label	Label description	Label position
	This label is attached to the ion source mount at the front side of the instrument. It indicates the presence of high voltage at the ion source.	

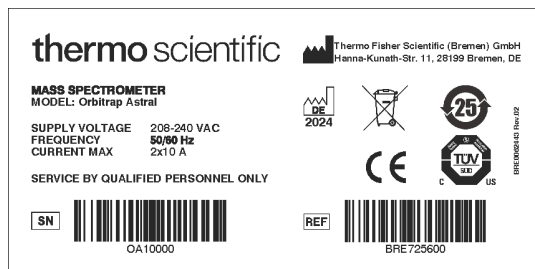
Label	Label description	Label position
	This label is attached to the ion source mount at the front side of the instrument. It indicates the presence of hot surfaces at the ion source mount.	
	This label is attached to the API probe. It indicates the presence of high voltage at the probe.	
	This label is attached to the ion source housing. It indicates the presence of hot surfaces at the ion source housing.	
	This label is attached to the ion source housing. It indicates that you must read the relevant user documentation before you use the ion source.	
	This label indicates the possible presence of toxic, carcinogenic, mutagenic, or corrosive/irritant chemicals.	This label is attached to the front side of the solvent waste container.
		Labels of this type are attached to several places on the instrument that are designed for user interaction. They remind you to read the manual before you operate the instrument.
		This label is attached to several places on the instrument that may be accessed by users. It indicates that only personnel that have received a special

Label	Label description	Label position
		training may service the instrument. User maintenance is restricted to operations that are described in this manual.
		This label is attached to several places on the instrument that may be accessed by users. It indicates that hazardous electric voltage is used in the instrument that can cause serious injury or even death. To make sure that the instrument is free from all electric current, always disconnect the power cords of the instrument before you remove the covers.
		This label is attached to the lid of the quadrupole chamber. It indicates that a crushing hazard is present when the lid is not in its resting position. Keep your hands clear when the lid is open.

6.2.1 Name plate

To identify the instrument correctly when you contact Thermo Fisher Scientific, always have the information from the name plate (also called rating plate) available. The name plate is attached to the rear side of the instrument below the main power switch. It contains the serial number, which is important in any type of communication with Thermo Fisher Scientific.

Figure Name plate (example)



6.3 Intended use

The Orbitrap Astral mass spectrometer is a stand-alone instrument with an atmospheric pressure ionization (API) source for liquid chromatography (LC) mass spectrometry (MS) high-throughput applications.



Observe the following usage guidelines when you operate the Orbitrap Astral mass spectrometer:

- The instrument is designed to be placed in the laboratory. It is not designed for outdoor usage.
- The instrument is designed to be used exclusively with API sources and probes that are approved by Thermo Fisher Scientific.
- The instrument is designed for use as general lab equipment. It is not designed for use in diagnostic or medical therapeutic procedures.

If the instrument is used in a manner that is not specified by Thermo Fisher Scientific, the protection that is provided by the instrument could be impaired. Thermo Fisher Scientific assumes no responsibility and will not be liable for instrument damage and/or operator injury that might result from using the instrument with other API sources and probes.

6.3.1 Notice on the susceptibility to electromagnetic transmissions

The instrument is designed to operate in a basic electromagnetic environment. Do not use radio frequency transmitters, such as mobile phones, in close proximity to the instrument.

6.3.2 Qualification of the personnel



Personnel that install or operate the Orbitrap Astral mass spectrometer must have the following qualifications:

- Electrical Connections

The electrical installation must be made by qualified and skilled personnel (electrician) according to the appropriate regulations (for example, cable cross-sections, fuses, grounding connection). Refer to the *Orbitrap Astral Preinstallation Requirements Guide* for the specifications.

- Installation

Only employees of Thermo Fisher Scientific or personnel who act on behalf of Thermo Fisher Scientific are allowed to install the Orbitrap Astral mass spectrometer.

- General Operation

The Orbitrap Astral mass spectrometer is designed to be operated by qualified laboratory personnel. Before they start, all users must be instructed about the hazards that are presented by the instrument and by the used chemicals. The users must be advised to read the relevant Safety Data Sheets (SDSs).

- Decommissioning

Only employees of Thermo Fisher Scientific or personnel who act on behalf of Thermo Fisher Scientific are allowed to decommission the Orbitrap Astral mass spectrometer. For information about decommissioning third-party components (for example, the forepump), refer to the manuals that came with these components.

6.3.3 Permitted materials

The Orbitrap Astral mass spectrometer is designed to be operated with these materials:

- Nitrogen gas: Used for the API sheath gas, the API auxiliary/sweep gas, the C-Trap trapping gas, the reagent carrier gas of the IC source, and the HCD collision gas
- Fluoranthene: Used as the calibrant species for the IC source
- Calibration compounds, samples

Polar or less polar chemical compounds—soluble in water or appropriate organic solvent. Compounds can have different molecular sizes. Compounds come from environmental specimens of different origins. The origin can be from living or non-living matter. Examples are sugars (carbohydrates) from plant material, proteins or peptides

from animal or human cell lines, synthetic polymers deriving from an organic synthesis. Small molecules may come from a honey sample when screening for or quantifying pesticides or from an organic synthesis.

- Solvents, additives

Typically, mixtures of water and organic solvents are applied. Organic solvents can be methanol, acetonitrile, or isopropanol. Also chloroform or dichloromethane are solvents applied when these are needed to solubilize the compounds of interest and to prepare them ready-to-use for the ion source interface.

Common additives to the solvent mixtures are acids such as trifluoroacetic acid or formic acid (positive ion mode) or a base such as ammonium hydroxide (negative ion mode).

6.4 Electric safety precautions



WARNING

High voltage

High voltages that can cause an electric shock are used in the instrument. Observe these safety precautions when you operate or do maintenance on the instrument:

- The instrument is properly grounded in accordance with regulations when it is shipped. You do not need to make any changes to the electrical connections or to the instrument's chassis to ensure safe operation.
- There are no customer serviceable parts inside the instrument. Do not remove any housing or protective cover except it is permitted elsewhere in this manual. When you leave the system, make sure that all protective covers and doors are properly connected and closed, and that heated areas are separated and marked to protect unqualified personnel.
- Do not switch on the instrument if you suspect that it has incurred any kind of electrical damage. Instead, disconnect the power cords of the mass spectrometer and the forepump and contact a Thermo Fisher Scientific field service engineer for a product evaluation. Do not try to use the instrument until it has been evaluated. Electrical damage might have occurred if the system shows visible signs of damage, or has been transported under severe stress.
- Do not place any objects on top of the instrument—especially not containers with liquids—unless it is requested by the user documentation. Leaking liquids might get into contact with electronic components and cause a short circuit.

6.4.1 Safety protection provided by the instrument

The Orbitrap Astral system provides protection according to several isolation and safety standards for electronic instruments. Refer to the *EU Declaration of Conformity* for information about the relevant standards.

IEC Degree of Protection: IP20

The instrument is resistant to solid foreign bodies from 12.5 mm in diameter (accidental finger contact), but it is not protected against the ingress of water or other liquids.

IEC Protection Class: I

The instrument has a protective ground connection and a basic insulation between accessible parts and the protective earth. A proper protective ground connection is compulsory.

IEC Overvoltage Category: II

The Orbitrap Astral system has a plug and socket connector or a fixed connection supplied from the electrical system of a building.

IEC Pollution Degree: 2

The instrument must be used in an environment where only non-conductive pollution occurs. Occasionally, temporary conductivity caused by condensation is to be expected.

6.4.2 Install a residual current device (RCD) in the laboratory building

The instrument and peripheral must be protected by a residual current device (RCD) with a tripping current of maximum 30 mA. If an RCD is available in your laboratory, the instrument and peripheral must be included in that fuse circuit. If no RCD is available, Thermo Fisher Scientific provides an in-line RCD box to separately protect the instrument and peripheral. In this case, contact your local service office.



CAUTION

Risk of electric shock

Under extremely rare conditions, accessible parts of the instrument might come into contact with the mains voltage. Grounding therefore prevents users and the instrument from damage. In rare cases, an additional protection via GFCI or RCD is necessary. Supply your instrument with an in-line RCD. If the RCD cannot be switched On, call the certified Thermo Fisher Scientific service representative to get support.



WARNING

Risk of electric shock

If the RCD switches Off and cannot be switched On again, or if the RCD repeatedly switches Off, disconnect the RCD from the power socket and call the certified Thermo Fisher Scientific service representative to get support.

An RCD is also called ground fault circuit interrupter (GFCI). When it senses an imbalance between the outgoing and incoming current, this device instantly breaks the electric circuit to prevent serious harm from an outgoing electric shock.

NOTICE

For the installation of an RCD in the building, the local installation requirements must be followed.

If an insulating type transformer, insulating type UPS or insulating type power conditioner is used, the RCD must be connected between the insulating device and the instrument. The in-line RCD box can be used here as well:

- in-line RCD box for North America
- in-line RCD box for other countries



TIP

For more information, refer to the *Safety information for the RCD kit*.

6.5 In case of an emergency



WARNING

Electric current

Electric shock hazard. Placing the main power circuit breaker switch in the Off (O) position may not be sufficient for safe work on the instrument.



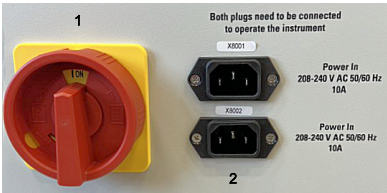
To allow shutting off the mass spectrometer in an emergency, free access to the main power switch at the rear side of the instrument must be possible at any time.

Shut down the system in case of an emergency

Procedure

1. Set the main power switch with a quarter turn counterclockwise to the OFF (O) position.
All power to the MS, including the internal vacuum pumps, is shut off. Also, all power to any devices that are supplied with power by the MS (for example, switching valves, syringe pump) is shut off.

Figure Mains switch with mains supply connectors



No.	Description	No.	Description
1	Main Power switch	2	Mains supply connectors

2. Disconnect the two power cords of the mass spectrometer to ensure that the instrument is free from all electric current.
3. The forepump is only switched off with the relay control cable. To disconnect a forepump from its electric power supply, unplug the power cord.
The pump shuts down automatically when the communication with the mass spectrometer stops. This may take some minutes.
4. Switch off the computer with its On/Off switch.

Result

The instrument is now in a safe state.

6.6 Residual hazards

Users of the Orbitrap Astral mass spectrometer must pay attention to these residual hazards.



WARNING

High voltage

If you touch liquid that leaks from the probe sample inlet while the mass spectrometer is in operation, you might receive an electric shock. Do not tighten the probe sample inlet fitting to eliminate a liquid leak while the mass spectrometer is in operation.



WARNING

Suffocation hazard

A significant amount of the nitrogen that is introduced into the API source can potentially escape into the laboratory atmosphere. Accumulation of nitrogen gas could displace sufficient oxygen to suffocate personnel in the laboratory. Make sure that the laboratory is well ventilated. Always operate the mass spectrometer and the API source with the attached drain tubing assembly that connects the source housing drain to a dedicated exhaust system. Local regulations may make a risk assessment for the workplace necessary.



CAUTION

Hazardous chemicals

Samples, solvents, and detergents might contain toxic, carcinogenic, mutagenic, or corrosive/irritant chemicals. Avoid exposure to potentially harmful materials. Always wear protective clothing, gloves, and safety glasses when you handle solvents or samples. Also contain waste streams and use proper ventilation. Refer to your supplier's Safety Data Sheet (SDS) for proper handling of a particular compound.



CAUTION

Hot parts

During operation of the mass spectrometer, the ion transfer tube and the sweep cone can reach temperatures up to 450 °C. This might cause severe burns upon touching or ignite combustible material. The external surface of the ion source housing can reach temperatures up to 100 °C. This is hot enough to cause skin burns.

Do not touch the ion source housing when the mass spectrometer is in operation. Let the ion source cool down as described in this manual before you remove the ion source housing. Do not touch the ion source mount immediately after you have removed the ion source housing. Keep combustible materials away from the ion source mount. Do not let the instrument stay unattended when the housing is not mounted to the source.

To ensure safety and proper cooling, always operate the MS with its covers in place. This is also necessary to comply with product safety and electromagnetic interference regulations.

6.7 Personal protective equipment

Appropriate safety clothing must be worn at all times while you operate the instrument, particularly when you handle hazardous material.

This manual can only give general suggestions for personal protective equipment (PPE), which protects the wearer from hazardous substances. Refer to the Safety Data Sheets (SDSs) of the chemicals handled in your laboratory for advice on specific hazards or additional equipment.

Eye protection

The type of eye protection required depends on the hazard. For most situations, safety glasses with side shields are adequate. Where there is a risk of splashing chemicals, goggles are necessary.

Protective clothing

When the possibility of chemical contamination exists, protective clothing that resists physical and chemical hazards should be worn over street clothes. Lab coats are appropriate for minor chemical splashes and solids contamination, while plastic or rubber aprons are best for protection from corrosive or irritating liquids.

Gloves

For handling chemical compounds and organic solvents, Thermo Fisher Scientific recommends white nitrile clean room gloves from Fisher Scientific or Unity Lab Services.

For handling hot objects, gloves made of heat-resistant materials (for example, leather) should be available.

Safety boots and shoes

Safety boots and shoes which are penetration-resistant and feature protective toecaps can protect employees from falling objects, vehicles and other heavy loads, for example when moving the instrument.

6.8 End of life of the instrument

Please follow these guidelines when you dispose of the instrument.

6.8.1 Recycling guide for the instrument

Please dispose of the third party components, if present, separately referring to the recycling guides of the producers:

- Forepump(s)
- Syringe pump

The rest can be recycled as AVV 16 02 13* General WEEE with hazardous components.

6.8.2 Dispose of packaging material

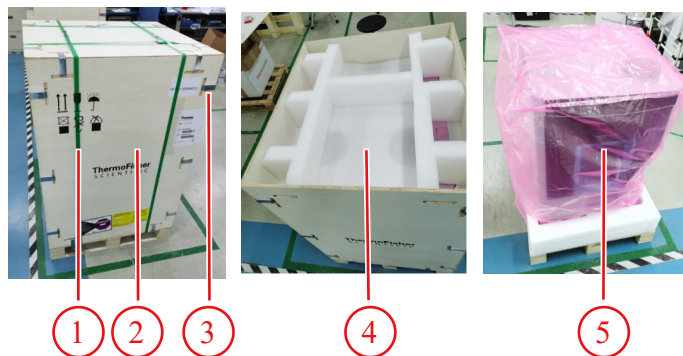
Upon receiving the instrument, we advise you to properly dispose of any packaging material that is used to secure the instrument during the shipping process.

Refer to the international recycling codes defined by the European commission and dispose of any packaging materials accordingly. The packaging of your instrument may differ.

Table International recycling codes

Code	Example
	The wooden crate 2 (shipping box of the instrument)
	All bands 1 and metal clips 3 (made of stainless steel, for example) arranged at the wooden crate
	All damping foam inlays 4 used to fix the instrument within the wooden crate (made of Polythene, for example)
	All plastic covers, bags and foils 5 (made of PE, for example)

Figure Packaging materials to be disposed of correctly



No.	Description	No.	Description
-----	-------------	-----	-------------

1	Bands	4	Foam inlays
2	Wooden crate	5	Bags
3	Clips		

For information on waste disposal sites, refer to your local waste provider.

Installation

Meeting the necessary conditions for the operating environment will ensure the continued high performance of your Orbitrap Astral system. Prepare your laboratory for the installation of the Orbitrap Astral system by ensuring all requirements that are specified in the *Orbitrap Astral Preinstallation Requirements Guide* are met. This guide also provides comprehensive information to assist in planning and preparing your lab site.

Contents

- Place the instrument
- Place the forepump
- Laboratory conditions
- Set up the instrument hardware and adjust the system parameters
- Configure the ion source
- Connect frequently used devices
- Obtain the calibration solution

7.1 Place the instrument

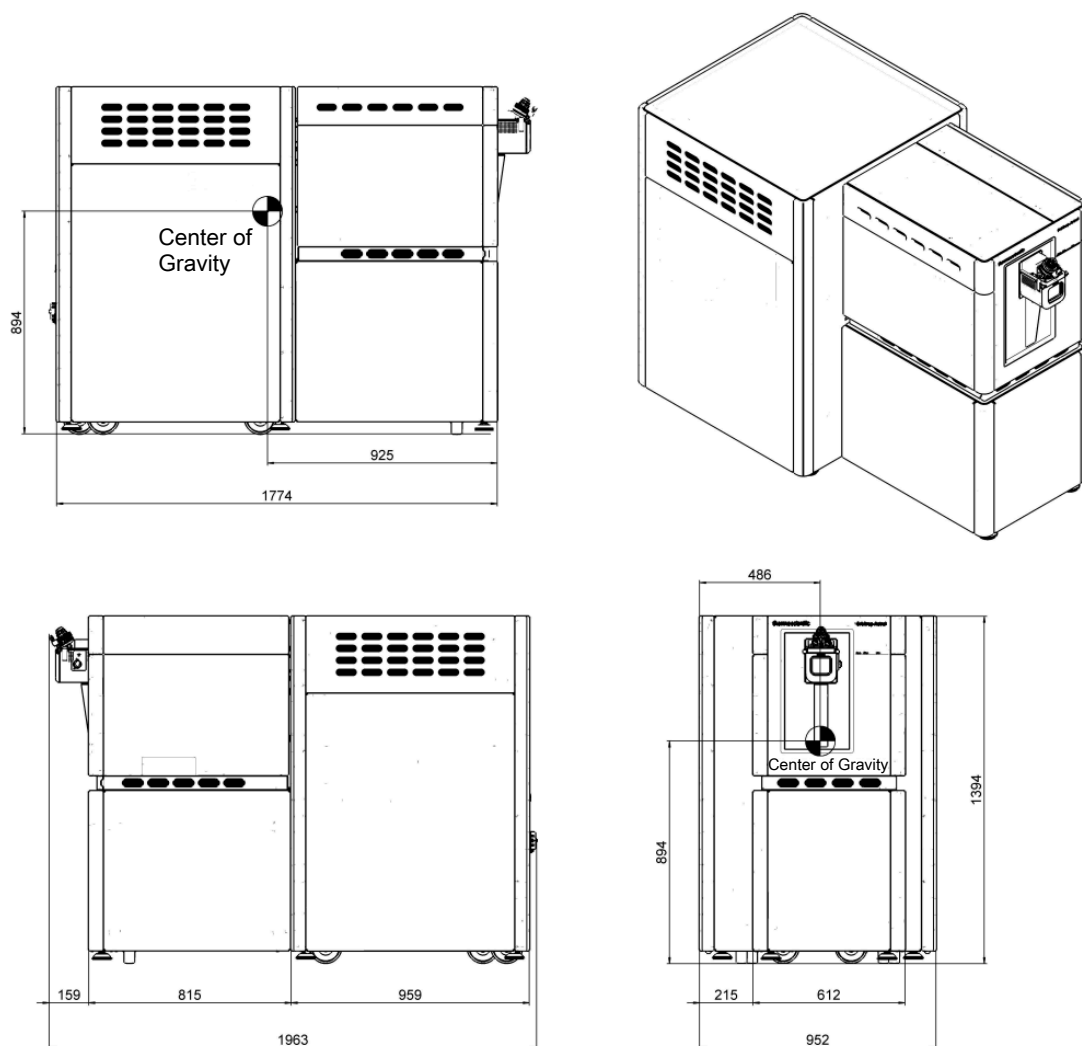
This section provides information that helps you positioning the instrument in the laboratory.

7.1.1 Instrument dimensions

Table Dimensions and approximate weight of Orbitrap Astral mass spectrometer

Length	Width	Height	Weight ¹⁾
1963 mm	952 mm	1394 mm	758 kg
77 1/4 in.	37 1/2 in.	54 7/8 in.	1671 lb

Figure Dimensions of Orbitrap Astral mass spectrometer



7.1.2 Move the instrument

The Orbitrap Astral mass spectrometer is supported by six height-adjustable feet with a diameter of 80 mm each. Six casters at the bottom side of the instrument facilitate positioning it at the intended place in the laboratory. The casters touch the ground when you release the height-adjustable feet.

The casters in the center are fixed for stability when all other casters are rotated inwards. Due to this the instrument rotates around the center and cannot be pushed sideways.



CAUTION

Risk of injury

The instrument is transported in a wooden box. The instrument is rolled out of the box to its final position on casters. When you move the instrument out of the box, you might get hurt by wood splinters or nails. Additionally, crush injuries could result if control over the

instrument is lost. Wear steel-reinforced safety shoes and work gloves during the installation.



CAUTION

Heavy objects

The mass spectrometer and the forepump might move uncontrollably and cause injuries. Wear steel-reinforced safety shoes and gloves during installation or maintenance.

At least 2 persons are required to unpack the instrument and move it to the lab!



NOTE

Before you move the instrument from one place to another, all participating personnel must carefully read and follow the instructions given in this manual.

Procedure

1. Make sure that the six height-adjustable feet are lifted.
2. Carefully roll the instrument on its casters to its final place.
3. When the instrument is in its final location, lower and fasten the six supporting feet.

Result

This lifts the instrument, and the casters hang in the air slightly above the floor.

7.1.3 Floor requirements

The weight of the Orbitrap Astral mass spectrometer (basic system, that is, with forepump but without options) is about 760 kg. The weight is distributed on six height-adjustable feet with a diameter of 80 mm each. The center of gravity lies 894 mm above the floor.

The floor of your laboratory must be able to accommodate the weight of

- the installed Orbitrap Astral MS, including the forepump
- data system
- LC
- any options that are added to the system:

The floor must be flat with a maximum slope of 0.5°.

7.1.4 Clearance

The connections for Ethernet, gas, and electric power are located at the rear side of the instrument. The exhaust lines for the forepump and the solvent waste container are also led out here.

Make sure that you have the following minimum clearances:

- 500 mm (19 $\frac{11}{16}$ in.) between the top of the system and any shelves above it.

Some service work on the Astral module requires opening the top lid. Then, it might be necessary to move the instrument to a place in the laboratory with more top clearance.

- 300 mm (13 $\frac{13}{16}$ in.) between the left, right, and rear side of the instrument and any other components for operating the instrument.
- 800 mm (31 $\frac{1}{2}$ in.) between the left and right side of the instrument and any other components for performing maintenance.



To ensure that the mass spectrometer can be shut off in case of an emergency, free access to the main power switch at the rear side of the instrument must be possible at any time.

NOTICE

Do not block the ventilation slots of the mass spectrometer. Make sure that items cannot fall behind the instrument, inhibit airflow, and cause the system to overheat.

7.1.5 Workbenches

To set up a typical LC/MS system, Thermo Fisher Scientific recommends that you have a minimum of two workbenches. The table lists the recommended minimum surface dimensions for each workbench.

Table Minimum workbench surface dimensions

Equipment	Surface
Data system	120 × 100 cm (47 $\frac{1}{4}$ × 39 $\frac{3}{8}$ in.)
LC system	65 × 61 cm (25 $\frac{9}{16}$ × 24 in.)

The workbench for the LC system must be capable of supporting the weight of the any option (liquid chromatograph, for example) and stand in a secure and level position.

NOTE

Only workbenches with four legs provide sufficient stability for the mass spectrometer. The workbench top must be dry and clean (free of grease). Thermo Fisher Scientific recommends that you use a workbench with a skidproof top.

Place the mass spectrometer, the data system workbench and the LC workbench adjacent to each other to prevent strain on the interconnecting Ethernet communications cables. The Ethernet port of the mass spectrometer is located at the rear side of the instrument.

7.2 Place the forepump

The Orbitrap Astral mass spectrometer is shipped with a dry pump with two vacuum ports. Install the pump on the floor below the Astral compartment. Before you place the pump, Thermo Fisher Scientific strongly recommends that you consider the information contained in Vibration.



CAUTION

Heavy object

Because of its weight, the forepump might move uncontrollably and cause injuries. Wear steel-reinforced safety shoes when you move the pump during the installation.

Use suitable lifting equipment when you move a pump during the installation. For detailed instructions (maximum tilt angle, for example), refer to the manufacturer's documentation that is shipped with the pump.

Figure Forepump



Because its noise level is lower than that of standard oil-based forepumps, the dry pump is shipped without a noise reduction cover.

Table Space and load requirements of dry pump

Height		Width		Length		Weight	
mm	in.	mm	in.	mm	in.	kg	lb

Height		Width		Length		Weight	
344	13 ⁹ / ₁₆	308	12 ¹ / ₈	654	25 ³ / ₄	78	172

7.2.1 Connect the forepump

The forepump is connected to the mass spectrometer and the laboratory in various ways.

Vacuum hoses

The large vacuum hose of the forepump has an inner diameter of 38 mm (1 ½ in.). Connect this vacuum hose to the large vacuum port of the mass spectrometer. The small vacuum hose of the forepump of the Orbitrap Astral MS has an inner diameter of 19 mm (¾ in.). Connect this vacuum hose to the small vacuum port of the MS. To avoid a vacuum leak, the hose clamp should be positioned only about 2 mm away from the end of the vacuum port, where the metal tube is not curved.

Both vacuum hoses have a length of 0.8 m (31 ½ in.). They are made of reinforced material. Because of their large bending radii, the actual lengths of the vacuum hoses are significantly shorter.

Exhaust hose

Lead out the exhaust hose of the forepump through the recess at the bottom of the rear side of the instrument. Connect it to the exhaust system of the laboratory.

NOTICE

Do not vent the drain tubing (or any vent tubing connected to the solvent waste container) to the same fume exhaust system to which you have connected the forepump. Otherwise, the analyzer optics might become contaminated.

The maximum length for the exhaust hose of the forepump is 10 m.

Power supply for the forepump

Connect the power supply cord of the forepump to a wall outlet. See Power supply for specifications of the power supply.

NOTE

To prevent an unwanted operation of the forepump, connect the MS and the pump to the power supplies in the correct sequence:

- When you install a system, first connect the switch cable between the instrument and the pump, then connect the power supply cords.

- When you deinstall or service a system, first disconnect the power supply cords, then disconnect the switch cable between the instrument and the pump.

Contact closure signals

The forepump is switched on and off by a contact closure signal from the mass spectrometer.

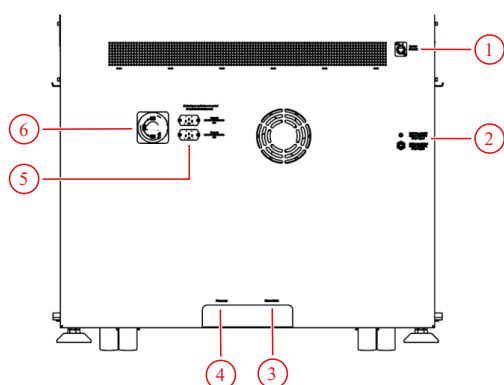
7.3 Laboratory conditions

This section gives an overview of important requirements for the laboratory where the Orbitrap Astral mass spectrometer is placed. For details, refer to the *Orbitrap Astral Preinstallation Requirements Guide*.

7.3.1 Rear side connectors

The rear side of the instrument provides switches and connections for powering the instrument and communication with peripherals.

Figure Connections at rear side of mass spectrometer



No.	Description	No.	Description
1	Ethernet port (labeled System Interlink)	4	Forepump exhaust outlet
2	Gas inlets: C-Trap & HCD (top), Source & Vent (bottom)	5	Main power connectors
3	Waste/Drain exhaust outlet	6	Main power switch

7.3.2 Power supply

Table Power Requirements

Mass spectrometer	
Nominal voltage	2 × 208–240 V AC (± 10%), 50/60 Hz, single phase
Fuse ¹⁾ _{.....}	2 × 15/16 A (tripping characteristic C)
Operation	apparent power: 1000 VA; effective power: 900 W
Bakeout	apparent power: 2500 VA; effective power: 2000 W
Forepump	
Nominal voltage	208–240 V AC, 50/60 Hz, single phase
Power	650 W
Fuse ¹⁾ _{.....}	15/16 A (tripping characteristic C)
Data system (computer, monitor, and Ethernet switch) and LC	
Nominal voltage	100–240 V AC, 50/60 Hz, single phase
Power	100 W
Fuse ¹⁾ _{.....}	15/16 A
Syringe pump and switching valves	
The mass spectrometer provides electric power for the syringe pump and the switching valve(s).	

The power consumption of the instrument depends on the active operating mode. The values in the following table include the power consumption of the forepump.

Table Power consumption depending on instrument mode

Mode	Consumption [W]
Low-Flow mode / Standby	1400
High-Flow mode	1500
Bake-out	2800

Two power connectors for the mains supply are located next to the main power switch at the rear side of the instrument. Use separate wall outlets to supply electrical power for the forepump, the data system, and the devices of the LC/MS system if applicable.

NOTE

The instrument is equipped with detachable mains cords. Use only mains supply cords with an adequate rating. All spare parts and consumable items used must be approved by Thermo Fisher Scientific.

Both power connectors must be connected to wall outlets in the laboratory to operate the instrument.

Thermo Fisher Scientific provides country-specific power cords for the mass spectrometer, forepump, data system, monitor, and Ethernet switch. They are approximately 2.5 m (8 ft) long. One power cord fits into a standard IEC 60320 C19 socket on the dry pump. The other cords fit into standard IEC 60320 C13 sockets on the mass spectrometer and the other system components. Plug the cords into wall outlets.

NOTE

The instrument is tested and certified for a line voltage between 208 V and 240 V. In some areas of the world, voltage sags during high use periods might decrease the working voltage. For nominal voltages below 208 V or for areas with working voltages below 187 V due to voltage sags, you must protect your instrument by using a suitable power conditioner or an uninterruptible power supply (UPS).



WARNING

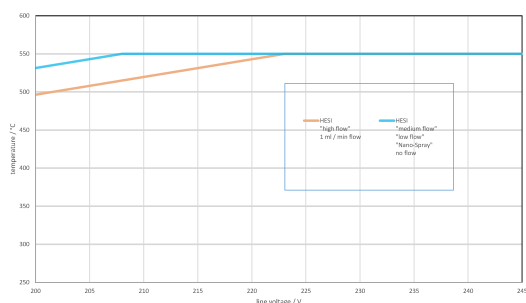
Electric shock hazard

Incorrect usage of these ports might endanger personnel. Read and understand this manual to prevent harm to the operator and to protect equipment against damage. To make sure that the instrument is free from all electric current, always disconnect the power cords of the mass spectrometer and the forepump.

7.3.2.1 Voltage dependency of the ion source heater

The vaporizer of the ion source operates with line voltage. If the main supply voltage drops below its nominal value, it may no longer be possible to reach the maximum temperature of the heater. The figure shows the derating curve of the maximum vaporizer temperature in dependence of the line voltage and the operating conditions for the HESI source.

Figure Maximum vaporizer temperature vs. line voltage



7.3.3 Gas supply

The mass spectrometer requires nitrogen gas for the API sheath gas, API auxiliary/sweep gas, reagent carrier gas, C-Trap bath gas, and HCD collision gas.

The fragment ions that are generated in the ion-routing multipole and the ion processor are generally quite reactive. High purity nitrogen contains notably more water and oxygen than ultra-high purity nitrogen, which gives rise to gas-phase adducts and related products. These species would deteriorate the MS/MS spectral assignments. For the ion-routing multipole and the ion processor, ultra-high purity (UHP, 99.999%) nitrogen is therefore mandatory.

For the reagent carrier gas, ultra-high purity nitrogen is also mandatory.

The gas consumption strongly depends on the type of analysis the instrument is used for. It is essential that the gas be delivered with the necessary pressure and purity.

Table Nitrogen supply requirements

Gas type	Pressure	Purity	Connector Size	Maximum Flow	Consumption
Source gas	0.6 ±	99%	6 mm	45 NL/min	65 000 NL/day
Reagent carrier gas, C-Trap bath gas, HCD collision gas	0.05 MPa (87 ± 7 psi)	99.999% (mandatory)	1/16 in.	0.07 NL/min	100 NL/day

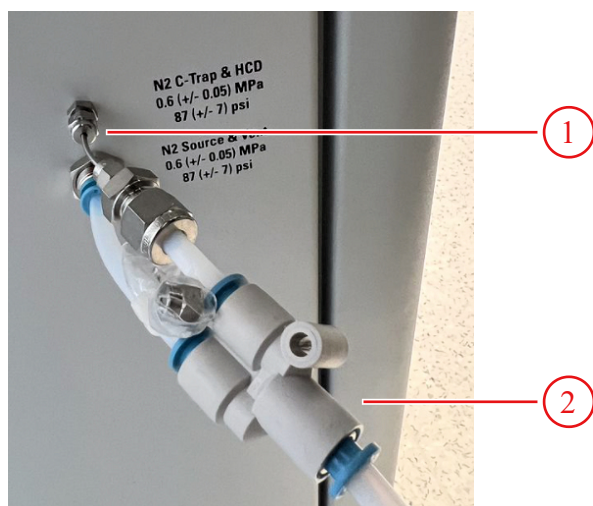
NOTE

Your laboratory gas supply might provide nitrogen also for other consumers or peripherals. In this case, make sure that the supply for the MS always meets the requirements for pressure, purity, and stability described in this topic.

The MS can operate reliably only when the pressure of the source gas stays within the required limits. If your laboratory gas supply provides nitrogen also for other consumers, then you must install a pre-regulator in the gas line that leads to the source gas port of the MS.

The gas inlet ports are at the rear side of the instrument. The port for source gas & vent gas (bottom) has a press-in fitting for a 6 mm hose. The port for C-Trap gas & HCD collision gas (top) has a Swagelok™-type fitting for 1/16 in. tubing.

Figure Gas connections at the rear instrument side



No.	Description	No.	Description
1	Gas inlets: C-Trap & HCD (top), Source & Vent (bottom)	2	Y-piece

NOTE

Do not connect other gases than nitrogen to the mass spectrometer! This might affect instrument performance. The maximum pressure for the gas inlet is 0.65 MPa (6.5 bar, 94 psi).

The gas consumption strongly depends on the type of analysis the instrument is used for.

Table Typical nitrogen consumption of LC/MS applications

Application	LC Flow Rate (approximate)	Source Settings [arbitrary units]	Consumption [NL/min]	
			Source gas ¹⁾	HCD collision gas
Nano-Spray		sheath gas = 0, aux gas = 0,	0	0.07 ²⁾³⁾

Application	LC Flow Rate (approximate)	Source Settings [arbitrary units]	Consumption [NL/min]	
			Source gas ¹⁾	HCD collision gas
		sweep gas = 0		
Low flow	5 µL/min	sheath gas = 5, aux gas = 0, sweep gas = 0	7	
Medium flow	200 µL/min	sheath gas = 35, aux gas = 10, sweep gas = 0	15	
High flow	1 mL/min	sheath gas = 60, aux gas = 20, sweep gas = 5	28	

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The following table shows the estimated nitrogen consumption for different calibration types at ambient pressure. Low-flow settings are used (source gas consumption: 7 NL/min).

Table Estimated Nitrogen Consumption for a Calibration

Action	Estimated duration	Estimated consumption
Manual spray stability optimization	30 min	195 NL
Mass calibration	1 min	5 NL
Full system calibration ⁴⁾	2 hours	800 NL

4)4)
.....

TIP

When you place the MS in Standby condition, do not switch off the nitrogen supply of the laboratory. An automatic source background flow of nitrogen prevents the backstreaming of contaminants and dust into the instrument:

NL/min: 7, NL/day: 10,000

You can supply the nitrogen for your instrument from one source (single supply) or two sources (dual supply). See Gas supply for an overview. Gas supply shows an instrument that is completely supplied

with UHP nitrogen (single supply). Make sure that the maximum flow rate of the nitrogen source of your choice matches the requirements of your applications.

Table Gas supply properties for single supply and dual supply

Supply Type	Nitrogen Purity		Nitrogen Source (Examples)
Single Supply	>99.999% purity (<0.001% oxygen content)		LN2 evaporator, cylinder bundle, PSA nitrogen generator
Dual Supply	Collision gas:	>99.999% purity (5.0)	LN2 evaporator, cylinder bundle, PSA nitrogen generator
	Source gases:	Required: >95% (<5% oxygen content)	Membrane nitrogen generator
		Recommended: >99% (<0.5% oxygen content)	LN2 evaporator, cylinder bundle, PSA nitrogen generator

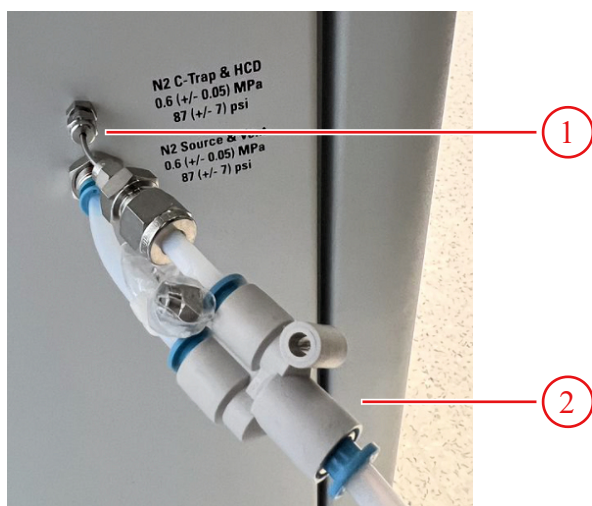
TIP

A nitrogen generator can expel a considerable amount of water (about 10 L per day) from its drain port, dependant on the ambient humidity. Connect the drain tubing to a suitable drain connection or container. If a container is used, it should be emptied at regular intervals. The container must not have an air tight seal because water and air are expelled at pressure. Fix the drain ending of the tube firmly to prevent it from vibrating during draining.

7.3.3.1 Connect the gas supply

The instrument is prepared for single supply. It comes with a Y-piece that has a 6 mm press-in fitting for the hose coming from the nitrogen source. A short segment of 6 mm hose connects the Y-piece to the source gas port of the instrument. A 6 mm-to-1/16 in. adapter connects the Y-piece to the HCD collision gas port of the instrument.

Figure Gas connections at the rear instrument side



No.	Description	No.	Description
1	Gas inlets: C-Trap & HCD (top), Source & Vent (bottom)	2	Y-piece

If you wish to use two nitrogen sources of different purities, you can connect them to the instrument with 6 mm hoses. Use the Y-piece as adapter to connect the hose for the UHP nitrogen to the HCD collision gas port of the instrument. A 6 mm plug is shipped with the Y-piece to close the unused tube. Thus, you do not need to provide 1/16 in. stainless steel tubing.

Connect a single nitrogen source to the mass spectrometer

Procedure

1. Connect an appropriate length of Teflon™ hose to the UHP nitrogen source in the laboratory. The Installation Kit contains 10 m (33 ft) of a suitable hose (OD 6 mm). The connection for the Teflon hose to the nitrogen gas supply is not provided in the kit. You have to supply this part.
2. Insert the opposite end of the Teflon hose into the press-in fitting of the Y-piece that is attached to the gas ports at the left side of the instrument. The Y-piece splits the nitrogen flow to source gas and trapping gas.
3. To connect the hose, align the Teflon hose with the opening in the fitting and firmly push the hose into the fitting until the hose is secure.

Connect two nitrogen sources to the mass spectrometer

Procedure

1. Connect the 6 mm hose that comes from the *high-purity nitrogen* source to the *bottom nitrogen inlet* (labeled N2 Source & Vent) of the instrument.
2. Connect the tubing that comes from the *ultra-high purity nitrogen* source to the *top nitrogen inlet* (labeled N2 C-Trap & HCD) of the instrument.
 - If you wish to use a 6 mm hose for the UHP nitrogen, connect it to the Y-piece. Replace the short 6 mm tubing of the Y-piece by the plug. Connect the 6 mm-to-1/16 in. adapter to the nitrogen inlet.
 - If you wish to use stainless-steel tubing for the UHP nitrogen, you have to provide this. Connect it directly to the 1/16 in. nitrogen inlet.

7.3.4 Laboratory temperature

Keep the laboratory room temperature between 18 and 27 °C (64 and 81 °F).

NOTICE

Do not put the mass spectrometer below an air duct, near windows, or near heating and cooling sources. Temperature fluctuations of 0.5 °C or more over a 10 minutes period can affect instrument performance.

If the temperature in the laboratory has changed for more than 2 °C since the last mass calibration, Thermo Fisher Scientific strongly recommends that you repeat the mass calibration.

TIP

The bakeout time required to reach the operating vacuum in the UHV chamber increases with the laboratory temperature. For a faster time to the end UHV vacuum, we recommend keeping the laboratory temperature at the lower end of the range.

7.3.5 Humidity

The relative humidity of the operating environment must be between 20 and 80%, with no condensation. Thermo Fisher Scientific recommends the installation of a temperature/humidity monitor in your laboratory. This makes sure that the laboratory is always within the specifications for temperature and humidity.

NOTE

Operating the mass spectrometer at very low humidity might cause the accumulation and discharge of static electricity, which can shorten the life of electronic components. Operating the mass spectrometer at high humidity might cause condensation, oxidation, and short circuits, and will also block the filters on the cooling fans.

A significant change of the humidity in the laboratory can affect the mass accuracy. If the humidity in the laboratory has changed significantly since the last mass calibration, Thermo Fisher Scientific recommends that you repeat the mass calibration.

7.3.6 Ventilation and fume exhaust

The vacuum waste and the solvent waste must be vented separately, and the wastes must be collected and disposed of properly.

Table System waste and exhaust requirements

Requirement type	Flow rate	Value
Pump exhaust requirements	Inrush flow rate	10 L/min
	Continuous flow rate	7 L/min
Ion source exhaust requirements	Maximum flow rate	45 L/min

Please read the following safety guidelines for ventilation and exhaust.



WARNING

Suffocation hazard

A significant amount of the nitrogen that is introduced into the API source can potentially escape into the laboratory atmosphere. Accumulation of nitrogen gas could displace sufficient oxygen to suffocate personnel in the laboratory. Make sure that the laboratory is well ventilated. Always operate the instrument with the drain tubing connected to a waste container that is connected to a dedicated fume exhaust system. Local regulations may make a risk assessment for the workplace necessary.



WARNING

Hazardous chemicals

The source exhaust might contain noxious chemicals. It will contain traces of the samples and solvents that you are introducing into the mass spectrometer. Potential health hazards of these compounds include chemical toxicity of solvents, samples, and buffers, as well as

biohazards of biological samples. To prevent contamination of the laboratory, always operate the instrument with the drain tubing connected to a waste container that is connected to a dedicated fume exhaust system.



WARNING

Hazardous chemicals

The forepump eventually exhausts much of what is introduced into the mass spectrometer. The effluent of the forepump might contain noxious chemicals. Connect the exhaust port of the forepump to an exhaust gas line that leads out of the building or to an adequate exhaust system.

NOTICE

Do not vent the drain tubing (or any vent tubing connected to the waste container) to the same fume exhaust system to which you have connected the forepump. Otherwise, the analyzer optics might become contaminated.

NOTICE

Organic compounds that are present in laboratory air can contaminate the LC/MS system. They may be the source of background signals in the mass spectrum. Take precautions to keep the laboratory air clean. Prevent airborne contaminants from entering the solvents.

7.3.7 Dust-free environment

Particulate matter might contaminate the samples, the sample introduction, and the ion source. It might increase the level of chemical background in the mass spectrometer. The air in your laboratory must not contain excessive dust, smoke, or other particulate matter. For reference, the air should contain fewer than 35×10^6 particles per cubic meter (1×10^6 particles per cubic foot) in excess of 5 μm .

NOTICE

Dust is most critical when you perform instrument maintenance. We recommend keeping the instrument under vacuum and the source housing mounted with the automatic source background flow activated at all times.

Open the instrument only when you have to perform maintenance in the instrument. Otherwise, keep the instrument closed.

7.3.8 Vibration

Floors must be free of vibration caused, for example, by equipment in adjoining locations. Excessive vibration can lead to a reduced lifetime of mechanical components and potentially can have an impact on performance. We recommend to maintain the vibration levels below the levels for office spaces as defined in ISO 9241-6:1999.

NOTICE

Because of the natural vibration of a forepump during operation, it must not have any mechanical contact to the mass spectrometer with the exception of the vacuum hose. Otherwise, the vibration might affect instrument performance.

7.3.9 Altitude

The Orbitrap Astral mass spectrometer is designed for indoor use at an altitude of up to 3000 m (10,000 ft) above sea level. To check whether a peripheral (for example, LC or printer) is suitable for the intended altitude, refer to the peripheral's manual or contact Thermo Fisher Scientific.

7.3.10 Airborne noise emission

The A-weighted emission sound pressure level created by the mass spectrometer at work stations does not exceed 70 dB(A).

7.4 Set up the instrument hardware and adjust the system parameters

NOTE

Only employees of Thermo Fisher Scientific or personnel who act on behalf of Thermo Fisher Scientific are allowed to install the mass spectrometer.

The mass spectrometer and other devices shipped with it are installed by a Thermo Fisher Scientific field service engineer. During the installation, the service engineer will demonstrate the basics of equipment operation and routine maintenance.

If you wish to connect the mass spectrometer to other LC equipment, see Connect frequently used devices for advice. Also refer to the manuals that came with the LC equipment.

For detailed information about changing the instrument parameters and developing experiments for the instrument, refer to the Tune Help.

NOTE

New instruments may show a slight mass drift after the installation due to factors such as electronics or temperature stabilization. We recommend to follow the default (daily) mass calibration schedule as advised by the Instrument Control Software in the initial weeks after the instrument is installed. This may also occur with an instrument that is put back into operation after it has not been used for an extended period of storage.

7.4.1 Connect the data system and the mass spectrometer

Place the data system workbench and the LC/MS workbench adjacent to each other to prevent strain on the interconnecting Ethernet communications cables. The Ethernet port of the mass spectrometer is located at the rear side of the instrument. Connect the Ethernet cables from the Ethernet switch to the data system computer's LC/MS Ethernet network card and the mass spectrometer.

7.5 Configure the ion source

The API source can be configured to operate in several API modes, which include electrospray ionization (ESI) for calibration only, heated electrospray ionization (HESI), atmospheric pressure chemical ionization (APCI), and atmospheric pressure photoionization (APPI). A separate nanospray ionization (NSI) source is necessary to operate in NSI mode.

7.5.1 Connect the source housing drain to the solvent waste container

The mass spectrometer internally routes the solvent waste from the bottom of the API source to the source drain port. During the initial installation of the mass spectrometer, a Thermo Fisher Scientific field service engineer installs the solvent waste system. The components of the solvent waste system are contained in the Installation Kit.

Equip your lab with at least two fume exhaust systems:

- The analyzer optics become contaminated if the drain/waste tubing and the exhaust tubing from the forepump connect to the same fume exhaust system. Route the exhaust tubing from the forepump to a dedicated fume exhaust system.
- Vent the solvent waste container to a dedicated fume exhaust system. The exhaust system for the API source must accommodate a flow rate of up to 45 L/min (95 ft³/h).

The waste container is installed on the left side of the compartment below the Orbitrap module. Mount the container with its holder to the frame.

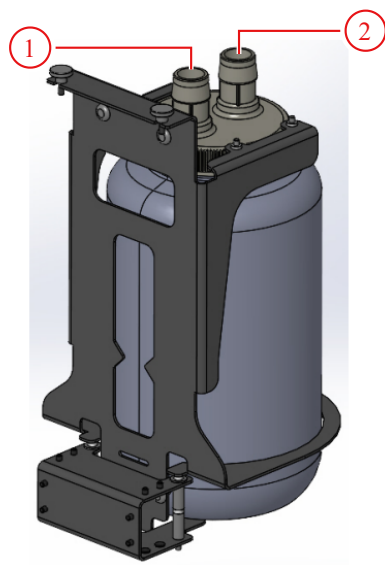
Figure Mounted waste container



NOTE

To prevent the laboratory from being accidentally contaminated by solvent wastes spills, mount the waste container to the instrument frame.

Figure Solvent waste container



No.	Description	No.	Description
1	Connection to source drain	2	Connection to exhaust

Connect the solvent waste container to the mass spectrometer

Procedure

1. Use the provided Tygon™ tubing (1 in. ID, 1 $\frac{3}{8}$ in. OD) to connect the solvent waste container to the drain/waste port of the instrument.
2. Use the provided black spiral tubing to connect the waste container to the external exhaust system. Make sure that your lab exhaust system can accommodate this tubing (1 in. ID, 1 $\frac{1}{4}$ in. OD) without using reducing connectors along the solvent waste path.
3. To prevent solvent waste from backing up into the instrument, make sure that all tubing is above the level of liquid in the waste container.



CAUTION

Tripping hazard

After completing all connections, route the tubing so that they are not a trip hazard.

TIP

The cap adapter insert of the solvent waste container has a small vent. If the exhaust port of the laboratory is always at negative pressure, Thermo Fisher Scientific recommends that you remove the cap that covers the vent. This makes the atmosphere in the solvent waste container independent of the fume exhaust system.

7.5.2 Change the ion source housing

For information about changing the ion source housing, see Remove and reinstall the API source housing. Also refer to the manuals that came with the API source and the API probe. When you change the ion source housing, pay attention to the general safety guidelines that follow.



WARNING

Hazardous chemicals

The source exhaust might contain noxious chemicals. It will contain traces of the samples and solvents that you are introducing into the source. Potential health hazards of these compounds include chemical toxicity of solvents, samples, and buffers, as well as biohazards of biological samples. To prevent contamination of the laboratory, always operate

the instrument with the drain tubing connected to a solvent waste container that is connected to a dedicated fume exhaust system.



CAUTION

Hot parts

During operation of the mass spectrometer, the ion transfer tube and the sweep cone can reach temperatures up to 450 °C. This might cause severe burns upon touching or ignite combustible material. The external surface of the ion source housing can reach temperatures up to 100 °C. This is hot enough to cause skin burns.

- Do not touch the ion source housing when the mass spectrometer is in operation. Let the ion source cool down as described in this manual before you remove the ion source housing.
 - Do not touch the ion source mount immediately after you have removed the ion source housing.
 - Keep combustible materials away from the ion source mount.
 - Do not let the instrument stay unattended when the housing is not mounted to the source.
-

7.5.3 Change the API probe

The mass spectrometer is shipped with the Thermo Scientific OptaMax NG™ API source housing and a combined HESI/APCI probe. See Maintenance of the API probe for instructions about installing and removing this API probe.

For information about changing to other API probes, refer to the manuals that came with the API source and the API probe.

NOTE

Only qualified laboratory personnel are allowed to change the API probe.

7.6 Connect frequently used devices

This section provides information on how to connect the mass spectrometer to external devices that are frequently used with the instrument. It also describes the specifications for the peripheral control connections. For more detailed information, refer to the manuals that come with the devices.

7.6.1 Connect the API probe to the mass spectrometer

All the wiring and gas plumbing for the API source are internal. This means you can install or remove the API source or change the ionization mode (HESI, APCI, or APPI) by changing the spray insert—all without the use of tools.

Procedure

1. Install the API source housing and the API probe onto the mass spectrometer as described in the manuals that came with the API source and the API probe.
2. Install the liquid capillaries between the switching valve, the LC system, the syringe pump, and the grounding union, as appropriate for your application.

7.6.2 Connect the inlet plumbing

TIP

This manual only describes making plumbing connections for ESI/MS and APCI/MS. For information about making plumbing connections for other probe types (H-ESI probe, for example), refer to the manuals that came with these probes.

Sample can be introduced into the API source of the Orbitrap Astral mass spectrometer with a switching valve or syringe pump in the following manner:

- The syringe pump is often used to introduce calibration solution for automatic tuning and calibrating in ESI mode. You can also use this technique to introduce a solution of pure analyte at a steady rate in ESI mode, for example, for determining the structure of an unknown compound.

For direct infusion, connect the syringe pump directly to the API source. To push sample into the ion source, set the rate at which the syringe pump depresses the plunger of the syringe.

- You can also use a union Tee to direct samples from the syringe pump into an LC flow (without a column), which then enters the mass spectrometer. This technique is used to introduce sample at a steady rate and at higher solvent flow rates; it is used especially for tuning in ESI or APCI on an analyte of interest. You can also use this technique to introduce a solution of pure analyte at a steady rate in ESI or APCI.

For high-flow infusion, connect the syringe pump and the outlet of an LC pump to two legs of a union Tee, and then connect the third leg of the union Tee to the ion source. To

introduce sample into the ion source, set the rate at which the syringe pump depresses the plunger of the syringe and the flow rate of the solvent stream produced by the LC pump.

- You can introduce samples from a syringe into the loop of the injector valve. You can then use the switching valve to introduce the sample into an LC flow, which then enters the mass spectrometer. This technique is used in ESI or APCI to introduce pure analytes into the mass spectrometer. It is useful when you have a limited quantity of pure analyte.

For loop injection (flow injection analyses), connect the solvent flow from an LC pump to port 2 of the switching valve, a sample loop to ports 1 and 4 of the valve, and a loop filler to port 5 of the valve. To introduce sample into the ion source, load sample into the sample loop through the loop filler, and then switch the position of the injection valve, allowing the solvent stream to backflush the contents of the sample loop into the ion source.

- You can also use an LC autosampler to introduce samples into an LC flow. This technique is also used in ESI or APCI to introduce a slug of *pure analyte* into the LC flow and then into the mass spectrometer.

Furthermore, you can perform LC/MS experiments with an LC autosampler to introduce a *mixture* onto an LC column. This technique is used with ESI or APCI to separate the analytes before they are introduced sequentially into the mass spectrometer.

For automated injections with an autosampler, connect the outlet from a liquid chromatography system that contains an autosampler to port 2 of the switching valve.

Then set up an autosampler to make automated injections into the solvent flow produced by the LC pump.

7.6.2.1 Set up the inlet for direct infusion

To calibrate the mass spectrometer, use the syringe pump to infuse a calibration solution into the ion source that is equipped with the calibration ESI probe. A suitable syringe pump is shipped with the instrument. For instructions about establishing power supply and communication between syringe pump and mass spectrometer, see Establish power supply and communication for the syringe pump.

To introduce sample solution with the syringe pump, connect an infusion line between the syringe pump and the grounding union that is held by the grounding bar of the OptaMax NG™ API source.

To connect the syringe to the grounding union, follow these procedures:

1. Set up the syringe
2. Connect an infusion line to the grounding union



CAUTION

Hazardous chemicals

Samples and solvents might contain toxic, carcinogenic, mutagenic, or corrosive/irritant chemicals. Avoid exposure to potentially harmful materials. Always wear protective clothing, gloves, and safety glasses when you handle solvents or samples. Also contain waste streams and use proper ventilation. Refer to your supplier's Safety Data Sheet (SDS) for proper handling of a particular solvent.



CAUTION

Pinch hazard

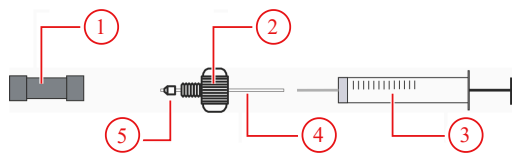
Do not place fingers between the pusher block and end block while the syringe pump is running.

7.6.2.2 Set up the syringe

Procedure

1. To fill the syringe, connect it to the LC union, and insert it into the syringe pump.
2. Fill a clean, 500- μ L syringe with your sample solution.
3. Connect a 4 cm (1.5 in.) length of Teflon™ tubing (0.03 in. ID $\times$ 1/16 in. OD) with a fingertight fitting (for a 10-32 receiving port and 1/16 in. OD tubing) and a ferrule to the LC union.

Figure Connect the syringe and the LC union



No.	Description	No.	Description
1	LC union	4	Teflon tube
2	Fingertight fitting	5	Ferrule
3	Syringe		

4. Insert the needle of the syringe into the segment of Teflon tube. Make sure that the needle tip of the syringe fits readily into the opening in the free end of the Teflon tubing. If necessary, you can make the opening at the end of the tubing larger.
5. Place the syringe into the syringe holder of the syringe pump.
6. Squeeze the release button on the syringe pump handle and slowly push the handle forward until it just contacts the syringe plunger.

7.6.2.3 Connect an infusion line to the grounding union



Make sure that the grounding union is installed. Operating the ion source without installed grounding union can cause a safety hazard.

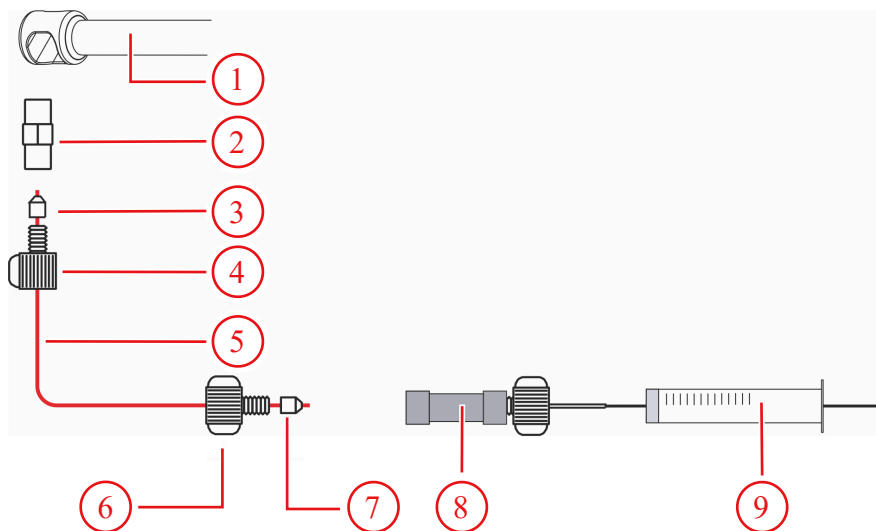
Procedure

1. Connect a section of red PEEK™ tubing (infusion line) with a fingertight fitting and ferrule (for a 10-32 conical receiving port and 1/16 in. OD tubing) to the free end of the LC union.
2. Connect the other end of the infusion line with a fingertight fitting (for a 10-32 port and 1/16 in. OD tubing) and a ferrule to the grounding union.

Result

The figure shows the connection between the grounding union and the LC union made with red PEEK™ tubing and fingertight fittings.

Figure Connecting the infusion line to the LC union and the grounding union



No.	Description	No.	Description
1	Grounding bar of the API source	6	Fingertight fitting
2	Grounding union, 10-32 internal ports, 0.010-in thru-hole	7	Ferrule
3	Ferrule	8	LC union
4	Fingertight fitting	9	Syringe
5	Red PEEK™ infusion line		

7.6.2.4 Establish power supply and communication for the syringe pump

Before you start connecting the syringe pump, make sure that it is safely placed in the accessories cabinet at the top right side of the mass spectrometer.

Procedure

1. Use the USB cable of the syringe pump to connect it to a USB port in the accessories cabinet of the mass spectrometer. This allows controlling the syringe pump by the instrument software.

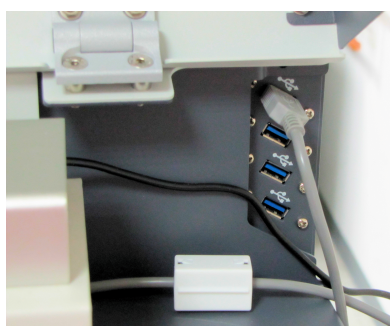


Figure USB ports

2. Connect the power cord of the syringe pump to a peripheral supply port of the MS in the following order:
 - a. Plug the 24 V power cord of the external power supply into the power port at the rear side of the syringe pump.
 - b. Plug the 230 V power cord of the power supply into a peripheral supply port at the bottom front side of the Astral module.

Figure Power supply for syringe pump and switching valves



3. Place the power supply of the syringe pump into one of the holders at the rear side of the Orbitrap module.



WARNING

Electric shock hazard

Incorrect usage of switches or contact ports might endanger personnel. Read and understand this manual to prevent harm to the operator and to protect the equipment against damage.

Refer to the Help for details about controlling the syringe pump with the instrument control software. Refer also to the manual that came with the syringe pump for further information.

7.6.2.5 Establish power supply and communication for the switching valve(s)

TIP

The IDEX™ MX Series II™ switching valves support both, USB and contact closure communication. The valve itself must be properly configured. Use the software tool provided by the IDEX corporation to configure it to USB communication before the first use. If it is not recognized at first use, please fully disconnect and unpower the valve and reconnect it again. To use the USB configured valve for other systems using contact closure operation, the valve must be reconfigured for contact closure.

Prerequisites

- ✓ Before you start connecting the switching valve(s), make sure that they are safely placed in the accessories cabinet at the top right side of the mass spectrometer.

Procedure

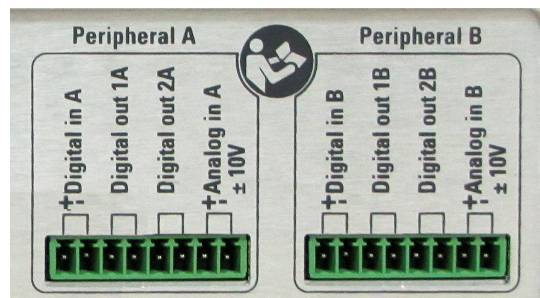
1. Connect the USB cable to the connector at the back of the switching valve. The USB ports of the mass spectrometer are located in the accessories cabinet of the mass spectrometer.
2. The mass spectrometer provides electric power for the switching valve(s):
 - a. Plug the Universal Power Supply male barrel connector into the female port at the rear side of the switching valve.
 - b. Plug the female connector of the power cord into the Universal Power Supply.
 - c. Connect the power cord of the switching valve to a peripheral supply port at the bottom front side of the Astral module.
 - d. Place the power supply of the switching valve into one of the holders at the rear side of the Orbitrap module.

Refer to the Help for details about controlling a switching valve with the instrument control software. Refer also to the manual that came with the switching valve for further information.

7.6.3 User I/O connections

This section describes the specifications for the identical control connections for two peripherals. Location and function of the peripheral control connections are described in Control panel.

Figure Peripheral control connectors



7.6.3.1 Output specifications

The mass spectrometer outputs correspond to the status functions listed below. The outputs are potential-free solid state relay contacts, which are closed when the status indicated by the name is true.

These interchangeable outputs are available:

- Digital Out 1 (Pins 3 and 4)
- Digital Out 2 (Pins 5 and 6)

Output specifications shows the circuit diagram and Output specifications lists the specifications of the peripheral control output.

Figure Output equivalent schematic

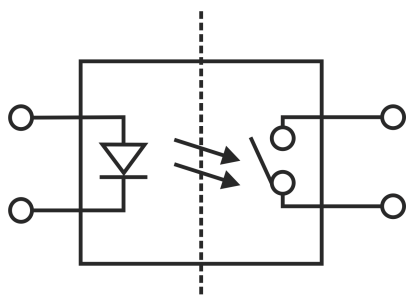


Table Output circuit specifications

Parameter	Value
Contact closed	$R_{\text{on}} < 1 \, \Omega$
Contact open	$R_{\text{off}} > 1 \, \text{G}\Omega$ (typically)
Current	$I_{\text{max}} = 0.6 \, \text{A}$
Voltage	$V_{\text{max}} = 60 \, \text{V}$
Power	$P_{\text{max}} = 720 \, \text{mW}$

7.6.3.2 Input specifications

The mass spectrometer provides a digital input and an analog input to connect external devices.

The *Digital In* input (Pins 1 and 2) is an input with internal pull-up resistor for connecting external relay contacts or open collector transistors. Start is triggered with the falling edge of input voltage. Digital In input equivalent schematic shows the circuit diagram and Input specifications lists the specifications of the peripheral control output.

Figure Digital In input equivalent schematic

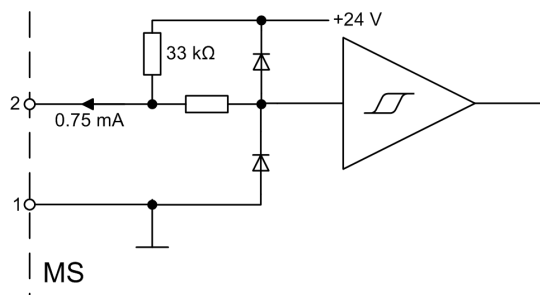


Table Digital In input circuit specifications

Parameter	Value
Low level input voltage	$U_{in} < 0.8 \text{ V @ } 0.75 \text{ mA}$
High level input voltage	$U_{in} > 2.0 \text{ V}$

The *Analog In* input (Pins 7 and 8) is a differential input without isolation for connecting external analog voltages. Analog input equivalent schematic shows the circuit diagram and Input specifications lists the specifications of the peripheral control output.

Figure Analog input equivalent schematic

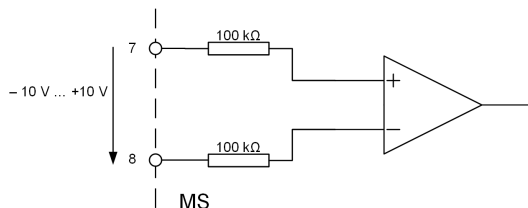


Table Analog input circuit specifications

Parameter	Variable	Value
Input Voltage	U_{in}	$-10 \text{ V ... } +10 \text{ V}$
Input Resistance	R_{in}	$>200 \text{ k}\Omega$ (differential)
Input Resistance	R_{in}	$>100 \text{ k}\Omega$ (ground)

7.7 Obtain the calibration solution

The Thermo Scientific™ Pierce™ FlexMix™ Calibration Solution is a mixture of 16 highly pure, ionizable components (mass range: 40 to 3000 m/z). It is designed for both positive and negative ionization calibration with the Tune application for the Orbitrap Astral mass spectrometer and other Thermo Scientific instruments that are powered by Orbitrap™ technology.

The Pierce FlexMix Calibration Solution features include:

- Improved mass accuracy—extended mass range improves mass accuracy
- Positive and negative mode calibration—single formulation with extended mass ranges for both ionization calibration modes
- Highly validated—high-quality, rigorously-tested formulation with lot-specific certificates of analysis
- Ready-to-use—reference standard may be directly loaded into a syringe and injected into the instrument
- Stable—store at room temperature for up to 18 months

The Pierce FlexMix Calibration Solution is a ready-to-use formulation that has been optimized for both positive and negative ionization calibration of instruments. The extended mass range (40 to 3000 m/z) improves sensitivity and mass accuracy of Orbitrap mass spectrometers. Additionally, the FlexMix Calibration Solution is manufactured at an ISO 9001 facility and each lot is quality controlled with strict specifications. The stable solution is provided in a leak-proof, high-purity PTFE bottle.

Table Available calibration solution packages

Product Name	Unit Size	Product Number
Pierce™ FlexMix™ Calibration Solution	10 mL	A39239

You can order the calibration solution from www.thermofisher.com or www.fishersci.com. The calibration solution is shipped at ambient temperature. Keep it in a dry, cool and well-ventilated place. Keep it in properly labeled containers. Store it in accordance with local regulations.



CAUTION

Hazardous chemicals

Samples and solvents might contain toxic, carcinogenic, mutagenic, or corrosive/irritant chemicals. Avoid exposure to potentially harmful materials. Always wear protective clothing, gloves, and safety glasses when you handle solvents or samples. Also contain waste streams and use proper ventilation. Refer to your supplier's Safety Data Sheet (SDS) for proper handling of a particular solvent.

7.7.1 Solvents and modifiers

For the installation of the instrument, LC/MS grade methanol and water must be available. Solvent modifiers might also be necessary during the installation of some systems. The following solvents are useful in operating and maintaining your instrument:

- Methanol
- Acetonitrile
- Water
- Isopropanol
- Acetic acid (modifier)
- Formic acid (modifier)

Some solvent impurities are transparent to UV/VIS detectors. Therefore, some LC/MS grade solvents might contain contaminants that interfere with the performance of the mass spectrometer. To operate your instrument, choose high purity solvents with minimum contamination.

You can order LC/MS grade consumables from Thermo Fisher Scientific, which are sold under its Fisher Scientific brand. Visit www.fishersci.com. Use only Optima™ LC/MS grade chemicals for operating your system.

NOTE

Do not filter solvents. Filtering solvents can introduce contamination.

7.7.2 Calibration ESI probe

A low-flow ESI probe is shipped with the instrument. Use it exclusively for calibrating the instrument. Do not use it for samples! Keep the calibration probe clean (by flushing it after every use with isopropanol) and store it safely in the sprayer container.

Operation

This chapter describes the checks and cleaning procedures of the Orbitrap Astral system that you should perform every day to ensure the proper operation of your system.

Contents

- Safety guidelines for operation
- Before operating the system
- Set the system in Standby condition
- Shut down the system
- Start up the system after a shutdown
- Introduce sample into the mass spectrometer
- Reset the system
- After operating the system

8.1 Safety guidelines for operation

When you operate the Orbitrap Astral system, pay attention to these general safety guidelines.



WARNING

Hazardous chemicals

The forepump eventually exhausts much of what is introduced into the mass spectrometer, including the small amount of oil vapor that mechanical pumps can emit. The effluent of the forepump might contain noxious chemicals. The connection to an adequate exhaust system is mandatory!



WARNING

Hazardous chemicals

The source exhaust might contain noxious material. It will contain traces of the samples and solvents that you are introducing into the source. Potential health hazards of these compounds include chemical toxicity of solvents, samples, and buffers, as well as biohazards of biological samples. To prevent contamination of the laboratory, always operate the instrument with the drain tubing connected to a solvent waste container that is connected to a dedicated fume exhaust system.



CAUTION

Risk of eye injury

The thin, sharp, flexible capillaries are difficult to see against the light. There is a risk of being stabbed or cut when you do work with the capillaries. Wear safety glasses to prevent eye injuries!



CAUTION

Hot surface

During operation, the forepump is hot and some surfaces reach high temperatures. Even during operational pauses, the pump might be hot. Switch off the pump and let it cool to room temperature before performing any intervention. Wear heat protective gloves when working on the pump that is “still warm from operation.”



To ensure safety and proper cooling, always operate the instrument with its covers in place. This is also necessary to comply with product safety and electromagnetic interference regulations.

8.2 Before operating the system

Every day before you start analyses, make sure that the instrument is ready for operation with these procedures:

1. Check the forepump
2. Check the nitrogen supply
3. Check the system vacuum levels
4. Check the disk space on the data system
5. Check the mass accuracy of the instrument

8.2.1 Check the forepump

Before use, make sure that the forepump is connected to the power supply.

8.2.2 Check the nitrogen supply

Check the nitrogen supply on the regulator of the nitrogen gas tank or the liquid nitrogen boil-off tank. Make sure that you have sufficient gas for your analysis. If necessary, replace the tank. Verify that the pressure of nitrogen that reaches the mass spectrometer is at 0.6 ± 0.05 MPa (6 ± 0.5 bar, 87 ± 7 psi). If necessary, adjust the pressure with the tank pressure regulator.



WARNING

Fire hazard

Air or oxygen in the ion source could produce a fire when it is combined with volatile solvents and high voltages. Introduce combustible samples and solvents into the instrument only with the nitrogen supply connected to the ion source. Make sure that the nitrogen supply is sufficient for daily operation.

8.2.3 Check the system vacuum levels

For proper performance, the mass spectrometer must operate at acceptable vacuum levels. Operating the system at poor vacuum levels can cause reduced sensitivity and tuning problems. Check your system for air leaks and check the system vacuum levels before you start the first acquisition. Make sure that the Status LED at the front of the mass spectrometer is green, which indicates that the instrument is in operating mode. If the LED is not green, check warnings and messages. See Vacuum leak for instructions. In the Tune window, a green square (■) indicates that the readback value is good.

8.2.4 Check the disk space on the data system

Periodically verify that your hard disk drive has enough free space for data acquisition. The amount of available disk space is shown in the Disk Space dialog box.

Procedure

1. From the Home Page window (which is available by choosing Start > Programs > Thermo Xcalibur > Xcalibur), choose Actions > Check Disk Space to open the Disk Space dialog

box. The Disk Space dialog box lists the following:

- Current drive and directory (for example, C:\Xcalibur\system\programs)
- Number of MB that are available (free) on the current drive
- Percentage of the current drive that is available
- Total capacity of the current drive

2. To select another disk drive so that you can determine its disk space, click Directory.

3. When you have completed this procedure, choose OK to close the dialog box.

If necessary, you can free space on the hard disk by deleting obsolete files and by moving files from the hard disk drive to a backup medium. First, copy files to the backup medium. After you have copied the files, you can delete them from the hard disk.

8.2.5 Check the mass accuracy of the instrument

Thermo Fisher Scientific recommends that you check the mass accuracy before you start to use the instrument. If the instrument indicates in the Tune window that the calibration parameters are not optimal, Thermo Fisher Scientific recommends that you calibrate the mass spectrometer. Refer to the Help for instructions.

TIP

Calibration parameters are instrument parameters that affect the mass accuracy and resolution. They are independent of the sample.

Accurate results can be obtained only if the system is properly calibrated.

8.3 Set the system in Standby condition

Do not shut down the Orbitrap Astral system completely if you are not going to use it for a short time, such as overnight or over the weekend. In that case, you can keep the system in Standby condition.

Thermo Fisher Scientific recommends that you keep the mass spectrometer in Standby condition overnight to provide the best mass accuracy next day.

Procedure

1. Wait until data acquisition, if any, is complete.
2. Switch off the flow of sample solution from the LC (or other sample introduction device).

For instructions on how to operate the LC from the front panel, refer to the manual that came with the LC.



3. In the Tune window, click the **System Standby** button () to put the instrument in Standby condition. The Scanning LED on the front panel of the mass spectrometer changes to yellow when the system is in Standby condition.
4. Keep the LC power On.
5. Keep the autosampler power On.
6. Keep the data system power On.
7. Keep the Orbitrap Astral main power circuit breaker switch in the On position.

NOTE

Dust can be sucked in through the ion inlet system. To reduce this risk to a minimum, keep an ion source housing installed on the mass spectrometer also during off times and standby times.

TIP

When you place an Orbitrap Astral MS in Standby condition, do not switch off the nitrogen supply of the laboratory. An automatic source background flow of nitrogen prevents backstreaming of contaminants and dust into the instrument: 7 LN/min; 10,000 LN/day.

8.4 Shut down the system

The Orbitrap Astral system must not be shut down completely if you are not going to use it for a short time, such as overnight or over a weekend. This section describes how to shut down the system for a maintenance or service procedure.

NOTE

Power to the mass spectrometer is removed abruptly when you place the main power circuit breaker switch in the Off (O) position. Although no component in the system is harmed, this is not the recommended shutdown procedure to follow.

Procedure

1. Wait until data acquisition, if any, is complete.

2. Switch off the flow of sample solution from the LC (or other sample introduction device).
For instructions on how to operate the LC from the front panel, refer to the manual that came with the LC.



3. In the Tune window, click the **System Off** button () to put the instrument in Off condition. All high voltages are shut off, as are the sheath and auxiliary gas.
4. Put the main power circuit breaker switch of the mass spectrometer in the Off position.
When the rotational speed of the TMPs are reduced sufficiently, power to the vent valve solenoid is shut off. The vent valve opens and the manifold is vented with nitrogen.
The forepump automatically shuts down when the communication with the mass spectrometer stops. This may take some minutes.



WARNING

High voltage

Hazardous electric voltage that can cause an electric shock is used in the instrument. To make sure that the instrument is free from all electric current, always disconnect the power cords of the mass spectrometer and the forepump.



CAUTION

Hot parts

During operation of the mass spectrometer, the ion transfer tube and the sweep cone can reach temperatures up to 450 °C. This might cause severe burns upon touching or ignite combustible material. The external surface of the ion source housing can reach temperatures up to 100 °C. This is hot enough to cause skin burns.

Do not touch the ion source housing when the mass spectrometer is in operation. Let the ion source cool down as described in this manual before you remove the ion source housing. Do not touch the ion source mount immediately after you have removed the ion source housing. Keep combustible materials away from the ion source mount. Do not let the instrument stay unattended when the housing is not mounted to the source.

If you plan to do routine maintenance or preventive system maintenance on the mass spectrometer only, you do not need to switch off the LC, the autosampler, or the data system. In this case, the shutdown procedure is completed. However, if you do not plan to operate the system for an extended period, you might want to switch off the LC, the autosampler, and the data system.

NOTICE

Dust can be sucked in through the ion inlet system. To reduce this risk to a minimum, keep an ion source housing installed on the mass spectrometer also during off times and standby times.

TIP

An instrument that is shut down still consumes nitrogen because the vent valve is connected to the nitrogen supply of the laboratory. Keeping on the nitrogen flow prevents humidity from contaminating the vacuum system of the mass spectrometer. You may however switch off the nitrogen if your supply is limited.

8.5 Start up the system after a shutdown

To start up the Orbitrap Astral mass spectrometer after it has been shut down (and vented), you must do the following:

1. Start up the instrument
2. Set up conditions for operation

Start up the Orbitrap Astral system

Procedure

1. Make sure that the main power circuit breaker switch at the rear side of the MS is in the Off (O) position.
2. Make sure that the power cords of all components of the LC/MS system and the forepump are plugged into the wall outlets.
3. Switch on the data system and the monitor as described in the manuals that came with them. Wait until the operating system of the computer is completely loaded.
4. Open the Tune application by choosing Start > All Apps > Thermo Instruments > model x.x > model Tune.

TIP

The data system must be running before you start up the instrument. The instrument will not operate until software is received from the data system.

5. Switch on the nitrogen flow at the tank, if it is off.

6. Place the main power circuit breaker switch of the MS in the On (I) position.
The forepump and the turbomolecular pumps are started.
7. When the vacuum system is switched on, the following occurs:
 - a. After the main switch is switched On, the pumps of the mass spectrometer are run up.
The Pirani gauge monitors the pressure at the forepump.

In a short time, the instrument must indicate a significant pressure decrease. If the threshold pressure is not reached after a preset time, or if the rotation speed of the TMPs has not reached at least 80% within 15 minutes, the complete vacuum system is switched off.
 - b. When the TMPs have exceeded 90% of their nominal rotation speed, the instrument will try to start an automatic bake-out.

NOTE

The instrument will start an automatic bake-out when the turbomolecular pumps are at nominal speed. This is typically 10–15 minutes after power up of the instrument. The data acquisition PC needs to be up and running and connected to the instrument.

- Orbitrap analyzer: 10 hours heating plus 3 hours cooling speed
 - Astral analyzer: 5 hours heating, 6 hours cooling
-

8. The Cold Cathode Gauges are switched on at the beginning of the cooling phase. They are then used to monitor the vacuum in the Orbitrap analyzer and the Astral analyzer.
9. The Status LED on the system panel changes to green when all the following conditions are met:
 - In the Tune application, all LEDs are green.
 - Analyzer temperature is below 45 °C.
 - The TMP frequencies have exceeded 90% of their maximum rotation speed.
10. When the vacuum measured by the Cold Cathode Gauges is better than 1E-8 mbar, the power supplies of the high voltage electronics and the tube heater are switched on.
11. If you have an LC or autosampler, start it as is described in the manual that came with the LC or autosampler. If you do not have either, set up your mass spectrometer for operation as described below.

Set up your Orbitrap Astral mass spectrometer for operation

Procedure

1. Make sure that the Vacuum LED on the front of the mass spectrometer is green, which indicates that the pressure gauges are within their threshold values. In the Tune window, a green square (■) indicates that the readback value is good.
2. Compare the current values of the pressures in the Tune application with the values listed in Start up the system after a shutdown. If the current values are higher than normal, there might be an air leak. Also compare the values of the other parameters in the instrument status window with values that you recorded previously.
3. Make sure that the gas pressure is within the operational limits:
Nitrogen: 0.6 ± 0.05 MPa (6 ± 0.5 bar, 87 ± 7 psi)
4. Continue to set up the instrument for operation. See Before operating the system for additional information.

Table Typical pressure values

Name in Tune application	Typical values
Forevacuum Pressure	about 3.5 mbar
UHV Pressure	< 8E-10 mbar
Astral UHV Pressure	<1E-8 mbar

8.6 Introduce sample into the mass spectrometer

A Thermo Fisher Scientific field service engineer has installed the mass spectrometer and the equipment for introducing sample solution and/or calibration solution (LC or syringe pump) and demonstrated their usage. If you wish to connect the MS to other LC equipment, see Connect frequently used devices for advice. Also refer to the manuals that came with the LC equipment.

NOTE

New instruments may show a slight mass drift after the installation due to factors such as electronics or temperature stabilization. We recommend to follow the default (daily) mass calibration schedule as advised by the Instrument Control Software in the initial weeks after the instrument is installed. This may also occur with an instrument that is put back into operation after it has not been used for an extended period of storage.

Consider the following safety guidelines when you introduce sample into the mass spectrometer.



WARNING

High voltage

If you touch liquid that leaks from the probe sample inlet while the mass spectrometer is in operation, you might receive an electric shock. Do not tighten the probe sample inlet fitting to eliminate a liquid leak while the mass spectrometer is in operation.



WARNING

Hazardous chemicals

Sample solution that contains noxious compounds might drip out of the ion source housing. Always operate the instrument with the drain tubing connected to a solvent waste container that is connected to a dedicated exhaust system.



CAUTION

Hazardous chemicals

Samples and solvents might contain toxic, carcinogenic, mutagenic, or corrosive/irritant chemicals. Avoid exposure to potentially harmful materials. Always wear protective clothing, gloves, and safety glasses when you handle solvents or samples. Also contain waste streams and use proper ventilation. Refer to your supplier's Safety Data Sheet (SDS) for proper handling of a particular solvent.



CAUTION

Pinch hazard

Do not place fingers between the pusher block and end block while the syringe pump is running.

NOTICE

While using corrosive fluids like acetonitril, any leakage from the capillaries connected to the source, syringe pump or switching valve might lead to corrosion of the sheet metal's surface. It is recommended to mitigate potential leakages by using a convenient tray fit for your purpose.

8.6.1 Perform mass spectrum measurements

The mass spectrometer is controlled by a PC running the Xcalibur™ software suite. The software controls all aspects of the instrument. The main software elements are the control of ion detection and

the control of the Orbitrap analyzer. Refer to the Help for information.

Consider the safety guidelines that follow when you perform measurements with the mass spectrometer.



WARNING

Suffocation hazard

A significant amount of the nitrogen that is introduced into the API source can potentially escape into the laboratory atmosphere. Accumulation of nitrogen gas could displace sufficient oxygen to suffocate personnel in the laboratory. Make sure that the laboratory is well ventilated. Always operate the mass spectrometer and the API source with attached drain tubing assembly that connects the source drain to a dedicated exhaust system. Local regulations may make a risk assessment for the workplace necessary.



WARNING

Fire hazard

Air or oxygen in the ion source could produce a fire when they are combined with volatile solvents and high voltages. Introduce combustible samples and solvents into the instrument only with the nitrogen supply connected to the ion source. Make sure that the nitrogen supply is sufficient for daily operation.

TIP

Organic compounds that are present in laboratory air can contaminate the LC/MS system. They may be the source of background signals in the mass spectrum. Take precautions to keep the laboratory air clean. Prevent airborne contaminants from entering solvents.

8.7 Reset the system

If communication between the mass spectrometer and data system computer is lost, then it may be necessary to reset the system with the reset button of the mass spectrometer.

Procedure

1. Make sure that the mass spectrometer and the data system computer are both powered on and that the mass spectrometer is in Standby condition. See Set the system in Standby condition for instructions about placing the instrument to Standby condition.

2. To reset the mass spectrometer, press the reset button on the control panel. When you press the reset button, the following occurs:
 - a. An interruption of the embedded computer causes the CPU to reboot.
 - b. After several seconds, the Status LED changes to yellow to indicate that the data system and the instrument are starting to establish a communication link.
 - c. After several more seconds, the Status LED changes to green to indicate that the data system and the instrument have established a communication link. Software for the operation of the instrument is then transferred from the data system to the instrument.

Result

After three minutes, the software transfer is complete. The System LED is either green to indicate that the instrument is functional and the high voltages are on, or yellow to indicate that the instrument is functional and it is in Standby condition.

8.8 After operating the system

After operating the mass spectrometer, follow these procedures:

1. Flush the sample transfer line, the metal needle, and the API probe
2. Clean the sweep cone and the ion transfer tube
3. Empty the solvent waste container

TIP

Thermo Fisher Scientific recommends that you keep the mass spectrometer in Standby condition overnight to provide the best mass accuracy next day.

8.8.1 Flush the sample transfer line, the metal needle, and the API probe

Flush the sample transfer line, the metal needle, and the API probe at the end of a working day or a series of measurements (or more often if you suspect they are contaminated) by flowing a 50:50 methanol/distilled water solution from the LC through the API source.

TIP

You do not need to flush the inlet components (sample transfer line, sample tube, and spray insert) daily. However, if a mass spectrum shows unwanted contamination peaks, flush the inlet components.

Procedure

1. Wait until the data acquisition, if any, is complete.
2. Open the Tune application by choosing Start > All Apps > Thermo Instruments > model x.x > model Tune.
3. From the Tune application window, click the System On button () to toggle it from Standby to On. The voltages and gas flows to the API source are switched on.
4. Set up the HESI probe as follows:
 - a. In the Tune application window, display the HESI Source window.
 - b. In the HESI Source window, enter 30 in the Sheath Gas Flow Rate text box.
 - c. In the HESI Source window, enter 5 in the Aux Gas Flow Rate text box.
 - d. In the HESI Source window, enter 0 in the Sweep Gas Flow Rate text box.
 - e. In the HESI Source window, enter 0 in the Spray Voltage text box.
 - f. Click Apply.
5. Set up and start a flow of 50:50 methanol/water solution from the LC to the API source:
 - Set the Flow Rate to a value that is typical for your experiments.
 - Set the solvent proportions to 50% methanol and water.



CAUTION

Hazardous substance

Methanol is highly flammable and noxious. Keep away from heat, sparks, and flame. Avoid accidental exposure. Use with adequate ventilation. When possible, dilute it with 50% water. Refer to your supplier's Safety Data Sheet (SDS) for information about the correct handling of methanol.

6. Let the solution flow through the sample transfer line, the metal needle, and the API probe for 15 minutes. After 15 minutes, switch off the flow of liquid from the LC to the API source. Keep the API source (including the APCI vaporizer, the sheath gas, and the auxiliary gas) on for additional 5 minutes. Stop the LC pump.

7. After 5 minutes, place the system in Standby condition.

8.8.2 Clean the sweep cone and the ion transfer tube

Clean the sweep cone (or spray cone) and the ion transfer tube on a regular basis to prevent corrosion and to maintain optimum performance of your API source. A good practice is to remove and clean the sweep cone and the ion transfer tube at the end of each operating day—after you have flushed the sample transfer line, the metal needle, and the API probe with a 50:50 methanol/water solution from the LC. If you are operating the system with nonvolatile buffers in your solvent system or with high concentrations of sample, you might need to clean the sweep cone and the ion transfer tube more often. It is not necessary to vent the system to clean the sweep cone and the ion transfer tube.

Procedure

1. Switch off the flow of liquid from the LC (or other sample introduction device) to the API source.
2. From the Tune application window, click the System Off button to switch off the mass spectrometer.



CAUTION

Hot parts

The APCI vaporizer typically operates at 350 to 450 °C and the ion transfer tube typically operates at 250 to 400 °C. This might cause severe burns upon touching. Let the ion source cool down as described in this manual before you touch or remove the vaporizer or the ion transfer tube.

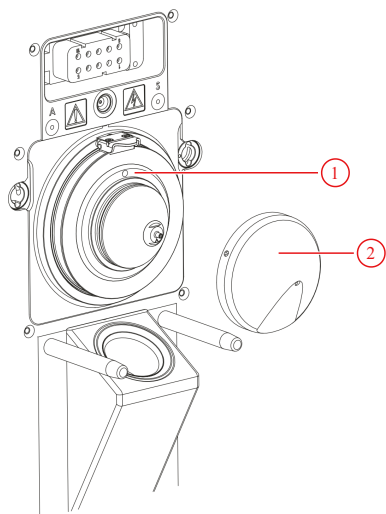
3. Remove the API source from the front of the mass spectrometer, as described in Remove the API source housing.
4. Use a spray bottle filled with 50:50 methanol/water solution and a lint-free industrial tissue (Kimwipes™, for example) to clean contaminants from the accessible surfaces of the ion source chamber.

TIP

To prevent contamination of the ion optical elements, do not clean the sweep cone or the ion transfer tube with solvent when they are attached to the system.

5. Remove the sweep cone (if it is installed):
 - a. Put on a new pair of lint- and powder-free gloves.
 - b. Grasp the outer ridges of the sweep cone and pull the cone straight off the API cone seal.

Figure Sweep cone removed from the MS mount assembly (illustration similar)



No.	Description	No.	Description
1	API cone seal	2	Sweep cone

6. Clean the ion transfer tube and the sweep cone (if it is installed). See Remove, clean, and reinstall the ion transfer tube for instructions.
7. Reinstall the ion transfer tube.

NOTE

A protective mechanism prevents the air from entering the vacuum manifold. It requires that you insert the ion transfer tube with a quick and steady motion into the spray cone.

8. Reinstall the sweep cone: Carefully align the gas inlet on the sweep cone with the sweep gas supply port in the API cone seal. Firmly press the sweep cone into position. Note the location of the sweep gas supply port in the API cone seal. The gas inlet on the sweep cone is placed in this port.

TIP

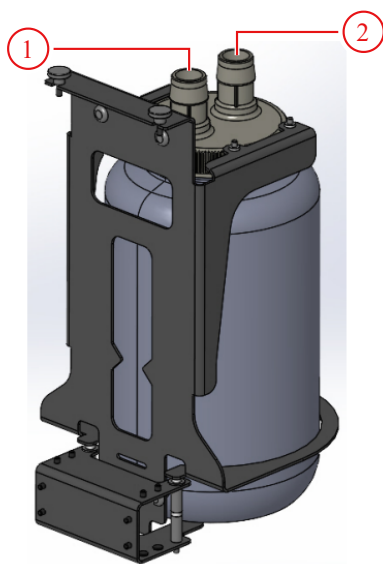
If the spray cone sits either too tight or too loose, adjust the three ball head spring force screws at the spray cone.

9. Reinstall the API source as described in Install the ion source housing.

8.8.3 Empty the solvent waste container

The solvent waste container is installed on the left side of the compartment below the Orbitrap module. It can be accessed after removing the panel.

Figure Solvent waste container



No.	Description	No.	Description
1	Connection to source drain	2	Connection to exhaust

Procedure

1. Check the solvent level in the solvent waste container on a daily basis.
2. Empty the solvent waste container if it is necessary. Dispose of the solvent waste in accordance with national and local regulations.

Troubleshooting

This chapter provides information about identifying and solving common problems with the Orbitrap Astral mass spectrometer.

Contents

- Safety guidelines for troubleshooting
- Fault table
- Main power failure
- Vacuum leak
- Failure of the UHV chamber heating control
- Turbomolecular pump is not running
- Turbomolecular pump cannot reach full rate
- Failure of the source heaters / tube heaters

9.1 Safety guidelines for troubleshooting

When you troubleshoot the Orbitrap Astral system, pay attention to these general safety guidelines.



WARNING

High voltage

High voltages that can create an electric shock are used in the instrument. Do not remove protective covers from PCBs. Opening the instrument housing is only allowed for maintenance purposes by Thermo Fisher Scientific personnel. To make sure that the instrument is free from all electric current, always disconnect the power cords of the mass spectrometer and the forepump.

NOTE

Service by the customer must be performed only by trained qualified personnel and is restricted to servicing mechanical parts. Service on electronic parts must be performed only by Thermo Fisher Scientific field service engineers.

Do not try to repair or replace any component of the system that is not described in this manual without the assistance of your Thermo Fisher Scientific field service engineer.

9.2 Fault table

If malfunctions on the mass spectrometer occur, you will find possible causes and instructions for repair in the table.

Table Troubleshooting

Problem	Possible Causes	Remedy/Information
Tune application shows bad vacuum	System was vented because of a mains supply failure	Vacuum leak
	Vacuum leak	
Temperature of UHV chamber is higher than expected	UHV chamber heating control has failed	Failure of the UHV chamber heating control
Turbomolecular pump is not running	TMP is switched off because of overheating	Turbomolecular pump is not running
Temperature of ion source components is higher than expected	Tube heating control or ion source heater control has failed	Failure of the source heaters / tube heaters
Temperature of ion source components is lower than expected	Line voltage has dropped (brownout)	Failure of the source heaters / tube heaters
Fore vacuum pressure too low	Ion transfer tube is blocked	Clean or exchange the ion transfer tube.
Full MS sensitivity is reduced		

9.3 Main power failure

A main power failure has the same consequence as switching off with the main power circuit breaker switch. If the power is available again, the system starts up automatically: the pumps are switched on and the vacuum is created.

NOTE

The instrument will start an automatic bake-out when the turbomolecular pumps are at nominal speed. This is typically 10–15 minutes after power up of the instrument. The data acquisition PC needs to be up and running and connected to the instrument.

Duration:

- Orbitrap analyzer: 10 hours heating plus 3 hours cooling speed
- Astral analyzer: 5 hours heating, 6 hours cooling

If the log file of the data system shows a reboot of the system and the pressure reading in the Tune application shows a bad vacuum, this indicates that the system was vented. In case of frequent but short power failures, Thermo Fisher Scientific recommends that you install an uninterruptible power supply (UPS). If main power failures occur frequently while the system is not attended (for example, in the night), Thermo Fisher Scientific recommends that you install a power fail detector.

9.4 Vacuum leak

For proper performance, the Orbitrap Astral system must operate at acceptable vacuum levels. You can check the current pressure values in the Tune application.

Check the vacuum levels

Procedure

1. Make sure that the Vacuum LED on the front of the mass spectrometer is green, which indicates that the pressure gauges are within their threshold values. In the Tune window, a green square (■) indicates that the readback value is good.
2. Compare the current values of the pressures in the vacuum manifold with the values listed in Vacuum leak. If the current values are higher than normal, there might be an air leak.
3. If the value for the UHV Pressure is high (above 1E-9 mbar), bake out the instrument for about 10 hours or more.
4. If the pressure remains high, the system might have an air leak. See below for instructions.

Table Typical pressure readings

Gauge type	Name in Tune application	Typical values
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Gauge type	Name in Tune application	Typical values
Pirani	Forevacuum Pressure	about 3.5 mbar, depends on temperature of heated ion transfer tube
Pirani	IRM Pressure	1.1E-2 mbar ¹⁾ (on/standby mode), 0E0 (off mode) ²⁾
Cold cathode	UHV Pressure	< 8E-10 mbar
Pirani	Ion Processor	1.0E-2mbar
Cold Cathode	Astral UHV Pressure	<1E-8 mbar

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Check the system for major air leaks

Procedure

1. Listen for a rush of air or a hissing sound inside the mass spectrometer.
2. Check the vacuum system for loose or disconnected fittings, improperly positioned O-rings, contaminated (by debris) O-rings, or open valves.
3. If you suspect an air leak in the forevacuum region, check the vacuum tube for cracks or slits.
4. Check the clamp that fixes the vacuum tube to the forevacuum port (elbow) of the mass spectrometer. It might be loose because the forepump was shifted or the vacuum tube is twisted.
5. Check that the hose clamps are not placed in a curved region of the vacuum connections.
6. Make sure that the top lid of the instrument is properly seated.

NOTICE

If the vacuum tube suddenly falls off, the forevacuum port opens completely. When the TMP is at full rate in that moment, the resulting shock wave from atmospheric pressure to high vacuum might destroy it.

Fix an air leak

Procedure

1. Shut down the system.
2. Make a visual inspection of the vacuum system and vacuum lines for leaks.

3. Check each fitting and flange on the system for tightness, and tighten the fittings or flanges that are loose.
4. Do not tighten fittings indiscriminately. Pay particular attention to fittings that have been changed recently or to fittings that have been subjected to heating and cooling.
5. Make sure that the O-rings are clean, and that they and the cover plates of the vacuum manifold are properly positioned.

If you cannot find the location of the leak or tighten the leak yourself, contact your Thermo Fisher Scientific field service engineer.

9.5 Failure of the UHV chamber heating control

During general operation of the mass spectrometer, the temperature of the UHV chamber is not regulated. Only during a system bake-out, electric power is supplied to the heating elements of the UHV chamber. If the mass spectrometer does not operate as expected, use the Tune application for error diagnosis. Refer to the Tune Help for details.

If the UHV chamber heating control fails, which protects the system from overheating, shut down the mass spectrometer. To prevent permanent damage to components of the mass spectrometer, Thermo Fisher Scientific recommends that you call a Thermo Fisher Scientific field service engineer.

9.6 Turbomolecular pump is not running

A turbomolecular pump may be switched off because of one of the following reasons:

- Turbomolecular pump is blocked.
- Failure of fans in turbomolecular pump.
- Turbomolecular pump starts up too slow and shuts down after 15 minutes.
- System shuts down 70 seconds after power up.

The turbomolecular pump switches off automatically if it is at risk of getting too hot. This protection prevents the outbreak of a fire and minimizes the risk of destructing the pump.

If the mass spectrometer does not operate as expected, use the Tune application for error diagnosis.

In case of startup issues, check the fore vacuum readbacks. In many cases, a leak causes too high pressure to start up the instrument.

In case of an overheated turbomolecular pump, shut down the mass spectrometer. To prevent permanent damage to components of the mass spectrometer, Thermo Fisher Scientific recommends that you call a Thermo Fisher Scientific field service engineer.

9.7 Turbomolecular pump cannot reach full rate

After the mass spectrometer was started, a TMP might not reach the full rate (Pfeiffer SplitFlow XL: 1000 Hz, Pfeiffer HiPace 80 Neo: 1500 Hz). The TMP then stops in about 20 minutes after start.

A possible reason might be that the internal time limit (currently 15 minutes) for the TMP startup is reached. In many cases, a leak causes too high pressure to start up the instrument:

- Double check if the lid of the vacuum chamber is properly seated.
- Check if the inner cage is fully inserted.
- Check if the vacuum hoses are properly connected.

Try to start the mass spectrometer again. If the failure of the TMP startup occurs a second time, first check for a vacuum leak and then refer to the other troubleshooting section.

9.8 Failure of the source heaters / tube heaters

A failure of the source heater control or ion transfer tube heater control might lead to an overheating of ion source parts. Non-metallic parts of the ion source (gaskets, for example) are made of non-flammable materials, thus preventing any ignition.

If the mass spectrometer does not operate as expected, use the Tune application for error diagnosis. Refer to the Tune Help for details.

A lasting contrast between the target temperatures set in the Tune application and the observed temperature of hardware components indicates a failure of the heater control. But see also Voltage dependency of the ion source heater for information about the dependency of the transfer tube temperature on the mains supply voltage.

In case of a failure of the source heater control or the tube heater control, shut down the mass spectrometer. To prevent permanent damage to components of the mass spectrometer, Thermo Fisher Scientific recommends that you call a Thermo Fisher Scientific field service engineer.

Maintenance

The users of the Orbitrap Astral mass spectrometer must perform different routine maintenance procedures to ensure optimum performance of the instrument.

Contents

- Safety guidelines for maintenance
- Guidelines for maintenance
- Inspection- and servicing plan
- Tools and supplies
- Maintenance of the vacuum system
- Maintenance of the fan filters
- Maintenance of the API source
- Maintenance of the internal calibrant discharge source
- Maintenance of the LC systems or the autosamplers
- Consumables
- Thermo Fisher Scientific Service

10.1 Safety guidelines for maintenance

When you perform maintenance on the Orbitrap Astral system, pay attention to these general safety guidelines:



WARNING

High voltage

High voltages that can cause an electric shock are used in the instrument. Do not remove protective covers from PCBs. Opening of the instrument housing is only allowed for maintenance purposes by Thermo Fisher Scientific personnel. To make sure that the

instrument is free from all electric current, always disconnect both the power cords of the *mass spectrometer and the forepump*.



CAUTION

Hazardous chemicals

Samples, solvents, and detergents might contain toxic, carcinogenic, mutagenic, or corrosive/irritant chemicals. Avoid exposure to potentially harmful materials. Always wear protective clothing, gloves, and safety glasses when you handle solvents or samples. Also contain waste streams and use proper ventilation. Refer to your supplier's Safety Data Sheet (SDS) for information about the correct handling of a particular compound.



CAUTION

Risk of eye injury

The thin, sharp, flexible capillaries are difficult to see against the light. There is a risk of being stabbed or cut when you do work with the capillaries. Wear safety glasses to prevent eye injuries!

NOTE

It is the customer's responsibility to maintain the system properly by performing the system maintenance procedures on a regular basis.

Service by the customer must be performed only by trained qualified personnel and is restricted to servicing mechanical parts. Service on electronic parts must be performed only by Thermo Fisher Scientific field service engineers.

Do not try to repair or replace any component of the system that is not described in this manual without the assistance of your Thermo Fisher Scientific field service engineer.

Thermo Fisher Scientific assumes no responsibility and will not be liable for instrument damage and/or operator injury that might result from any servicing other than that contained in this manual or related manuals.

10.2 Guidelines for maintenance

Preventive maintenance must start with installation, and it must continue during the warranty period to maintain the warranty. Thermo Fisher Scientific offers maintenance and service contracts. Contact your local Thermo Fisher Scientific representative for more information.

Accurate results can be obtained only if the system is in good condition and properly calibrated.

To prepare the work area, do the following:

- Make sure that the surrounding area is neat and clean. It should be free from clutter and unrelated objects.
- Prepare a clean work surface by covering the area with lint-free paper or a large sheet of clean aluminum foil.
- Have nearby the necessary tools, supplies, and replacement parts (when applicable).

For optimal results, follow these guidelines when you perform the procedures in this chapter:

- Always wear a new pair of lint- and powder-free gloves when handling internal components. Do not reuse the gloves after you remove them because the surface contaminants on them recontaminate clean parts. See Personal protective equipment for a specification for the gloves.
- Always place the components on a clean, lint-free work surface.
- Dirty tools can contaminate your system. Keep the tools clean and use them exclusively for maintenance and service work at the mass spectrometer.
- Do not insert a test probe (for example, an oscilloscope probe) into the sockets of female cable connectors on PCBs.
- Do not overtighten a screw or use excessive force.

NOTICE

Make sure that you do not introduce any scratches or surface abrasions while you handle the internal components. Even small scratches can affect performance if they are close to the ion transmission path. Do not use tools, such as metal pliers, that might scratch these components.

10.3 Inspection- and servicing plan

Routine and infrequent maintenance procedures to be performed by the user are listed in the table. For a list of maintenance procedures that must be performed by Thermo Fisher Scientific personnel, see Services to be performed by Thermo Fisher Scientific service only.

Table User maintenance procedures

MS Component	Procedure	Frequency
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MS Component	Procedure	Frequency
Instrument	System bake-out	After the system was vented (even partially by a short power outage).
	Check warning labels on instrument	Annually
	Leak check gas lines	Annually
	Check condition of tubings and hoses	Annually
	Check the condition of cables	Annually
Cooling fans	Check fan filters	Every four weeks.
	Clean fan filters	If necessary.
Ion optics	Clean flatapole focus lens	If necessary.
	Clean injection filter	If necessary.
	For instructions about performing maintenance on the quadrupole and the bent flatapole, refer to the <i>Orbitrap Exploris Performance Maintenance Manual</i> .	
IC source	Replace	If necessary.
API source	Flush (clean) sample transfer line, metal needle, and API probe	Daily
	Clean the sweep cone and the spray cone	Daily
	Remove and clean ion transfer tube	Daily, or if ion transfer tube bore is contaminated (chemical noise) or obstructed (pressure decrease).
	Replace ion transfer tube	If ion transfer tube bore is corroded.
	Clean ion funnel	Check bi-weekly. Clean as needed. ¹⁾
HESI probe	Replace needle insert	If metal needle is blocked.
APCI, ESI, or HESI probe	Replace metal needle	If metal needle is broken or obstructed.
Calibration ESI probe	Replace low-flow needle assembly	If metal needle is blocked.

10.3.1 Clean the surface of the instrument

Clean the outside of the instrument with a dry cloth. To remove stains or fingerprints on the surface of the instrument (panels, for example), slightly dampen the cloth (preferably made of microfiber) with distilled water.

NOTICE

Prevent any liquids from entering the inside of the instrument. Leaking liquids might get into contact with electronic components and cause a short circuit.

10.3.2 Check the warning labels

Safety warnings on the mass spectrometer and other devices of the LC/MS system must always be complete, clearly visible and legible.

In addition to the safety instructions that can be found throughout this manual, warning labels on the instrument inform the user about possible hazards (for example those caused by hot surfaces or high voltage). To protect all personnel coming near the instrument, annually make sure that all warning labels on the instrument are still present. If warning labels are missing or unreadable, contact a Thermo Fisher Scientific field service engineer for replacements. Also check the warning labels on the forepump and other devices of the LC/MS system.

10.3.3 Check the gas lines for leaks

Regularly leak check each gas line from the gas supply in the laboratory to the instrument. The instrument must be powered for the test.

Procedure

1. After you have closed all valves in the instrument, monitor the manometer of the gas regulator for some minutes.
2. If the pressure falls significantly (for example, the nitrogen pressure falls by more than 690 mbar [10 psi] in two minutes), then you should search for leaks in the gas line.
3. Search for leaks in the gas line (for example, with a conventional thermal conductivity-based leak detector, such as is widely used to check leaks in gas chromatography equipment).
4. If you detect a leak (which is usually at a connection), then verify the tightness of the connection. In case of doubt, replace it. Note that over-tightening a connection might also

cause a leak or worsen it.

5. When you cannot find a leak in the gas line, Thermo Fisher Scientific recommends that you call a Thermo Fisher Scientific field service engineer to check for gas leaks in the instrument.

10.3.4 Check the condition of tubings and hoses

Regularly inspect all tubings and hoses that are connected to the mass spectrometer for damage.

Procedure

1. Make sure that the gas lines are securely connected, free from kinks, and not trapped.
2. Make sure that the liquid can run freely down the drain tube, which must not contain kinks or liquid traps.
3. Visually inspect all tubing for leaks or signs of deterioration.
4. Visually inspect the surfaces below all liquid lines for signs of liquid.

Result

If you detect a leak or other damage to any tubing or hose, then you must replace it.

10.3.5 Check the condition of the cables and plugs

Regularly inspect all cables or cords that are connected to the instrument for damage. In particular, check the current-carrying cables for signs of scorching.

10.4 Tools and supplies

Very few tools are necessary to perform routine maintenance for the Orbitrap Astral mass spectrometer. You can remove and disassemble many of the components by hand. Tools and supplies lists the necessary chemicals, tools, and equipment for maintaining the instrument.



Avoid exposure to potentially harmful materials.

By law, producers and suppliers of chemical compounds must provide their customers with the most current health and safety information in the form of Safety Data Sheet (SDSs). The SDSs must be freely available to lab personnel to examine at any time. These data sheets describe the chemicals and summarize information on the hazard and toxicity of specific

chemical compounds. They also provide information on the proper handling of compounds, first aid for accidental exposure, and procedures to remedy spills or leaks. Read the SDS for each chemical you use. Store and handle all chemicals in accordance with standard safety procedures. Always wear protective gloves and safety glasses when you use solvents or corrosives. Also, contain waste streams, use proper ventilation, and dispose of all laboratory reagents according to the directions in the SDS.

NOTICE

To prevent corrosion, do not use nitric acid to clean metal parts.

Table Chemicals, tools, and equipment

Description	Comments
<i>Chemicals</i>	
Isopropanol, methanol, or ethanol	LCMS grade
Water	LCMS grade or ultrapure deionized water, for example Milli-Q™
Liquinox™	Mildly alkaline detergent
Citranox™	Strongly acidic detergent
MucasoI™ or Decon 90™	Strongly alkaline detergent
Nitrogen gas or compressed air	Clean and oil-free
The preferred chemical is indicated in bold letters. The other listed chemical(s) may be used alternatively.	
<i>Tools</i>	
Ion transfer tube removal tool	Provided in the Source Installation Kit
Screwdriver, slot head	
Screwdriver, Phillips #2 (M3)	
<i>Equipment</i>	
Forceps or tweezers	
Magnifying device	
Ultrasonic bath	Ideally heatable
Gloves	Lint-free and powder-free

Description	Comments
Chamois-tipped swabs	
Lint-free foam tip swabs	
Lint-free wipes	Cleanroom wipes
For example, 1 µm grain size diamond paste	For example, 1 µm grain size diamond paste
Polishing linen	For example, MICRO-MESH™ 6000 and 12000 universal mesh
Blow gun	Supplied with the Cleaning Kit

10.5 Maintenance of the vacuum system

This section describes user maintenance procedures for the vacuum pumps and the vacuum manifold of the Orbitrap Astral mass spectrometer.

This section only outlines the maintenance procedures for the turbomolecular pump (TMP) of the mass spectrometer. The manuals of the pump manufacturers give detailed advice regarding safety, operation, maintenance, and installation. Note the warnings and precautions contained in these manuals!

10.5.1 Maintenance of the forepump

We recommend a monthly visual inspection of the dry pump system. Check that the electrical supply cord, hoses and pipelines connected to the pump are in good condition and tightened.

The forepump does not require user maintenance. The frequency of the maintenance can be adjusted based on pump use. It is recommended that the pump should be serviced every 5 years. Maintenance must be done by the service centers of the pump manufacturer.



CAUTION

Hot surface

During operation, the forepump is hot and some surfaces reach high temperatures. Even during operational pauses, the pump might be hot. Switch off the pump and let it cool to room temperature before performing any intervention. Wear heat protective gloves when working on the pump that is “still warm from operation.”

NOTE

To prevent an unwanted operation of the forepump, connect the MS and the forepump to the power supply in the correct sequence:

- When you install a system, first connect the switch cable between the instrument and the forepump, then connect the power supply cords.
 - When you deinstall or service a system, first disconnect the power supply cords, then disconnect the switch cable between the instrument and the forepump.
-

10.5.1.1 Move the forepump



CAUTION

Heavy object

Because of its weight, the forepump might move uncontrollably and cause injuries. Wear steel-reinforced safety shoes when you move the pump during the installation.

The forepump is heavy (~80 kg). If lifting of the pump becomes necessary, only a suitable lifting device should be used to lift the pump at the three lifting eyes.

NOTE

A lifting device is the only officially recommended way of handling the pump.

Should people lift the pump, at least two persons should lift it. Pay attention to the position of each person when lifting! Straight back, flexed knees, and adapted large back belts, etc. are recommended. Be careful with your hands when you set down the pump to avoid squashing them!

NOTE

Do not lift or pull the pump at the cables!

10.5.2 Maintenance of the turbomolecular pumps

The pump manufacturer recommends that you change the operating fluid reservoir every four years at the latest. Depending on the operating conditions, changing in two-year intervals might be necessary. Changing of the TMP bearing is recommended every four years, at least.

To perform maintenance for the TMPs, it is necessary to remove the instrument housing and to partially disassemble the instrument. Therefore, a Thermo Fisher Scientific field service engineer must

be called if TMP maintenance is necessary. See also Services to be performed by Thermo Fisher Scientific service only.

10.5.3 Bake out the system

The system bake-out of the mass spectrometer removes unwanted gases or molecules (collected or remaining) from the high-vacuum region of the instrument. A bake-out is mandatory after the system has been vented for maintenance or service work in the analyzer region. You should bake out an instrument that has been vented for at least ten hours before you can start using it again.

If the system has been vented during a power failure, then it is necessary to bake out the system to attain the operating vacuum. Before you start the bake-out, make sure that the pumps are up and running at their operating speed. If you have just switched on the mass spectrometer, this will take about ten minutes.

Always check the fore vacuum pressure after switching on the instrument. It should be close to the value shown in Bake out the system. If the vacuum requirements are not met, the system might self-shutdown again or not start the bake-out procedure to protect itself from damage.

NOTE

The instrument will start an automatic bake-out for 10 hours heating plus 3 hours cooling when the turbomolecular pumps are at nominal speed. This is typically 10–15 minutes after power up of the instrument. The data acquisition PC needs to be up and running and connected to the instrument.

Perform a manual system bake-out

Procedure

1. When the vacuum pressures become normal, place the MS in Off mode in the Tune window.
2. Select the Diagnostics icon (), and then select System > Vacuum > Bake Vacuum Chamber.
3. Select Start. The MS indicates an active bake-out procedure by a white flashing Status LED and a white progress bar of the Scan LED.
4. After the bake-out is completed, do the following:
 - a. In the Tune window, place the MS in Standby condition.
 - b. Open the Status pane, click the downward arrow, and then choose By Board.
5. Check the readback values for the source and Orbitrap (UHV) pressure gauges as follows:

- a. Click Source, and then verify that the Forevacuum Pressure readback value is below the operating threshold limit (see Bake out the system).
- b. Click FT Vacuum, and then verify that the UHV Pressure readback value is below the operating threshold limit.

Result

In the Tune window, normal readback measurements show a green square (■).

Table Typical pressure readings after bake-out

Name in Tune application	Typical values
Forevacuum Pressure	3.5 mbar
IRM Pressure	1.1E-2 mbar
UHV Pressure	<E-10 mbar
Ion Processor	1.0E-2mbar
Astral UHV Pressure	<1E-8 mbar



CAUTION

Hot parts

If you abort a system bake-out, parts of the instrument can be hot. Touching hot parts of the instrument immediately after a bake-out might cause burns. Let the instrument cool for at least three hours before you start to operate it again.

TIP

When the instrument cools after the bake-out, warm air comes out of the venting slots of the instrument. This is a normal process.

10.6 Maintenance of the fan filters

Each of the ventilation slots at the front side, the left side, and the right side of the mass spectrometer is equipped with a fan filter.

NOTE

Do not block the ventilation slots of the mass spectrometer. Items might fall behind the instrument, inhibit airflow, and cause the system to overheat.

Check the fan filters

Check the fan filters every four weeks and clean them if they are dirty. Replacements for the fan filters are available from Thermo Fisher Scientific.

Procedure

1. Each fan filter bracket is plugged into the instrument frame. Beginning at the front side of the instrument, insert a finger into the recess in the bracket and pull at the fan filter bracket to open it.
2. Pull each fan filter out of the filter bracket and check it for dust. If the fan filters are covered with dust, continue with step 3. If the fan filters are clean, proceed to step 6.
3. Wash the fan filters in a solution of soap and water.
4. Rinse the fan filters with tap water.
5. Squeeze the water from the fan filters and let them air dry.
6. Reinstall the fan filters in the fan filter brackets.
7. Reinstall the filter brackets, finishing with the filter bracket at the instrument's front side.

10.7 Maintenance of the API source

Various routine maintenance procedures must be performed to ensure the optimum performance of the Atmospheric Pressure Ionization (API) source. Most of the procedures involve cleaning. This section also describes procedures for replacing the ion transfer tube. Refer also to the *OptaMax NG Ion Source User Guide* for additional information.

10.7.1 Prepare the API source maintenance

Before you start to clean the API source, read the information in the following topics carefully.

10.7.1.1 Cleaning environment

Before you open the system to perform cleaning, prepare a clean, sufficiently large surface that is covered by a clean metal tablet or a sheet of sturdy aluminum foil, free of dust and particles. Always wear clean gloves when you handle parts that are removed from the vacuum manifold. All tools that are used in the vacuum manifold should be clean, and free from oil, particles, and dust. All devices,

holders, lenses, etc. that are removed from the vacuum manifold should be placed on the clean surface. Keep the vacuum manifold closed when it is not actively being accessed. Likewise, cover removed parts to prevent dust accumulation when they are not actively being accessed.

10.7.1.2 Using abrasives

NOTE

Use abrasives only when it is absolutely necessary or as a last resort to remove problematic stains. When you use abrasives (polishing paste or MICRO-MESH™), either thoroughly wipe off the generated particles with lint-free wipes (first wet, then dry) or rinse them off with water. Most lenses are easy to wipe, while other ion optics parts should be rinsed. Always add some solvent to prevent dry particles from spreading on the cleaned surface.

10.7.1.3 Cleaning solutions

Prepare these cleaning solutions:

- 2% alkaline detergent in hot tap water (approximately 50 °C)
- 50:50 isopropanol/water
- 100% isopropanol

TIP

Isopropanol may be replaced by methanol or ethanol.

Ideally, three beakers are prepared with each solvent, although the flushing steps in 50:50 isopropanol/water and isopropanol may be done consecutively in the same beaker.

The ultrasonic bath may be heated to 50 °C. Make sure to operate in a well-ventilated area and to cover the beakers with some aluminum foil or a watch glass when using isopropanol.

10.7.1.4 Purge the optical parts

As the final step in each cleaning procedure, purge all ion optics parts with clean and oil-free nitrogen immediately after the last wipe with 100% isopropanol. Do not let solvents air-dry on the ion optics, especially water-containing ones. The part must be absolutely dry after the purging.

10.7.2 Frequency of cleaning

The frequency of cleaning the components of the API source depends on the types and amounts of samples and solvents that are introduced into the instrument. In general, for a given sample and ionization technique, the closer a mass spectrometer component is to the source of the sample, the more rapidly it becomes dirty.

- The sample transfer line, the metal needle, and the API probe should be cleaned at the end of each operating day to remove any residual salts from buffered mobile phases or other contamination that might have accumulated during normal operation.
- The ion transfer tube and the sweep cone of the API source must be removed and cleaned periodically.
- The ion funnel becomes dirty at a slower rate than the API probe, the sweep cone, and the ion transfer tube.

10.7.3 Remove and reinstall the API source housing

You must remove the API source housing before you perform maintenance on the ion optics and the ion source interface.

This section contains the following topics:

- Remove the API source housing
- Install the ion source housing

10.7.3.1 Remove the API source housing

You must remove the API source housing to get access to the sweep cone.



CAUTION

Hot parts

During operation of the mass spectrometer, the ion transfer tube and the sweep cone can reach temperatures up to 450 °C. This might cause severe burns upon touching or ignite combustible material. The external surface of the ion source housing can reach temperatures up to 100 °C. This is hot enough to cause skin burns.

Do not touch the ion source housing when the mass spectrometer is in operation. Let the ion source cool down as described in this manual before you remove the ion source housing. Do not touch the ion source mount immediately after you have removed the ion source

housing. Keep combustible materials away from the ion source mount. Do not let the instrument stay unattended when the housing is not mounted to the source.

NOTICE

Dust can be sucked in through the ion inlet system. To reduce this risk to a minimum, keep an ion source housing installed on the mass spectrometer also during off times and standby times.

Procedure

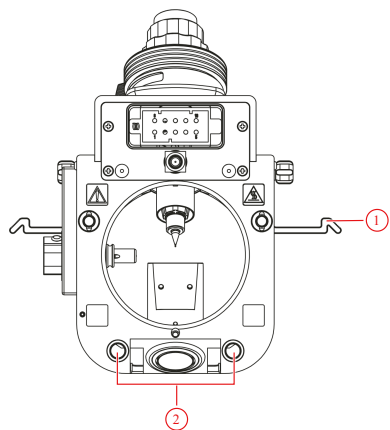
1. If the mass spectrometer was recently in operation, let the ion source housing cool down before you touch its external metal surface.
2. Set the tube temperature to 0 °C. Wait until the ion source has cooled down (approximately 30 minutes) before you perform work on it.
3. If a probe is connected to the source housing, disconnect the external liquid lines before you remove the source housing from the mass spectrometer.
4. Rotate the ion source housing locking levers 90 degrees downwards to release the ion source housing from the ion source mount assembly.
5. Remove the ion source housing by pulling it straight off the ion source mount assembly. Place the housing in a safe location for temporary storage.

10.7.3.2 Install the ion source housing

Procedure

1. Carefully align the two guide pin holes on the back side of the ion source housing with the ion source housing guide pins on the mass spectrometer. Carefully press the ion source housing onto the ion source mount.

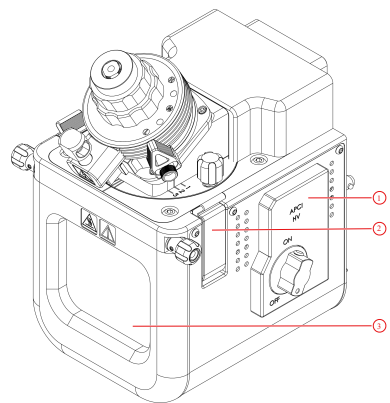
Figure OptaMax NG ion source with unlocked levers (back view)



No.	Description	No.	Description
1	Locking lever (unlocked)	2	Guide holes on the back of the housing

2. Rotate the ion source housing locking levers 90 degrees upwards to lock the ion source housing onto the ion source mount assembly.

Figure OptaMax NG API source (front view)



No.	Description	No.	Description
1	APCI components (built-in)	3	Window
2	Locking lever (locked position)		

Result

The ion source is now properly installed on the mass spectrometer.

TIP

Prevent solvent waste from backing up into the ion source and the mass spectrometer. Always make sure that liquid in the drain tube is able to drain to a waste container.

By default, the source gases will not drop below a minimum flow of 7 L/min to compensate for the suction through the transfer tube. This ensures a slightly positive pressure in the ion source to prevent solvent waste from being sucked back from the drain.

10.7.4 Maintenance of the API source housing

User maintenance is limited to cleaning the API source housing as necessary. Follow all safety precautions in the sections regarding the installation and removal of the housing. For any additional service that might be necessary, contact your local Thermo Fisher Scientific field service engineer.



CAUTION

Hazardous substance

Methanol is highly flammable and noxious. Keep away from heat, sparks, and flame. Avoid accidental exposure. Use with adequate ventilation. When possible, dilute it with 50% water. Refer to your supplier's Safety Data Sheet (SDS) for information about the correct handling of methanol.

Clean the API source housing

Procedure

1. Put on appropriate eye-wear and gloves.
2. After the API source has cooled down to room temperature, remove it from the mass spectrometer.
3. Below an appropriate fume hood, rinse the interior of the housing with UHPLC/MS-grade methanol.
4. Allow the housing to dry before you install it on the mass spectrometer.

10.7.5 Remove, clean, and reinstall the ion transfer tube

The bore of the ion transfer tube can become blocked by buffer salts or highly concentrated samples. The ion transfer tube can be removed for cleaning. You do not have to vent the system to remove the ion transfer tube.

If the pressure in the source region (as measured by the Pirani gauge) drops considerably below 2.5 mbar, you should suspect a blocked ion transfer tube. You can check the Pirani gauge pressure in

the Tune application.



WARNING

High voltage

High voltages that can cause an electric shock are used in the instrument. Switch off the instrument before you proceed.



CAUTION

Hot parts

During operation of the mass spectrometer, the ion transfer tube and the sweep cone might reach temperatures up to 450 °C. This might cause severe burns upon touching or ignite combustible material. The external surface of the ion source housing can reach temperatures up to 100 °C. This is hot enough to cause skin burns.

Do not touch the ion source housing when the mass spectrometer is in operation. Let the ion source cool down as described in this manual before you remove the ion source housing.

Do not touch the ion source mount immediately after you have removed the ion source housing. Keep combustible materials away from the ion source mount. Do not let the instrument stay unattended when the housing is not mounted to the source.

10.7.5.1 Maintenance of the ion transfer tube and the sweep cone

The Orbitrap Astral mass spectrometer has a high capacity transfer tube with a rectangular orifice, where the larger aperture allows for a higher ion flux into the interface.

Figure Ion transfer tube



See Parts for ion source for a list of the parts that are necessary when you perform maintenance of the ion transfer tube and the sweep cone. They are contained in the Installation Kit.

Remove the ion transfer tube and the sweep cone

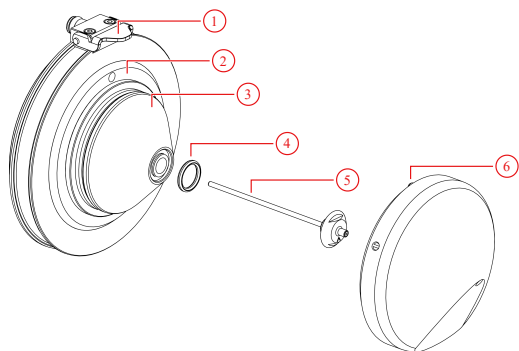


To prevent mechanical damage to the transfer tube, make sure that the tube temperature is well below 200 °C. Let the spray cone cool down for at least 60 minutes to allow safe handling by laboratory personnel. See the safety information in Remove, clean, and reinstall the ion transfer tube.

Procedure

1. Switch off the flow of liquid from the LC (or an other sample introduction device) to the API source.
2. In the Tune window, place the system in Off mode. Wait for at least 60 minutes to let hot components cool down.
3. Remove the API source housing from the front of the mass spectrometer.
4. Remove the sweep cone by grasping its outer ridges and pulling the cone straight off the API cone seal.

Figure Spray cone, O-ring, ion transfer tube, and sweep cone of the ion source interface



No.	Description	No.	Description
1	Release lever for the ion source interface	4	Vespel™ O-ring
2	API cone seal	5	Ion transfer tube
3	Spray cone	6	Gas inlet on the sweep cone

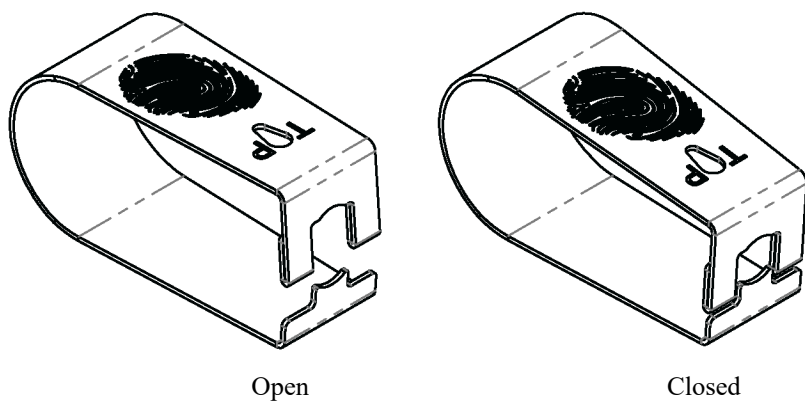
NOTE

Do not accidentally lift the release lever at the top of the ion source interface, which will vent the MS.

To avoid contaminating the ion transfer tube, do not touch its exposed entrance.

5. Grip the exposed tip of the ion transfer tube with the custom removal tool.

Figure Removal tool for the ion transfer tube



6. Rotate the ion transfer tube only 1/4-turn counterclockwise. Then, use the tool to remove the tube from the ion source interface.

NOTE

A protective mechanism prevents the air from entering the vacuum manifold. It requires that you drag the ion transfer tube with a quick and steady motion out of the spray cone.

Clean the ion transfer tube

Procedure

1. If the ion transfer tube is extremely contaminated, then start the cleaning with step 1a. If the ion transfer tube is normally contaminated, then start with step 2.
 - a. Overnight, sonicate the component in a 10% solution of Liquinox in tap water.
 - b. Rinse the component with tap water. Then, for 2 minutes force a strong stream of water through the orifice.
 - c. For 30 minutes, sonicate the component in tap water.
2. For 30 minutes, sonicate the component in a fresh 2% solution of CITRANOX™ in warm tap water.

Alternatively, for 30 minutes, sonicate the component in a 50:50 solution of methanol/water that contains 20% formic acid.



CAUTION

Hazardous substance

Formic acid is highly corrosive also to the skin, flammable, and noxious. Wear protective gloves, protective clothing, and eye protection. Keep away from heat, sparks, and flame. Avoid accidental exposure. Use with adequate ventilation. When possible, dilute it with water. Refer to your supplier's Safety Data Sheet (SDS) for information about the correct handling of formic acid.



CAUTION

Hazardous substance

Methanol is highly flammable and noxious. Keep away from heat, sparks, and flame. Avoid accidental exposure. Use with adequate ventilation. When possible, dilute it with 50% water. Refer to your supplier's Safety Data Sheet (SDS) for information about the correct handling of methanol.

3. Rinse the component thoroughly with tap water.

4. For 15 minutes, sonicate the component in deionized water.
5. In a beaker, rinse the component with isopropanol.
6. For 15 minutes, sonicate the component in isopropanol.
7. Dry the ion transfer tube with a strong stream of oil-free nitrogen gas to make sure that the solvent evaporates completely.

Replace the ion transfer tube if the bore becomes corroded or blocked.

Clean the sweep cone

Procedure

1. Clean the inside and the outside of the sweep cone with lint-free wipes that have been soaked in a 50:50 isopropanol/water solution.
2. Clean the inside and the outside of the sweep cone with lint-free wipes that have been soaked in 100% isopropanol.
3. If the cone is still stained after the previous steps, use MICRO-MESH™ 6000 to remove all stains and finish up with MICRO-MESH 12000.
4. Repeat steps 1 and 2.
5. Sonicate the cone in 100% isopropanol for five minutes.
6. Dry the cone with a strong stream of oil-free nitrogen gas to make sure that the solvent evaporates completely.

Clean the spray cone

NOTE

Do not let any solvents get into the opening for the ion transfer tube because it might prevent the proper operation of the ball valve in the cage.

Procedure

1. Clean the outside of the spray cone with lint-free wipes that have been soaked in a 50:50 isopropanol/water solution.
2. Clean the outside of the spray cone with lint-free wipes that have been soaked with 100% isopropanol.
3. If the cone is still stained after the previous steps, use MICRO-MESH™ 6000 to remove all stains and finish up with MICRO-MESH 12000.
4. Repeat steps 1 and 2.

5. Dry the cone with a strong stream of oil-free nitrogen gas to make sure that the solvent evaporates completely.

Clean the O-ring in the spray cone

Procedure

1. Remove and inspect the O-ring that is seated in the spray cone under the entrance end of the ion transfer tube. Clean it with methanol or replace it if necessary.
2. Reinsert the O-ring into the spray cone.

Reinstall the ion transfer tube and the sweep cone

Procedure

1. Put on a new pair of lint- and powder-free gloves.
2. Insert the ion transfer tube into the heater block. Rotate the tube while you insert it.

NOTE

When you reinstall the ion transfer tube into the heater block, take these precautions:

- Make sure that everything is properly aligned to prevent stripping the threads on the ion transfer tube.
- Do not bend the ion transfer tube.
- Insert the ion transfer tube with a quick and steady motion into the spray cone to prevent air from entering the vacuum manifold.

-
3. Align the gas inlet on the sweep cone with the sweep gas supply port on the ion source mount. Firmly press the sweep cone into the ion source mount.
 4. Reinstall the API source housing on the mass spectrometer.
 5. In the Tune window, place the system in Standby condition.

Result

If you have successfully cleared the ion transfer tube, then the Pirani gauge pressure should increase to a normal value (approximately 3.5 mbar). If you cannot clear the ion transfer tube by this method, then replace it.

10.7.6 Maintenance of the API probe

For the HESI probe, only minimum maintenance is necessary. If the metal needle is blocked, replace it. To replace the metal needle, you must partially disassemble the probe. Refer to the *OptaMax NG Ion Source User Guide* for instructions.



CAUTION

Hot parts

At typical operating temperatures between 350 and 450 °C, touching the vaporizer might cause severe burns. Let the HESI probe cool to room temperature (approximately 60 minutes) before you touch or remove it.

NOTICE

For best results, do not operate the HESI probe at elevated temperatures without solvent flow. Letting the HESI probe to run dry at elevated temperatures can cause blockage of the replaceable metal needle.

TIP

Wear a new pair of lint- and powder-free gloves when you handle HESI probe components.

10.7.6.1 Flush the sample transfer line and the metal needle

For best results, flush the sample transfer line, metal needle, and HESI probe for 15 minutes at the end of each working day (or more often if you suspect they are contaminated). Use a 50:50 methanol/distilled water solution from the LC system through the API source.



CAUTION

Hazardous substance

Methanol is highly flammable and noxious. Keep away from heat, sparks, and flame. Avoid accidental exposure. Use with adequate ventilation. When possible, dilute it with 50% water. Refer to your supplier's Safety Data Sheet (SDS) for information about the correct handling of methanol.

After 15 minutes, switch off the flow of liquid from the LC to the API source, but keep the API source on (including the sheath gas and auxiliary gas) for an additional 5 minutes.

10.7.7 Maintenance of the ion source interface

The ion source interface assembly includes the sweep cone, the ion transfer tube, the IC unit, and the ion funnel. The ion transfer tube has a finite lifetime. You must replace it if its bore becomes corroded.

It is good practice to flush the sweep cone and the bore of the ion transfer tube at the end of each working day with a 50:50 methanol/water solution.

For cleaning or replacement, you can remove the sweep cone and the ion transfer tube without venting the system.

10.7.7.1 Sequence of steps

To maintain ion source interface assembly components other than the sweep cone and the ion transfer tube, proceed in this sequence:

1. Shut down and vent the system.
2. Remove the API source housing.
3. Remove the API source interface assembly.
4. Clean the injection filter and the flatapole focus lens.
5. Clean the IC unit and the ion funnel.
6. Reinstall the API source interface assembly.
7. Reinstall the API source housing.
8. Start up the system.

The above steps are described in the topics that follow.

10.7.7.2 Remove the ion source interface

You must remove the ion source interface to access the IC unit and the ion funnel.



CAUTION

Hot parts

The ion source interface can become hot enough to cause severe burns when the mass spectrometer is in operation. Do not touch the ion source interface immediately after you have removed the ion source housing. Let the ion source interface cool until the temperature readback is <60 °C before you remove it from the mass spectrometer.

Procedure

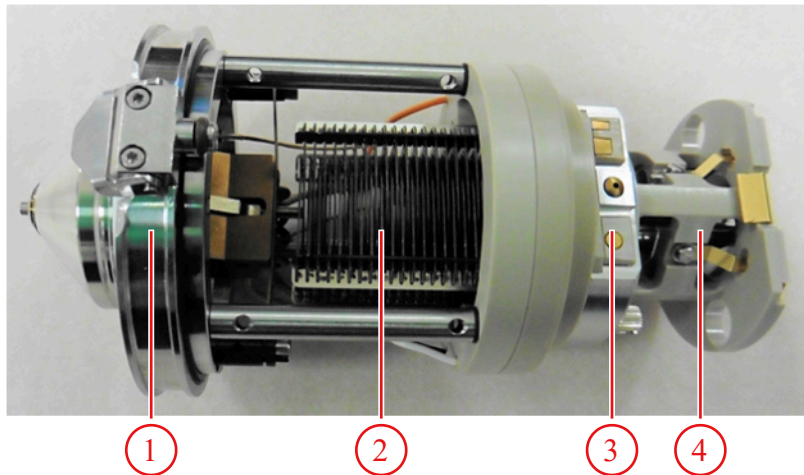
- 1. Shut down and vent the system.
- 2. Remove the ion source housing.
- 3. Put on a new pair of lint- and powder-free gloves.
- 4. Lift up the release latch until it pushes the ion source interface out by a few millimeters.
- 5. Grasp the ion source interface with your fingers, and then carefully pull it out of the vacuum manifold.



Figure Removing the ion source interface from the vacuum manifold

- 6. Place the assembly on a clean, lint-free surface.

Figure Ion source interface



No.	Description	No.	Description
1	Source cage	3	Internal calibration unit

-
7. Block the opening to the analyzer chamber to prevent dust contamination, for example with the PEEK™ accessory that comes with the instrument upon installation or a piece of aluminum foil.

10.7.7.3 Clean the injection filter and the flatapole focus lens

Remove the injection filter

Procedure

1. Put on a new pair of lint- and powder-free gloves.
2. Insert a flat head screwdriver through the holes in the PEEK™ holder of the injection filter. Loosen the screws that attach the injection filter to the ion source interface cage. In this stage, the screws should turn freely and be movable by a few millimeters.

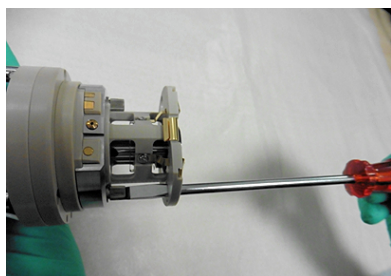


Figure Removing the injection filter

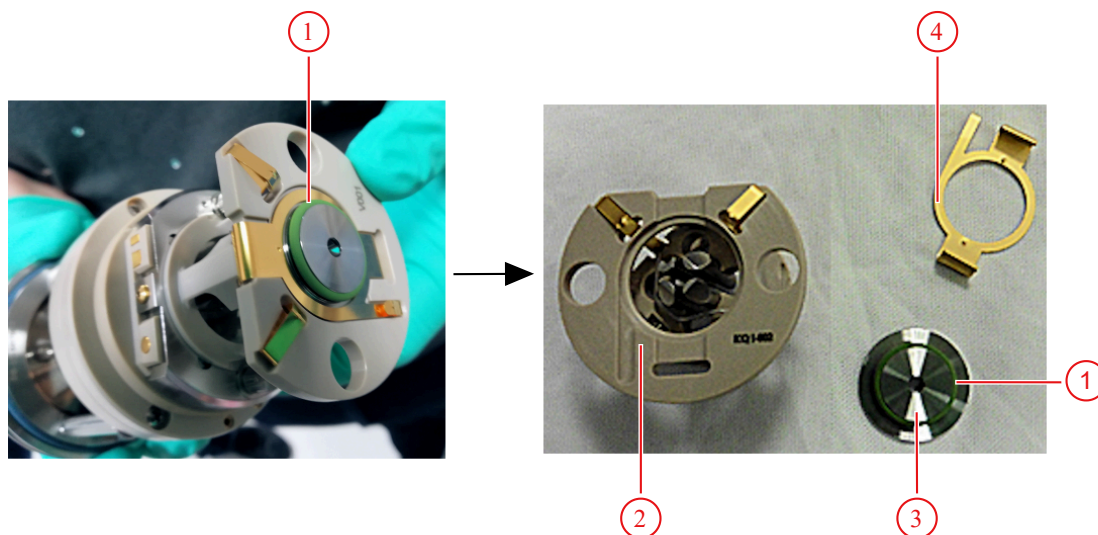
3. Remove the injection filter.
4. Put the components on a clean, lint-free surface.

Remove the flatapole focus lens

Procedure

1. Pull up on the top tab of the metal retainer for the flatapole focus lens until it is freed from the PEEK holder.
2. Unhook the bottom side of the retainer and remove the lens. Clean the body of the injection filter.
3. Remove the O-ring.

Figure Disassembling the injection filter



No.	Description	No.	Description
1	O-ring	3	Flatapole focus lens
2	Injection filter	4	Retainer

Clean the injection filter

Procedure

1. Sonicate the injection filter in a 2% solution of an alkaline detergent in water for 5 minutes.
2. Remove the injection filter from the beaker.
3. Clean the rod surfaces facing the ion beam with a lint-free swab that has been soaked in a 2% solution of an alkaline detergent in water. Wipe each rod 20 times (apply gentle pressure to the swab handle). Turn the injection filter around and repeat from the other side.
4. Sonicate the injection filter in a 2% solution of an alkaline detergent in water for 10 minutes.
5. Rinse the component thoroughly with warm tap water.
6. Sonicate the component in a 50:50 isopropanol/water solution for five minutes.
7. Sonicate the component in 100% isopropanol for five minutes.
8. Dry the component with a strong stream of oil-free nitrogen gas to make sure that the solvent evaporates completely.

9. With a magnifying device, inspect the component for any lint, particulates, and sample buildup or coatings. Use plastic tweezers or a similar tool to remove lint or particulates.

Clean the flatapole focus lens

Procedure

1. Clean the inside of the orifice with a conical swab that has been soaked in a 50:50 isopropanol/water solution.
2. Clean the lens orifice from both sides with a lint-free wipe that has been soaked in a 50:50 isopropanol/water solution.
3. Clean the lens orifice from both sides with a lint-free wipe that has been soaked in 100% isopropanol.
4. Dry the component with a strong stream of oil-free nitrogen gas to make sure that the solvent evaporates completely.

Reinstall the flatapole focus lens

Procedure

1. Reinstall the O-ring onto the lens.
2. Place the flatapole focus lens onto the injection filter assembly. Then hook the lower part of the holder clip under the holder. Slide it with both thumbs from the bottom to the top, so that it clips onto the top.

Reinstall the injection filter

With a flat head screwdriver, attach the injection filter back to the ion source interface cage with the screws. Use the pin for alignment.

NOTICE

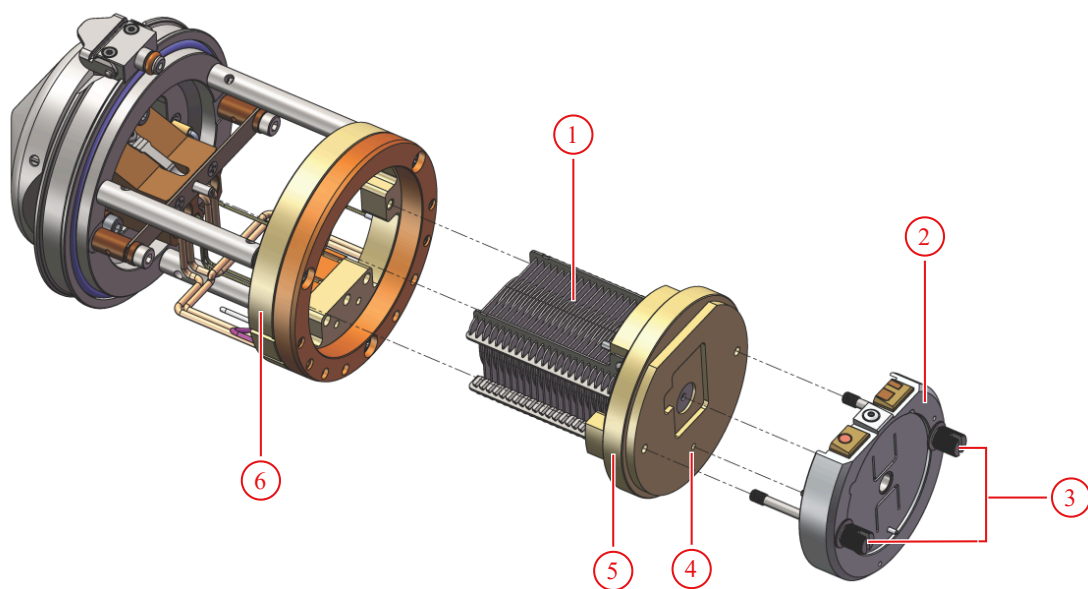
Sensitive part

Do not use excessive force when you tighten the screws. The PEEK holder of the filter is not designed to withstand mechanical stress and might break.

10.7.7.4 Clean the ion funnel

Remove the ion funnel and the IC unit from the ion source interface cage before you clean them.

Figure Exploded view of the ion source interface cage



No.	Description	No.	Description
1	Ion funnel	4	Lead pin socket
2	IC unit	5	Ion funnel holder
3	Thumbscrews	6	Ion source interface cage

Remove the ion funnel from the ion source interface cage

Procedure

1. Cover the area with lint-free paper or aluminum foil to prepare a clean work surface.
2. Put on a new pair of lint- and powder-free gloves.
3. Loosen and extend the two thumbscrews that secure the ion funnel to the ion source interface cage and the IC unit to the ion funnel.
4. Grasp the two thumbscrews and carefully pull the ion funnel straight out of the ion source interface cage. Place it on a clean, lint-free surface.
5. With the thumbscrews extended, continue turning them counterclockwise until they are only holding in the IC unit/reagent ion source assembly.
6. Pull on the two thumbscrews again to separate the IC unit /reagent ion source from the ion funnel.
7. Put the IC unit on a clean, lint-free surface.

Clean the ion funnel

NOTICE

Do not clean the lenses with abrasives, acidic or caustic substances, or detergents that are not stated in this manual.

Do not disassemble the ion funnel from its holder. Otherwise, you risk losing the correct alignment of the lens segments.

Procedure

1. Put on a new pair of lint- and powder-free gloves.
2. For 60 minutes, sonicate the ion funnel, with the PEEK holder facing down, in a 2% solution of detergent in tap water.

TIP

Which detergent (CITRANOX™ or Mucaso™) is the most effective depends on your MS application. Up to 10% detergent can be used in case of an extreme contamination.

3. Rinse the component thoroughly with warm, running tap water.
4. For 5 minutes, sonicate the component in deionized water.
5. If you used a high concentration of detergent in step 2, then repeat the steps 3 and 4 five times with fresh water.
6. For 10 minutes, sonicate the component in a 50:50 isopropanol/deionized water solution. If noticeable residues come off the component at this stage, then repeat the steps 2 to 6 with fresh solutions until this is not noticeable anymore.
7. For 10 minutes, sonicate the component in isopropanol.
8. Dry the component with a strong stream of oil-free nitrogen gas to ensure that the solvent evaporates completely. Start with purging all drill holes and narrow gaps.
9. With a magnifying device, inspect the component for any lint, particulates, and sample buildup or coatings.

Clean the IC unit

Follow the instructions in Clean the calibrant ion source.

Reinstall the ion funnel

Procedure

1. Put on a new pair of lint- and powder-free gloves.
2. Align the lead pin on the IC unit with the lead pin socket on the ion funnel (see Exploded view of the ion source interface cage), and then firmly press the unit until it snaps into place.
3. Tighten the two thumbscrews a few turns into the ion funnel.
4. Orient the ion funnel as shown in Exploded view of the ion source interface cage, and then carefully slide it into the ion source interface cage.
5. Tighten the two thumbscrews a few turns into the ion source interface cage.

Continue with Reinstall the ion source interface.

10.7.7.5 Reinstall the ion source interface

Prerequisites

- ✓ Before you install the ion source interface, inspect the O-ring for damage. Replace it if necessary.

Procedure

1. Orient the ion source interface with the release latch at the top (see figure).



Figure Inserting the ion source interface

2. Carefully insert the ion source interface into the vacuum manifold. Push until there is no or only a small gap left between the two.
3. Reinstall the ion transfer tube and lock it with a quarter turn.

4. Reinstall the ion source housing..
5. If necessary, reinstall the ion source probe and the LC liquid lines.
6. Start up the system.
7. Run the Ion Optics Connectivity Check from the Diagnostics pane in the Tune application.

Result

Now you can start using the instrument again.

10.8 Maintenance of the internal calibrant discharge source

Over time, the ion signal from the internal calibrant discharge source decreases as the internal components become dirty, need to be recalibrated, or both. The expected lifetime of the internal calibrant discharge source is six months to one year.

NOTICE

You can service (replace) the internal calibrant discharge source, but you cannot service the calibrant vial.

Replace the internal calibrant discharge source when the Orbitrap™ mass analyzer cannot lock to the internal calibrant mass and the S/N of the calibrant peak is less than 33 at a resolution setting of 240,000. (Assumes that the instrument has had a full calibration in positive mode, negative mode, and IC modes, and that the mass accuracy of the Orbitrap analyzer is within specification.)

10.8.1 Replace the internal calibrant discharge source

The internal calibrant discharge source is a consumable part. You may, however, clean the parts that are described in Clean the calibrant ion source. To replace the internal calibrant discharge source, follow these procedures:

Procedure

1. Shut down the mass spectrometer completely.
2. Remove the ion source interface.
3. Remove the internal calibrant discharge source.
4. Install the new internal calibrant discharge source.

5. Reinstall the ion source interface.

NOTE

Before you continue, read the Safety guidelines for maintenance and the Guidelines for maintenance. Also take the instructions in Maintenance of the ion source interface as guidelines.

TIP

Wear a new pair of lint- and powder-free gloves when you handle API source components.

10.8.1.1 Remove the internal calibrant discharge source

To remove the Internal Calibrant discharge source, follow these procedures:

Procedure

1. Remove the IC source heater interface.
2. Remove the Internal Calibrant discharge source.

Remove the IC source heater interface

Procedure

1. Remove the ion source interface.
2. Insert a flat head screwdriver through the holes in the PEEK holder of the injection filter and loosen the screws that attach the injection filter to the ion source interface cage. Remove the injection filter.

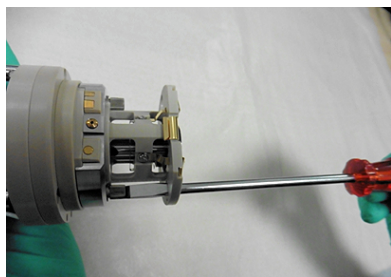
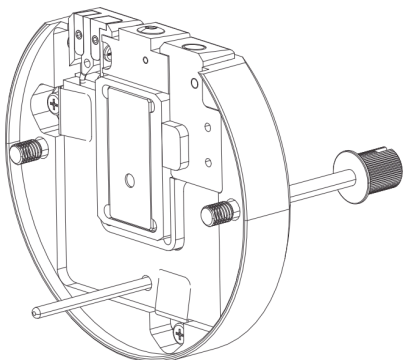


Figure Removing the injection filter

3. Continue to loosen the two thumbscrews and use them to carefully pull out the ion funnel assembly from the ion source interface cage.

4. Loosen the two thumbscrews even further and use them to pull out the IC source heater interface.
Hold the heater interface with the thumbscrews facing upward to prevent the calibrant ion source from falling out of the IC source heater interface.

Figure Front of the IC source heater interface

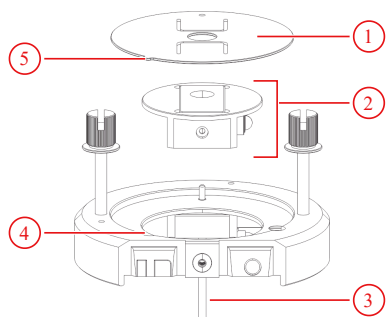


Remove the Internal Calibrant discharge source

Procedure

1. Hold the IC source heater interface with the thumbscrews facing up.
2. From the bottom, press upward with your fingers to remove the Internal Calibrant discharge source retainer and the Internal Calibrant discharge source.

Figure Calibrant ion source removed from the IC source heater interface



No.	Description	No.	Description
1	Internal Calibrant discharge source retainer	4	Orientation notch
2	Internal Calibrant discharge source	5	Orientation tab
3	Guide pin		

3. Dispose of the old Internal Calibrant discharge source according to the local regulations.

10.8.1.2 Install the new internal calibrant discharge source

To install the internal calibrant discharge source, follow these procedures:

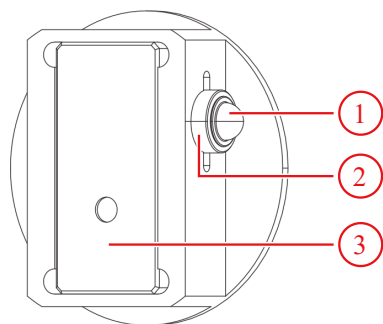
1. Install the new internal calibrant discharge source.
2. Reinstall the ion funnel and the IC source heater interface.
3. Reinstall the ion funnel assembly.

Install the new internal calibrant discharge source

Procedure

1. Make sure that the discharge cathode is fully inserted into the ceramic insulator.

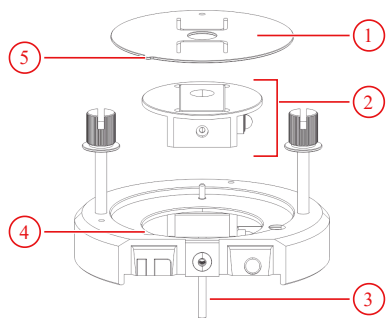
Figure Calibrant ion source (front view)



No.	Description	No.	Description
1	Discharge cathode	3	Anode lens/ion volume
2	Ceramic insulator		

2. Align the internal calibrant discharge source and the internal calibrant discharge source retainer as shown in the figure. Then reinstall them into the IC source heater interface.

Figure Calibrant ion source removed from the IC source heater



No.	Description	No.	Description
1	Internal Calibrant discharge source retainer	4	Orientation notch
2	Internal Calibrant discharge source	5	Orientation tab
3	Guide pin		

3. Make sure that the orientation tab of the internal calibrant discharge source retainer fits into the corresponding notch in the Internal Calibrant source heater interface.

Reinstall the ion funnel and the IC source heater interface

Procedure

1. Align the guide pin on the IC source heater interface with the guide pin socket on the ion funnel, and then firmly press the heater interface until it snaps into place.
2. Tighten the two thumbscrews a few turns into the ion funnel.
3. Carefully slide the ion funnel into the ion source interface cage.
4. Tighten the two thumbscrews a few turns into the ion source interface cage.

Reinstall the injection filter

With a slotted screwdriver, attach the injection filter back to the ion source interface cage with the screws.

10.8.2 Reinstall the ion source interface

Procedure

1. Orient the ion source interface with the release latch at the top.
2. Carefully insert the ion source interface into the vacuum manifold.

3. Reinstall the ion source interface housing.
4. Start up the system

10.8.3 Clean the calibrant ion source

When you remove the API source interface from the instrument to clean the internal components, Thermo Fisher Scientific recommends that you also clean the areas of the calibrant ion source as noted in this section.

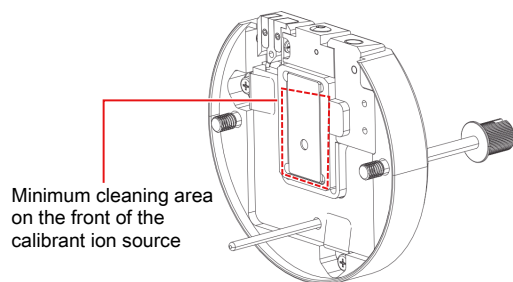
NOTE

You do not need to remove the calibrant ion source retainer or the calibrant ion source from the IC source heater interface to clean the designated areas.

Procedure

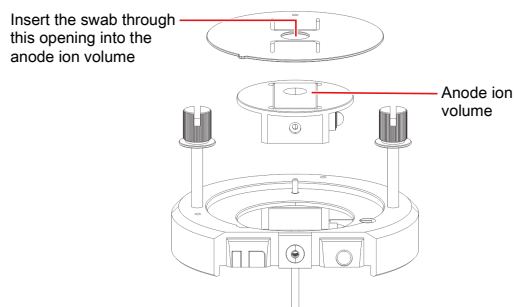
1. Remove the IC source heater interface.
2. Soak foam-tipped swabs in a 50:50 solution of methanol/water.
3. Clean the front of the IC source heater interface with the swabs, as shown in the figure.

Figure Front of the IC source heater interface



4. On the back of the IC source heater interface, insert a swab through the calibrant ion source retainer into the opening of the anode ion volume, and then clean inside the opening.

Figure Back of the IC source heater interface (exploded view)



5. Dry the component with nitrogen gas to make sure that all the solvent evaporates. Also purge through the gas connection at the top (facing front in the above figure).
6. Using a magnification device, inspect the opening for any residual lint or particulates.

NOTE

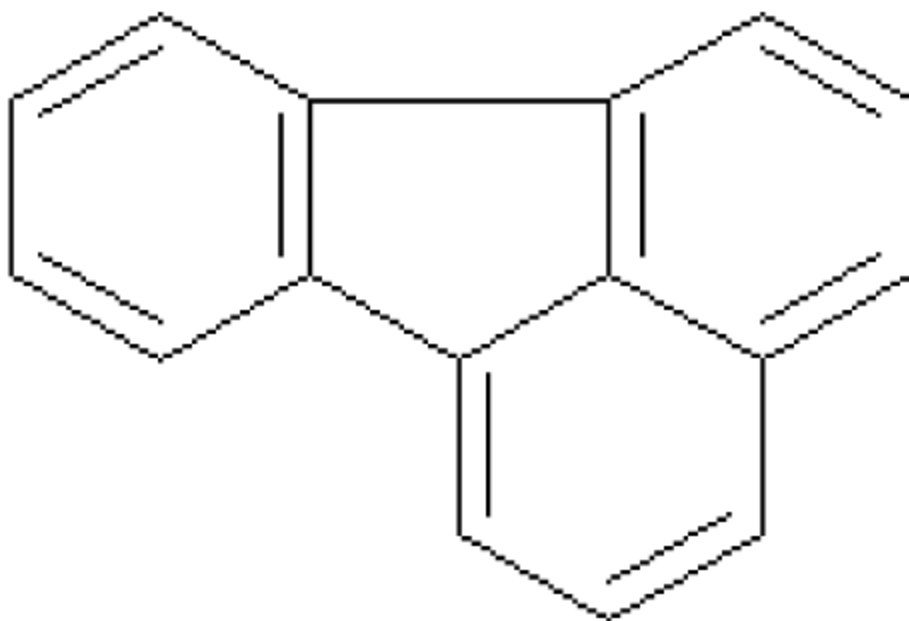
Inspect the inside surface and edges to confirm that no lint or particulates are present. Use plastic tweezers or a similar tool to remove the lint or particulate.

7. Reinstall the ion funnel and the IC source heater interface.
8. Reinstall the injection filter.
9. Reinstall the ion source interface.

10.8.4 Fluoranthene

The Orbitrap Astral/IC system uses fluoranthene as the calibrant species.

Figure Fluoranthene



The factory installs 0.15 g of fluoranthene, which is contained in a stainless steel vial, in the calibrant oven assembly. With continuous operation of the mass spectrometer, the calibrant is estimated to last for at least one year. Contact your local Thermo Fisher Scientific field service engineer when it is time to replace the calibrant vial. Only a Thermo Fisher Scientific field service engineer can order and replace the calibrant vial.

Fluoranthene is potentially hazardous. Use it in accordance with its Safety Data Sheet (SDS).

NOTE

Store and handle all chemicals in accordance with standard safety procedures. The Safety Data Sheet (SDS) describing the chemicals being used should be freely available to lab personnel for them to examine at any time. Safety Data Sheets (SDSs) provide summarized information on the hazard and toxicity of specific chemical compounds.

SDSs also provide information on the proper handling of compounds, first aid for accidental exposure, and procedures for cleaning spills or dealing with leaks. Producers and suppliers of chemical compounds are required by law to provide their customers with the most current health and safety information in the form of an SDS. Read the SDS for each chemical you use. Dispose of all laboratory reagents in the appropriate manner (see the SDS).

10.9 Maintenance of the LC systems or the autosamplers

For instructions on maintaining LCs or autosamplers, refer to the manual that comes with the LC or autosampler.

10.10 Consumables

For information on consumables such as fittings, nitrogen gas, or cleaning agents, refer to the *Orbitrap Astral Preinstallation Requirements Guide*, chapter *Consumables*.

For information on calibration- and test-chemicals or solvents, see Obtain the calibration solution.

See also Permitted materials for information about personnel safety.

10.11 Thermo Fisher Scientific Service

This section contains information concerning maintenance work that must be performed by Thermo Fisher Scientific personnel.

10.11.1 Returning parts



CAUTION

Hazardous chemicals

Hazardous material might contaminate certain parts of your system during analysis. To protect our employees, we ask you to adhere to special precautions when you send back parts to the factory for exchange or repair.

If hazardous materials have contaminated instrument parts, Thermo Fisher Scientific can only accept these parts for repair if they have been properly decontaminated.

Materials that might be toxic due to their structure and the applied concentration or that are reported in publications to be toxic are regarded as hazardous. Materials that will cause synergetic hazardous effects in combination with other materials present are also considered hazardous.

Parts that are contaminated by radioisotopes must not be returned to Thermo Fisher Scientific—neither under warranty nor within the exchange part program. If you are unsure whether parts of the system are possibly contaminated by hazardous material, make sure that the Thermo Fisher Scientific field service representative is informed before the representative starts to perform work on the system.

Your signature on the Health and Safety Form confirms that the returned parts have been decontaminated and that they are free of hazardous materials. You can download this form from the Help Portal. Instead, request a copy from the Thermo Fisher Scientific field service representative.

10.11.2 Services to be performed by Thermo Fisher Scientific service only

The table lists services that must be performed only by a Thermo Fisher Scientific field service engineer. Depending on the actual workload of your Orbitrap Exploris Series mass spectrometer, you might increase the maintenance frequency.

Table Thermo Fisher Scientific service procedures

MS Component	Procedure	Frequency
TMP	Change operating fluid reservoir	Every four years
	Change TMP bearing	Every four years

Replaceable parts

This chapter contains part numbers for replaceable and consumable parts for the mass spectrometer, data system, and kits. To ensure proper results in servicing the Orbitrap Astral system, order only the parts listed or their equivalent.

11.1 Ion source probes and parts

Refer also to the *OptaMax NG Ion Source User Guide* for more lists of replaceable parts for the OptaMax NG API source and available API probes. See also Obtain the calibration solution.

Item	Part number
APCI Probe Kit	OPTON-32101
APPI Probe Kit	OPTON-30350

11.1.1 Parts for ion source

Item	Part number
ESI Calibration spray probe with low flow needle	OPTON-20065
Low flow HESI needle	OPTON-30139
Tool, probe capillary length	80000-20957
Sweep cone	80100-20646
Ion transfer tube, high capacity	80500-20045
Removal tool for ITT letter box	BRE0061054
Graphite seal, for high capacity transfer tube	70005-20922

11.1.1.1 Source accessory parts

Item	Part number
Syringe 500 µL	365JL720
Syringe Adaptor Kit	70005-62011

11.1.1.2 Source drain parts

Item	Part number
Tubing, Tygon™, 1-3/8 in. OD, 1 in. ID, 3 m (10 ft)	00301-01-00020
Spiral coiled tube, PVC, ID=25 mm, OD=31 mm	BRE0011767
ASSY, WASTE BOTTLE, 4L	80100-60946

11.2 Instrument parts

11.2.1 Gas supply parts

Item	Part number
Teflon hose, OD 6 mm / ID 4 mm	0690280

11.2.2 Forevacuum parts

Item	Part number
nXL110iDE Dual-Inlet Enhanced Dry Pump Edwards	BRE0060500
Exhaust hose, 13 × 3.5, PVC	0690720
Spiral coiled tube, PVC ID=19 AD=26 L=0.8m	BRE0031483
Spiral coiled tube, PVC ID=38 AD=47 L=0.8m	BRE0067383 ¹⁾
KF/Hose adaptor; stainless steel KF40/ nipple, OD 38 mm	BRE0011722
Centering ring; NW 40, Viton/stainless steel	1168170
KF Clamping chain; synthetic, DN 40	1258190

Item	Part number
KF Elbow; stainless steel, DN 40	1258200
Tube clamp, 12-20 mm, W4	1005970

11.2.3 Ethernet switch and cables

Item	Part number
Switch	2108640
Patch cord, RJ45 SFTP, 5 m	2125930
Patch cord, RJ45 SFTP, 1m	2125920

11.2.4 Plug connectors (to connect other devices of the LC/MS system)

Item	Part number
COMBICON_MC1.5/8ST-3.81	2087270
KGG MC1.5/8 cable housing	2087280

11.2.5 Additional hardware

Item	Part number
Syringe pump Chemyx SKE10	BRE0011389
Divert/inject valve, Rheodyne MX Series II	00109-99-00046

Customer feedback

You are encouraged to report errors or omissions in the text or index. Send an email to the Technical Documentation at documentation.bremen@thermofisher.com. The PDF versions of our manuals allow adding comments with Adobe Acrobat Reader or other freely available PDF reader programs.

You can also send feedback about our product documentation through the Thermo Scientific Help Portal.

Procedure

1. Go to docs.thermofisher.com.
2. Enter the product name in the search bar and press ENTER (or select the search icon).
3. From the search results list, select a product manual or topic.
4. Select **Send Feedback**  .
The Feedback window opens.

TIP

If you are not signed in to the Thermo Scientific Help Portal, your default email application will generate an email with a basic template for providing your feedback.

5. Enter your feedback in the text box.
6. (Optional) Select **Drop Files to Attach, or Browse** to upload supporting documents and images to your submission.
7. Select **Send**.

Legal notices

Contents

- Copyright and trademarks
- FCC Compliance statement
- WEEE Compliance statement
- EU REACH statement
- EU Declaration of Conformity

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Original Operating Instructions

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Revision	Release date
B	June 2025
A	June 2023

General Laboratory Equipment. Not intended for use in diagnostic procedures.

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- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the receiver into an outlet on a circuit that is different from that to which the equipment is connected.
- Consult the dealer or an experienced radio or TV technician for help.

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This product is required to comply with the European Union's Waste Electrical & Electronic Equipment (WEEE) Directive 2012/19/EU. It is marked with the following symbol:

Figure WEEE symbol



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- Name of the manufacturer or distributor (where you purchased the product)
- Number of product pieces, and the estimated total weight and volume
- Pick-up address and contact person (include contact information)
- Appropriate pick-up time
- Declaration of decontamination, stating that all hazardous fluids or material have been removed from the product



This recycling program is not for biological hazard products or for products that have been medically contaminated. You must treat these types of products as biohazard waste and dispose of them in accordance with your local regulations.

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13.5 EU Declaration of Conformity

Figure EU Declaration of Conformity

Deutsch / Englisch German / English	Original
EU-Konformitätserklärung <i>EU Declaration of Conformity</i>	
ThermoFisher S C I E N T I F I C	
Thermo Fisher Scientific (Bremen) GmbH Hanna-Kunath-Str. 11 28199 Bremen, Germany	
Wir erklären hiermit, dass die folgenden Produkte <i>We hereby declare that the following products</i>	
Typ: <i>Type:</i>	Massenspektrometer <i>Mass Spectrometer</i>
Serie / Series: <i>(Modell / Model)</i>	Thermo Scientific Orbitrap Astral <i>Thermo Scientific Orbitrap Astral</i>
alle einschlägigen Anforderungen der folgenden Richtlinien erfüllen: <i>fulfill all the relevant requirements of the following directives:</i>	
Niederspannungsrichtlinie <i>Low Voltage Directive</i>	2014/35/EU <i>2014/35/EU</i>
Richtlinie über elektromagnetische Verträglichkeit <i>Electromagnetic Compatibility Directive</i>	2014/30/EU <i>2014/30/EU</i>
RoHS-Richtlinie <i>RoHS Directive</i>	2011/65/EU & (EU) 2015/863 <i>2011/65/EU and (EU) 2015/863</i>
Die folgenden einschlägigen harmonisierten Normen wurden zugrunde gelegt: <i>The following relevant harmonized standards were used:</i>	
DIN EN 61010-1:2020-03 <i>DIN EN 61010-1:2020-03</i>	DIN EN 61326-1:2013-07 <i>DIN EN 61326-1:2013-07</i>
Für die Zusammenstellung der technischen Unterlagen ist bevollmächtigt: <i>Person authorized to compile the technical file:</i>	
Rainer Bröring (Geschäftsführer) Thermo Fisher Scientific (Bremen) GmbH	
 Unterschrift <i>Signature</i>	Bremen, Juli 11, 2024 Datum <i>Date</i>
FORM-00500 Rev. 02	CMPLDOC00000263

Third party documents

This appendix contains excerpts from component suppliers' documents with safety-relevant information that may concern users of the Orbitrap Astral mass spectrometer.

14.1 Syringe pump

Observe the following safety guidelines when you operate the syringe pump.



CAUTION

Pinch hazard

Do not place fingers between the pusher block and end block while the pump is running.

Glossary

APCI source

Contains the APCI probe assembly, APCI manifold, and API stack.

API source

The sample interface between the liquid chromatograph (LC) and the mass spectrometer (MS).

Asymmetric Track Lossless analyzer (Astral)

A high-resolution accurate mass (HRAM) analyzer that shares some of the operational principles of other types of mass analyzers, such as ion trap, time-of-flight, and Orbitrap analyzers.

atmospheric pressure chemical ionization (APCI)

A soft ionization technique done in an ion source operating at atmospheric pressure. Electrons from a corona discharge initiate the process by ionizing the mobile phase vapor molecules. A reagent gas forms, which efficiently produces positive and negative ions of the analyte through a complex series of chemical reactions.

atmospheric pressure ionization (API)

Ionization performed at atmospheric pressure by using atmospheric pressure chemical ionization (APCI), electrospray ionization (ESI), or nanoelectrospray ionization (nanoESI or NSI).

atomic mass unit

Atomic Mass Unit (u) defined by taking the mass of one atom of carbon-12 as being 12u; unit of mass for expressing masses of atoms or molecules.

Automatic Gain Control™ (AGC)

Sets the ion injection time to maintain the optimum quantity of ions for each scan. With AGC on, the scan function consists of a prescan and an analytical scan.

auxiliary gas

The outer-coaxial gas (nitrogen) that assists the sheath (inner-coaxial) gas in dispersing and/or evaporating sample solution as the sample solution exits the APCI (optional), APPI (optional), ESI, or H-ESI spray insert. The mass spectrometer heats this gas to the user-specified vaporizer temperature.

centroid data

Data used to represent mass spectral peaks in terms of two parameters: the centroid (the weighted center of mass) and the intensity. The data is displayed as a bar graph. The normalized area of the peak provides the mass intensity data.

charge state

The imbalance between the number of protons (in the nuclei of the atoms) and the number of electrons that a molecular species (or adduct ion) possesses. If the species possesses more protons than electrons, its charge state is positive. If it possesses more electrons than protons, its charge state is negative.

collision energy

The energy used when ions collide with the collision gas.

collision gas

A neutral gas used to undergo collisions with ions.

collision-induced dissociation (CID)

A method of fragmentation where molecular ions are accelerated to high-kinetic energy and then allowed to collide with neutral gas molecules such as helium or nitrogen. The collisions break the bonds and fragment the ions into smaller pieces.

contact closure connection

The cable connection is from the external peripheral device to the mass spectrometer contact closure pins (Start In and Ground). The external device sends the contact closure (start) signal to the mass spectrometer.

divert/inject valve

A modular valve that can be plumbed as a divert valve to control the flow direction or as a loop injector for flow injection analysis.

electrospray ionization (ESI)

A soft ionization technique operating at atmospheric pressure. Ions are generated in solution and a high voltage is applied to generate small droplets that are then evaporated until all ions are in the gas phase.

Extensible Markup Language (XML)

A general-purpose markup language that is used to facilitate the sharing of data across different information systems, particularly via the Internet.

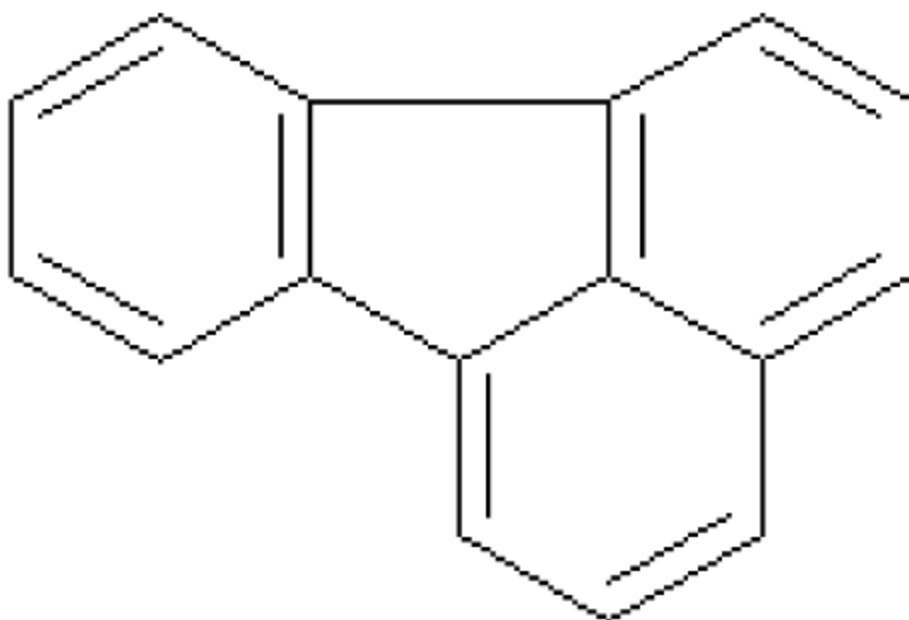
Field Service Engineer (FSE)

Refers to a team/individual who performs the on-premise installation and deployment. It can be Thermo Fisher Scientific IT staff or customer IT staff.

FlexMix

Calibration solution that is a mixture of 16 highly pure, ionizable components (mass ranges: 50 to 3,000 m/z), which is designed for both positive and negative ionization calibration.

fluoranthene



A reagent anion that is used in an electron transfer dissociation (ETD) experiment. This compound is not only used as reagent ion but also as internal calibration compound.

forevacuum pump, forepump

The pump that evacuates the foreline. A rotary-vane pump is a type of forepump. It might also be referred to as a backing, mechanical, roughing, or vacuum pump.

fragment ion

A charged dissociation product of an ionic fragmentation. Such an ion can dissociate further to form other charged molecular or atomic species of successively lower formula weights.

handshake

A signal that acknowledges that communication can take place.

heated-electrospray ionization (H-ESI)

A type of atmospheric pressure ionization that converts ions in solution into ions in the gas phase by using electrospray (ESI) in combination with heated auxiliary gas.

high-Field Asymmetric waveform Ion Mobility Spectroscopy (FAIMS)

An optional device for separating ions at atmospheric pressure. FAIMS provides ion separation by taking advantage of compound-dependent changes in ion mobility at high electric field strengths.

ID

inside diameter

ion optics

Focuses and transmits ions from the API source to the mass analyzer.

ion source

A device that converts samples to gas-phase ions.

ion-routing multipole (IRM)

The ion-routing multipole not only functions as a collision cell for HCD fragmentation, but also as an ion storage-routing device so ions can be routed to the linear ion trap and back to the Orbitrap via the IRM cell.

liquid chromatography / mass spectrometry (LC/MS)

An analytical technique in which a high-performance liquid chromatograph (LC) and a mass spectrometer (MS) are combined.

mass spectrometer

An instrument that ionizes sample molecules and then measures and analyses the ions according to their mass-to-charge ratio (m/z). The resulting mass spectrum is a characteristic pattern for the identification of a molecule.

mass spectrum

A graphical representation (plot) of measured ion abundance versus mass-to-charge ratio. The mass spectrum is a characteristic pattern for the identification of a molecule and is helpful in determining the chemical composition of a sample.

mass-to-charge ratio (m/z)

Used to denote the quantity formed by dividing the mass of an ion (in u or Da) by the number of charges carried by the ion. For example, for the ion $C_7H_7^{2+}$, $m/z=45.5$.

matrix-assisted laser desorption/ionization (MALDI)

A method of ionizing proteins where a direct laser beam is used to facilitate vaporization and ionization while a matrix protects the biomolecule from being destroyed by the laser.

multipole

A symmetrical, parallel array of (usually) four, six, or eight cylindrical rods that acts as an ion transmission device. An RF voltage and a DC offset voltage are applied to the rods to create an electrostatic field that efficiently transmits ions along the axis of the multipole rods.

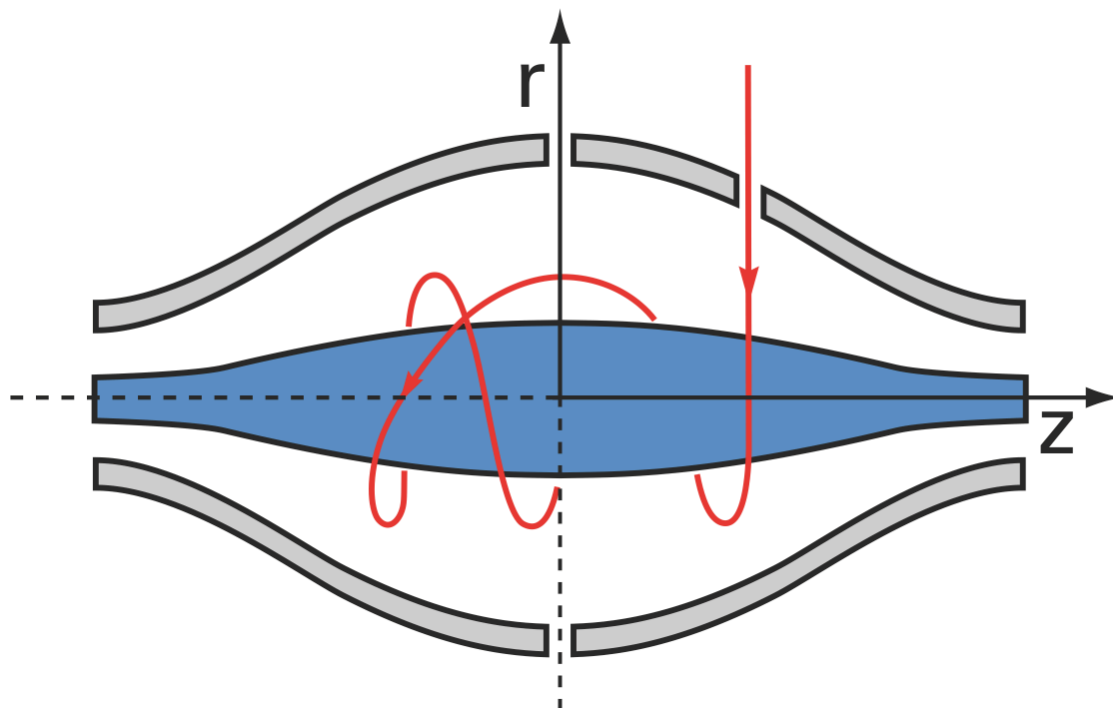
nanoelectrospray ionization (nano ESI or NSI)

A type of electrospray (ESI) that accommodates very low flow rates of sample and solvent on the order of 1 to 20 nL/min (for static nanospray) or 100 to 1,000 nL/min (for dynamic nanoelectrospray, also called nanoESI nanoLC gradient separation).

OD

outside diameter

Orbitrap mass analyzer (OT)



The Orbitrap™ mass analyzer consists of a spindle-shape central electrode surrounded by a pair of bell-shaped outer electrodes. Ions inside the mass analyzer orbit in stable trajectories around the

central electrode with harmonic oscillations along it. Two detection electrodes record an image current of the ions as they undergo harmonic oscillations. A Fourier transformation extracts different harmonic frequencies from the image current. An ion's mass-to-charge ratio m/z is related to the frequency f of its harmonic oscillations and to the instrumental constant k by: $m/z = k/f^2$

part number (P/N)

A part number (often abbreviated PN, P/N, part no., or part #) is an identifier of a particular part design or material used in a particular industry. Its purpose is to simplify reference to that item. A part number unambiguously identifies a part design within a single corporation,

quadrupole

A symmetrical, parallel array of four hyperbolic rods that acts as a mass analyzer or an ion transmission device. As a mass analyzer, one pair of opposing rods has an oscillating radio frequency (RF) voltage superimposed on a positive direct current (DC) voltage. The other pair has a negative DC voltage and an RF voltage that is 180 degrees out of phase with the first pair of rods. This creates an electrical field (the quadrupole field) that efficiently transmits ions of selected mass-to-charge ratios along the axis of the quadrupole rods.

reagent carrier gas

Ultra-high-purity nitrogen gas used to transfer the reagent to the reagent ion source that is regulated by the backpressure regulator.

resolution

The ability to distinguish between two points on the wavelength or mass axis.

RoHS

Restriction of Hazardous Substances. EU directive on the restriction of the use of certain hazardous substances in electrical and electronic equipment.

SDS

Safety Data Sheet

sheath gas

The inner coaxial gas (nitrogen), which is used in the API source to help nebulize the sample solution into a fine mist as the sample solution exits the ESI, H-ESI or APCI nozzle.

sweep gas

Nitrogen gas that flows out from the gap between the sweep cone and the ion transfer tube into the API source. Sweep gas aids in solvent declustering and adduct reduction.

syringe pump

A device that delivers a solution from a syringe at a specified rate.

turbomolecular pump (TMP)

A vacuum pump that provides a high vacuum for the mass spectrometer and detector system.

ultra high vacuum

Vacuum regime characterized by pressures lower than about 1×10^{-6} pascals (1.0×10^{-8} mbar; 7.5×10^{-9} Torr).

vacuum system

Components associated with lowering the pressure in the mass spectrometer. A vacuum system includes the vacuum manifold, pumps, pressure gauges, and associated electronics.

vent valve

A valve that allows the vacuum manifold to be vented to air or other gases. A solenoid-operated valve.

WEEE

European Union Waste Electrical and Electronic Equipment Directive. Provides guidelines for disposal of electronic waste.

XML (Extensible Markup Language)

A general-purpose markup language that is used to facilitate the sharing of data across different information systems, particularly via the Internet.

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- loop injector

M

- main power circuit breaker switch
 - function
 - function
 - location
 - location
 - location
 - usage
 - usage
 - usage
 - usage
- mains failure
- mains failure
- maintenance
 - turbomolecular pumps
 - vacuum system
- dry pump
- manual bakeout
- mass accuracy

- mass accuracy
- mass accuracy
- mass calibration
- mass calibration
- maximum vaporizer temperature
- mobile phones
- modifiers

N

- name plate
- nitrogen
 - connection to source
 - consumption
 - consumption
 - consumption
 - gas flow
 - inlet
 - pressure
 - pressure
 - purity
 - supply
- noise emission
- nominal voltage

O

- oil vapor
- operating fluid reservoir, of TMP

- Orbitrap analyzer
 - differential pumping
 - ion extraction
 - ion trajectories
 - lenses
 - measuring principle
 - principle
- ordering
 - calibration solution
 - replaceable parts
- overheat, avoiding
- Overvoltage Category
- oxygen, in the ion source
- oxygen, in the ion source

P

- packaging material
- part numbers
- particulate matter
- patents
- PDF files, of manuals
- performance
 - impediments
- peripheral supply port
- peripheral supply port
- peripherals
 - power supply

- power supply
- personal protective equipment (PPE)
- Pirani gauge
- placing, the instrument
- Pollution Degree
- power
 - failure
 - failure
 - frequency
 - fuses
- power conditioner
- power consumption
- power cords
 - forepump
 - forepump
 - instrument
- power fail detector
- power supply, for peripherals
- power supply, for peripherals
- syringe pump
- pressure, of nitrogen
- printed circuit boards
- product documentation
- Protection Class
- pumping, the system
- pumps, manuals

Q

- quadrupole

R

- rating plate
- reagents
- rebooting, of instrument
- API source housing
- removing
 - gases
 - stains
- reset button
- resetting, the instrument
- residual current device (RCD)
- resolution
- returning, parts
- parts
- parts
- room temperature

S

- Safety Data Sheet (SDS)
- safety glasses
- safety labels
 - instrument
- safety relay

- safety shoes
- serial number
- serial number
- sheath gas
- sheath gas
- sheath gas
- short circuits
- shutdown
- shutdown
- software
- solvent waste container
- solvent waste container
- solvent waste container
- solvent waste container
- solvent waste container
- solvents
 - for operating the instrument
 - recommendations
- source background flow
- source drain port
- source exhaust
- Source Installation Kit
- source vacuum pump
- spectrum
- split lens
- split lens
- spray cone
 - cleaning

- stains, removing
- standby condition
- static electricity
- Status LED
- Status LED
- sweep cone
 - cleaning
 - location
- switching on, the vacuum system
- switching valve
- switching valve
- switching valve
- syringe pump
 - description
 - power supply
 - usage
 - usage
- syringe, connecting to LC union
- system
 - bakeout
 - bakeout
 - rebooting
 - resetting
 - shutdown
 - standby
 - starting up
 - status LEDs
- System LED

T

- temperature
 - fluctuations
 - heated capillary
 - monitor
- third-party manuals
- TMP
 - maintenance
- trained personnel
- training
- transport RF multipole
- trapping gas
- trigger cable
- tripping characteristic
- turbomolecular pump
 - bearing
 - description
 - status

U

- uninterruptible power supply
- uninterruptible power supply
- USB ports

V

- vacuum

- deterioration
 - establishing
 - gauges
 - hoses
 - manifold
- vacuum system
 - maintenance
 - moisture
 - schematics
- vent prevent ball
- vent tubing
- vent tubing
- vent valve
 - function
 - location
- ventilation slots
- venting, waste drain tubing
- venting, waste drain tubing
- vibration
- voltage sags

W

- warning labels
 - location
- waste
 - container
 - drain tubing

- drain tubing
 - solvent
- weight, of instrument
- wheels
- window, at the ion source housing
- workbenches

X

- Xcalibur

Z

- Z-lens